

From Predictive Dynamics to Spectral Geometry

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Abstract

In Alain Connes' noncommutative geometry, geometry is encoded by an algebra, a Hilbert space, and a Dirac operator taken as fundamental spectral data. We show that these structures can instead emerge from a more primitive predictive dynamics. A compatible first-order variation on the induced history algebra canonically generates a real even Hodge–Dirac spectral triple, while the algebra alone determines neither its metric scale nor the full Dirac operator. The resulting spectral geometry is an information-reduced description: dynamically inequivalent predictive systems can share the same effective spectral data. Nevertheless, a finite commutator derivation determines a canonical centered Dirac implementer, with the remaining ambiguity recovered by the sharp finite number of independent real linear commutant coordinates, and on periodic three-dimensional lattice approximants the emergent spectral distance converges to a Riemannian metric reconstructed independently from the dynamics. At fixed vertex masses and symmetric conductances, the strictly positive stationary-current fibre is globally parametrized by the cohomology class of its edge affinities. Thus Connes-type spectral geometry can arise as an effective operator-geometric description of richer predictive dynamics, providing a concrete setting in which spectral structure is reconstructed rather than assumed at the microscopic entrance. Code and formalization information is given in the code-availability statement at the end of the main text.

Keywords. Noncommutative geometry; spectral triples; Renewal Geometry; predictive dynamics; Hodge–Dirac operators; Connes distance; stationary currents; spectral reconstruction.

Contents

1	Introduction	3
2	From predictive dynamics to spectral geometry	6
2.1	The Renewal Grand Tensor and the first-order differential interface	7
2.2	What the differential determines	8
2.3	Which spectral geometries can be reconstructed?	9
3	Independent metric reconstruction	10
3.1	An independent metric test	10
4	Spectral geometry as an information-reduced description	11
4.1	The stationary-current fibre	11
4.2	Positive memory and global monodromy	12
4.3	Componentwise coarse graining	12
5	Further mathematical extensions	13
5.1	From local spin discretization to a global convergence criterion	13
5.2	Indefinite memory beyond the positive branch	13
5.3	Joint realizations and completed inverse fibres	14
6	Interpretation and discussion	14
6.1	Spectral geometry as an effective description of predictive dynamics	14
6.2	Inverse fibres, scope, and open problems	16
	Mathematical Supplement	19

1 Introduction

Many spaces that arise naturally in mathematics and physics do not admit a satisfactory description as ordinary geometric spaces. Quotients may be singular, points may fail to capture the relevant observables, and the structures of interest may be encoded more faithfully by operators than by a classical manifold. In such situations the usual geometric language—built from points, coordinates, and functions on a well-behaved space—can cease to be adequate, even though an algebra of observables remains meaningful.

Noncommutative geometry was developed to address this problem. Connes’ programme replaces the requirement of an underlying point-set space by an operator-algebraic description in which an algebra plays the role of generalized coordinates. In the commutative case this remains compatible with ordinary geometry, since a suitable commutative algebra determines the underlying space through Gelfand duality. Allowing the algebra to become noncommutative extends the same principle to settings for which no classical space of points need exist.

The spectral formulation equips this generalized coordinate algebra with geometry. A compact spin manifold can be encoded by an algebra $\mathcal{A}$ represented on a Hilbert space $\mathcal{H}$, together with a Dirac operator D [1]. The representation supplies the operator-theoretic carrier, while D encodes differential and metric information. Under the reconstruction axioms for commutative spectral triples, this operator data recovers compact spin geometry [2, 3]; in the genuinely noncommutative case the same operator-geometric language continues to apply even when no ordinary underlying space exists.

Spectral triples consequently provide a common language for noncommutative spaces, differential calculi, index theory, and metric geometry. The Connes distance formula turns the commutator with D into a metric notion, making the Dirac operator simultaneously a differential and geometric object. Related work on compact quantum metric spaces formulates metric information at the level of an algebra equipped with a Lipschitz seminorm [4]. Rieffel’s quantum Gromov–Hausdorff distance and the later propinquity and spectral propinquity provide frameworks for convergence of quantum metric spaces and metric spectral triples [5, 6, 7], while real structures and gradings encode analogues of spin and orientation data.

A different structural question is whether the objects entering this spectral description must themselves be regarded as primitive. This is logically prior to spectral reconstruction: rather than asking which geometry is encoded by a given spectral triple, one asks whether the algebra, representation, differential structure, and Dirac data can arise from an underlying dynamics in which the persistent algebraic and Hilbert-space carrier is not initially prescribed, while the compatible differential is specified separately.

From predictive dynamics to spectral geometry. Renewal Geometry provides one setting in which this question can be posed operationally [8]. Its starting point is not a manifold, graph, causal order, algebra, Hilbert space, or spectral triple, but a finite operational process specified by composable intervention histories and their conditional future statistics. Histories that cannot be distinguished by any admissible future continuation are identified. The resulting future-equivalence quotient reconstructs the minimum persistent predictive carrier of the process rather than requiring such a carrier to be supplied independently.

Complete operational statistics then reconstruct a canonical positive history object, the *Renewal Grand Tensor*. We denote the Grand Tensor associated with a finite predictive system X by $\mathbb{T}_{\text{Grand}, X}$. Its role is pregeometric: it records which distinctions survive prediction, how represented histories compose, and how they overlap, before positions, distances, metrics, or continuum geometry have

been assigned. The corresponding history algebra and positive moment structure therefore arise from the operational process rather than being inserted as geometric input.

For the present paper, the relevant interface is the algebraic content carried by $\mathbb{T}_{\text{Grand},X}$. The Grand Tensor canonically supplies the future-minimal predictive carrier, the represented history algebra $\mathcal{A}_X^{\text{hist}}$, and its positive moment structure. On the finite algebra there is in particular an intrinsic normalized regular trace, which will provide one canonical choice of τ_X . To pass from this reconstructed algebraic object to spectral geometry, one then specifies a compatible first-order variation δ_X of the history algebra. Thus the interface used here is

$$\mathbb{T}_{\text{Grand},X} \longmapsto (\mathcal{A}_X^{\text{hist}}, \tau_X) \xrightarrow{\delta_X} (\mathcal{A}_X, \mathcal{H}_X, D_X, J_X, \gamma_X). \quad (1)$$

The first arrow is reconstructed directly from the operational statistics and is summarized in Section 2 and developed self-containedly in Section S6. The second is the Hodge–Dirac spectralization studied in this paper. The Grand Tensor is not identified with the displayed algebraic or spectral data: it retains the predictive carrier and complete positive history structure, including information that need not survive in the spectral image. Nor do we assume that the Grand Tensor or its algebra alone fixes δ_X ; the first-order variation is exactly the additional datum whose metric scale and Dirac ambiguity are analyzed below.

The same $\mathbb{T}_{\text{Grand},X}$ also supports the separate Lorentzian spacetime construction developed within the Renewal Geometry framework [8]. The present work asks instead whether Connes-type spectral geometry can arise from a compatible first-order variation of its reconstructed history algebra. The additional spacetime-specific assumptions of the Lorentzian construction are not used below; once a finite-dimensional unital C^* -algebra $\mathcal{A}$, a faithful tracial state τ , and a compatible real Hilbert-bimodule derivation δ are supplied, the finite spectralization results are ordinary operator-theoretic statements independent of the operational route by which $\mathcal{A}$ was obtained.

The resulting problem has both a forward and an inverse aspect. On the forward side one asks whether a first-order predictive differential canonically produces the structures of spectral geometry, and whether the resulting metric has an independent dynamical meaning. On the inverse side one asks which spectral geometries can be reached, what additional data are required to reconstruct a full Dirac operator, and whether different microscopic dynamics can have the same spectral image.

What lies before a spectral triple? Finite spectral geometry already has a substantial reconstruction and classification theory. Krajewski classified finite spectral triples in terms of matrix data and diagrams [9], while approximately finite C^* -algebras, weighted trees, ultrametric Cantor sets, fractals, stationary Bratteli diagrams, and aperiodic systems admit natural spectral-triple constructions from prescribed filtrations or hierarchical data [10, 11, 12, 13]. Related constructions attach spectral data to given C^* -dynamical systems, including crossed products, symbolic dynamics, and irreversible dynamics [14, 15, 16]. In these settings the algebraic or geometric carrier is already present before spectralization.

A closer operator-theoretic precedent comes from quantum Riemannian geometry and noncommutative potential theory. Majid constructs geometrically realized Dirac operators from a prescribed noncommutative coordinate algebra, differential calculus, quantum metric, and compatible bimodule connection [17]. Cipriani and Sauvageot realize tracially symmetric Dirichlet forms through closable derivations into Hilbert bimodules [18]; this calculus underlies Dirac constructions on fractals and spectral triples associated with Dirichlet, resistance, and sub-Markovian structures [19, 20, 21, 22]. These works are particularly close to the Hodge–Dirac layer used below, but they begin with the algebra and differential or Markov structure already specified.

The operational entrance has complementary precedents. Quantum combs and process-tensor formalisms encode complete multitime intervention statistics by positive causal process objects and describe memory-bearing dynamics without a Markov approximation [23, 24, 25]. Classically, observable operator models, predictive-state representations, and computational mechanics define state through future predictions rather than hidden microscopic labels [26, 27, 28, 29, 30, 31, 32]. The predictive construction used here combines this future-equivalence principle with multitime operational composition and positive process structure: the persistent carrier and represented history algebra are reconstructed only after future-invisible distinctions have been removed, and a compatible first-order differential is then spectralized on that reconstructed algebra.

The resulting question is whether the algebra, Hilbert-space carrier, differential structure, and Dirac data of noncommutative geometry can arise from an underlying predictive dynamics in which the persistent carrier is not initially prescribed and the differential input is made explicit. A nontrivial answer requires more than reproducing a desired spectral triple by definition: one must identify what the first-order differential actually fixes, which spectral geometries are obtainable under different reconstruction categories, whether the resulting metric has an independent dynamical meaning, and what information is lost under spectralization.

The last point is essential. If distinct predictive systems have the same spectral image, then a spectral triple is not, in general, a complete invariant of the dynamics from which it arose. It may instead describe an equivalence class of microscopically distinct systems that agree on their effective metric and differential structure. In this sense spectral geometry can be *information-reduced*: complete for the operator-geometric information it encodes while insufficient, in general, to reconstruct the richer underlying dynamics.

Our contribution. We first construct a canonical spectralization of finite differential data. A star-compatible derivation δ into a real Hilbert $\mathcal{A}$ -bimodule determines a minimal one-form carrier and a real even Hodge–Dirac spectral triple with exact order-zero and first-order relations. Two rigidity results identify what this construction does not determine. The algebra and faithful trace do not fix an absolute Dirac scale, and a fixed commutator derivation determines the full compatible Dirac operator only modulo a finite-dimensional self-adjoint commutant. The orthogonally centered representative is the unique minimum-Hilbert–Schmidt-norm implementer, and the dimension of this commutant is the sharp number of independent real linear coordinates required to distinguish the remaining Dirac ambiguity.

This distinction separates an *austere Hodge construction* from an enlarged *calibrated marked reconstruction*. The bare Hodge operator always has the unit zero-mode and therefore cannot realize arbitrary finite Dirac operators. When the exact $q_S = \dim_{\mathbb{R}} \mathfrak{c}_S$ scalar commutant anchors are retained, every finite spectral packet is reconstructible, and fewer independent real linear coordinates cannot distinguish the full affine Dirac fibre. Every separable compact-resolvent packet arises as a strong cofinal limit of finite marked screens. A stricter requirement, in which multiplication must be transported asymptotically in norm, is genuinely selective and is characterized exactly by spectral quasidiagonality.

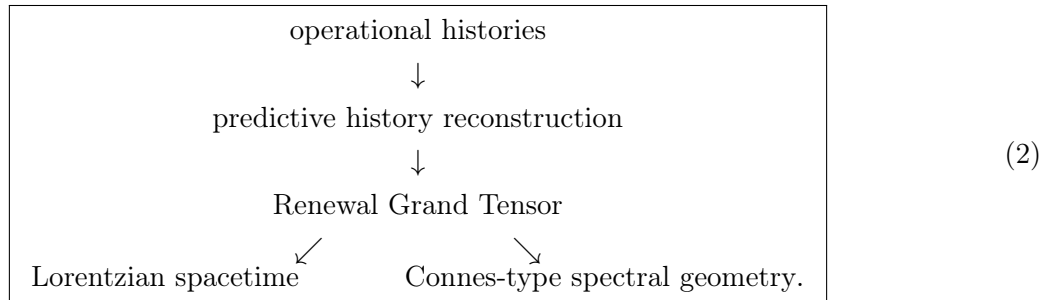
We next test whether the construction has independent geometric content. For stationary graph dynamics, the symmetric jump data determine a Hodge–Dirac operator whose square on zero-forms is -2 times the symmetric Markov generator and whose commutator seminorm is its local quadratic form. On periodic A_3 approximants, the corresponding Connes distance converges uniformly to a Riemannian metric reconstructed independently from the jump second moments. In the spin extension, the local positive-stencil and covariant Wilson construction gives the required chartwise estimates; on flat periodic spin tori these estimates close globally and give unconditional

norm-resolvent convergence, while on general closed Riemannian spin manifolds the same conclusion follows under the stated compatible stable-spin discretization hypotheses. The independent scalar metric reconstruction is unconditional.

Finally, we characterize several mechanisms by which spectralization loses information. Directed stationary current can vary at fixed metric spectral geometry, hidden dynamical memory can vary at fixed static spectral data, and locally identical spectral packets can differ by global marked monodromy. In their stated classes, these inverse fibres have intrinsic coordinates and compatible coarse-graining maps. Thus spectralization is not merely noninjective: the information it forgets has its own mathematical structure and transformation laws.

The two principal graph results are the uniform A_3 metric limit and the identification of the strictly positive stationary-current fibre with $H^1(G; \mathbb{R})$ through its affinity class. They form the mathematical core, together with the finite differential and operational reconstruction theory. The spin, rational-memory, and completed-fibre constructions are further extensions, each with its own additional hypotheses; they are not needed for those two graph theorems.

The resulting interpretation is that Connes-type spectral geometry can arise as an effective geometric description of a richer predictive dynamics rather than necessarily as its most primitive level. The independent Lorentzian construction [8] shows that Lorentzian spacetime and spectral geometry can both arise downstream of the same predictive structure, without either being required as the primitive entrance to the other. The two constructions therefore lead to the schematic picture



The Lorentzian branch is developed in [8]. The present paper develops the spectral branch: its canonical finite construction, its category-dependent image, its independent metric test, and its inverse fibres. The central question is therefore not only how spectral geometry can arise, but what a richer underlying dynamics retains that must be forgotten when it is compressed to a spectral description.

Reading the paper. The main text gives the predictive construction, the principal theorem statements, and their interpretation. Part I of the Mathematical Supplement contains the self-contained core proofs. Part II retains the additional spin, dilation, indefinite-memory, and joint-realization results. The shared notation and cross-references make the extensions available without making them prerequisites for the graph theorems.

2 From predictive dynamics to spectral geometry

The spectralization theorem is algebraically self-contained, but its motivating entrance is a finite predictive process. We retain the operational description and the Renewal Grand Tensor interface here; the full reconstruction, realization, and recognition arguments are given in Section S6. This separates the meaning of the construction from its technical proofs without changing the input data.

2.1 The Renewal Grand Tensor and the first-order differential interface

Fix a finite cutoff X , a finite typed event grammar, and the reachable histories. The conditional event law determines the future cylinder probabilities

$$p_X(w \mid h) = \prod_{j=1}^k p_X(a_j \mid ha_1 \cdots a_{j-1}), \quad w = a_1 \cdots a_k. \quad (3)$$

Histories ending at the same cut type are identified when every declared future continuation has the same conditional probability. Thus

$$h \sim_X h' \iff p_X(w \mid h) = p_X(w \mid h') \text{ for every admissible future } w, \quad Z_{X,x}^{\min} = H_{X,x} / \sim_X. \quad (4)$$

The branch probabilities are $\kappa_{X,a}([h]) = p_X(a \mid h)$, and the update $T_{X,a}([h]) = [ha]$ is defined only on branches with positive probability. This partial-transition convention matters: no conditional successor is inferred from a zero-probability extension. The quotient is the coarsest reachable deterministic predictive realization for the specified futures.

To obtain the operator-algebraic history object, the input is strengthened to complete multitime intervention tables on specified finite operator frames. The reconstructed Choi tensors must be positive and satisfy the causal prefix identities; nonnegative entries in a probability table alone do not replace these conditions. The external intervention ports are part of this operational input. The persistent memory representation is reconstructed using the coherent isometric realization that is minimal at every prefix.

Its reduced branches and their formal Hilbert–Schmidt adjoints generate a represented process algebra. The formal adjoints are algebraic operations, not necessarily executable reverse channels. Quotienting the two-sided ideal invisible to all retained readout contexts gives the future-separated history algebra $\mathcal{A}_X^{\text{hist}}$. Its intrinsic normalized regular trace is

$$\hat{\tau}_{X,\text{reg}}(a) = \frac{\text{Tr}_{\mathbb{C}}(L_a)}{\dim_{\mathbb{C}} \mathcal{A}_X^{\text{hist}}}, \quad \mathbb{K}_{X,r}(v, w) = \hat{\tau}_{X,\text{reg}}([v]^*[w]), \quad |v|, |w| \leq r. \quad (5)$$

The Gram kernels record the word relations. When two successive word spans have equal rank, the finite panel determines the multiplication, involution, and all source words; see Theorem S6.6. These are exact finite-data statements, rather than a statistical recovery claim from an incomplete or noisy table.

Definition 2.1 (Finite Renewal Grand Tensor). The finite Renewal Grand Tensor associated with the operational process X is

$$\mathbb{T}_{\text{Grand},X} = \left(\mathcal{A}_X^{\text{hist}}, \{ \mathbb{K}_{X,r} \}_{r \geq 0}, Z_X^{\min} \right). \quad (6)$$

It is a complete positive finite history-moment object: $Z_X^{\min}$ records which past distinctions remain predictively effective, $\mathcal{A}_X^{\text{hist}}$ records how future-visible represented histories compose, and the Gram hierarchy records their intrinsic positive overlaps.

The reconstruction in Proposition S6.7 therefore supplies this finite history object. Conversely, the boundary-relative realization and recognition results retain representation multiplicities: preparations and terminal readouts remain external contexts, and are not silently adjoined to the internal history algebra. The separate dimension-minimal recognition argument is not a uniqueness theorem for arbitrary physical implementations of the coherent process representation.

First-order differential interface. The Grand Tensor is logically upstream of spectral geometry, but it is important not to identify algebraic reconstruction with metric differentiation. From (6) we canonically obtain the history algebra and its positive finite-dimensional structure. We take

$$\mathcal{A} = \mathcal{A}_X^{\text{hist}}, \quad \tau = \widehat{\tau}_{X,\text{reg}}, \quad (7)$$

or, more generally, any declared faithful tracial state on $\mathcal{A}$. To spectralize this algebra one additionally specifies a compatible first-order variation. Abstractly, this is a Hilbert $\mathcal{A}$ -bimodule $\mathcal{K}$, an antiunitary involution $C_{\mathcal{K}}$ reversing the bimodule actions, and a complex-linear map

$$\delta : \mathcal{A} \longrightarrow \mathcal{K} \quad (8)$$

satisfying

$$\delta(\mathbf{1}) = 0, \quad \delta(ab) = a\delta(b) + \delta(a)b, \quad C_{\mathcal{K}}\delta(a) = \delta(a^*). \quad (9)$$

When this datum comes from a Renewal process, δ represents the declared first-order variation of the reconstructed predictive history algebra. The results below do not assume that $\mathbb{T}_{\text{Grand},X}$ or $\mathcal{A}$ alone uniquely determines this variation. This distinction is essential: the no-scale result below shows that rescaling δ changes the spectral metric while leaving the algebra and trace fixed.

The logical interface is therefore

$$\boxed{\text{operational histories} \longrightarrow \mathbb{T}_{\text{Grand},X} \longmapsto (\mathcal{A}_X^{\text{hist}}, \tau_X) \xrightarrow{\delta_X} (\mathcal{A}_X, \mathcal{H}_X, D_X, J_X, \gamma_X).} \quad (10)$$

The first arrows reconstruct predictive and algebraic structure; the final arrow is the Hodge–Dirac spectralization studied in this paper.

The zero-form carrier is

$$\mathcal{H}_0 = L^2(\mathcal{A}, \tau), \quad (11)$$

and the one-form carrier is the minimal quotient of the universal calculus generated by δ ,

$$\mathcal{H}_1(\delta) = \Omega_u^1(\mathcal{A}) / \text{Ker } T_\delta, \quad \Omega_u^1(\mathcal{A}) = \text{Ker}(m : \mathcal{A} \otimes \mathcal{A} \rightarrow \mathcal{A}). \quad (12)$$

On $\mathcal{H} = \mathcal{H}_0 \oplus \mathcal{H}_1(\delta)$, the induced derivation ∂ gives

$$D = \begin{pmatrix} 0 & \partial^* \\ \partial & 0 \end{pmatrix}, \quad \gamma = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \quad (13)$$

while reversal of universal one-forms gives the real structure J . Thus the spectral triple is constructed from a first-order differential on an algebra that has itself already been reconstructed from predictive histories.

2.2 What the differential determines

The finite Hodge construction is canonical once δ is fixed. It produces a real even spectral triple with exact order-zero and first-order relations. The spectral scale is not already hidden in $(\mathcal{A}, \tau)$: replacing δ by $s\delta$ leaves the algebra and trace unchanged but sends

$$D \longmapsto sD, \quad L_D \longmapsto sL_D, \quad d_D \longmapsto s^{-1}d_D. \quad (14)$$

Thus an absolute metric scale enters through the differential data.

There is a second, finite ambiguity. For a marked spectral packet $S = (\mathcal{A}, \mathcal{H}, \pi, D, J, \gamma)$ define

$$\mathfrak{c}_S = \{C \in \pi(\mathcal{A})'_{\text{sa}} : C\gamma + \gamma C = 0, JC = \varepsilon' C J\}. \quad (15)$$

All compatible self-adjoint Dirac operators implementing the same commutator derivation form exactly the affine space

$$D + \mathfrak{c}_S. \quad (16)$$

The orthogonally centered representative $D_{\text{der}} = D - P_{\mathfrak{c}_S} D$ is the unique minimum-Hilbert–Schmidt-norm implementer, and $q_S = \dim_{\mathbb{R}} \mathfrak{c}_S$ real linear scalar measurements are necessary and sufficient to distinguish every point of the fibre.

The austere Hodge construction has an additional visible restriction: $D(\mathbf{1}, 0) = 0$. It therefore cannot be surjective onto arbitrary finite Dirac operators. This is why the inverse statement below uses an enlarged calibrated marked reconstruction rather than the bare Hodge functor.

2.3 Which spectral geometries can be reconstructed?

At the finite marked level, the represented algebra and symmetry marks are retained together with the commutator derivation and exactly the independent real linear coordinates required to resolve its compatible commutant fibre. If $q_S = \dim_{\mathbb{R}} \mathfrak{c}_S$, then q_S independent real linear coordinates are sufficient and, in general, necessary. At the strong compact level, finite spectral screens converge strongly on fixed vectors, resolvents, commutators, and matrix coefficients. At the norm-multiplicative spectral-compression level, multiplication must additionally be transported asymptotically in norm.

Theorem 2.2 (Finite Hodge–Dirac reconstruction and commutant coordinates). *The following statements hold.*

- (i) *Every finite real differential datum $(\mathcal{A}, \tau, \delta)$ determines the real even Hodge–Dirac spectral triple above, functorially under trace- and differential-preserving $*$ -isomorphisms.*
- (ii) *The algebra and trace do not determine the Dirac scale, and a fixed commutator derivation determines the full compatible Dirac operator exactly modulo $\mathfrak{c}_S$.*
- (iii) *For the enlarged calibrated marked reconstruction,*

$$\begin{aligned} \text{EssIm}(\mathfrak{S}_{\text{fin}}^{\text{cal}}) &= \mathbf{Spec}_{\text{fin}}, \\ \text{EssIm}(\mathfrak{S}_{\text{cpt}}^{\text{strong, cal}}) &= \mathbf{Spec}_{\text{cpt}}^{\text{sep}}, \\ \text{EssIm}(\mathfrak{S}_{\text{cpt}}^{\text{norm, cal}}) &= \mathbf{Spec}_{\text{cpt}}^{\text{sqd}} \subsetneq \mathbf{Spec}_{\text{cpt}}^{\text{sep}}. \end{aligned} \quad (17)$$

Thus every finite spectral packet is reconstructible from its represented commutator data plus exactly q_S independent real linear commutant coordinates; every separable compact-resolvent packet is a strong cofinal limit of finite anchored screens; and norm-multiplicative spectral-compression realizability is equivalent to spectral quasidiagonality.

The distinction between the austere Hodge image and the anchored reconstruction is essential. The commutator derivation fixes a canonical centered implementer, while the q_S anchors recover precisely the finite component invisible to commutators; no full generator germ is retained. The complete definitions, sharp minimality statement, and essential-image proof are given in [Section S1](#) of the mathematical supplement.

At the compact-resolvent level, “compression” refers to the represented matrices and the convergence conditions of Definition S1.9. The compressed observables are generally an operator system rather than a representation of the original algebra, and are not asserted to inherit every real first-order axiom. The strong and norm-multiplicative assertions are separated in Theorem S1.11.

3 Independent metric reconstruction

Producing an operator called D is not by itself evidence that geometry has emerged. A stronger test is available when the same finite dynamics determines a metric by a route that does not use the Dirac operator.

3.1 An independent metric test

For a stationary finite graph, let $m_x > 0$ be the stationary masses, $q_{xy} = m_x k_{xy}$ the stationary fluxes, and

$$c_{xy} = \frac{q_{xy} + q_{yx}}{2}, \quad j_{xy} = \frac{q_{xy} - q_{yx}}{2}. \quad (18)$$

The symmetric conductances determine the graph Hodge–Dirac operator $D_{m,c}$. Exactly,

$$D_{m,c}^2|_{\ell^2(V,m)} = -2\mathcal{L}_s, \quad \|[D_{m,c}, f]\|^2 = \max_x \frac{1}{m_x} \sum_y c_{xy} |f(y) - f(x)|^2. \quad (19)$$

Thus the Dirac square on zero-forms is -2 times the symmetric Markov generator, and its commutator seminorm is the local quadratic form of that generator.

For the periodic root graphs, let $W_0 = \{x \in \mathbb{R}^4 : \sum_i x_i = 0\}$, $\Lambda_{A_3} = W_0 \cap \mathbb{Z}^4$, and $V_h = h\Lambda_{A_3}/\Lambda_{A_3}$, where $h = d^{-1}$ and $d \geq 3$. The twelve oriented roots are $\Phi_{A_3} = \{e_i - e_j : i \neq j\}$. With masses h^3 and root-edge rates $1/(8h^2)$, their jump covariance is

$$Q_h = \sum_{\alpha \in \Phi_{A_3}} \frac{(h\alpha) \otimes (h\alpha)}{8h^2} = I_{W_0}, \quad g = Q_h^{-1}. \quad (20)$$

This calculation specifies the flat metric without using D_h . The commutator normalization then gives

$$L_h(f)^2 = \max_{x \in V_h} \frac{1}{8h^2} \sum_{\alpha \in \Phi_{A_3}} |f(x + h\alpha) - f(x)|^2, \quad (21)$$

with the following uniform metric conclusion.

Theorem 3.1 (Uniform A_3 metric recovery). *The pure-state Connes distance satisfies*

$$\sup_{x,y \in V_h} |d_h^{\text{Connes}}(x,y) - d_g(x,y)| \longrightarrow 0, \quad (22)$$

where d_g is the flat geodesic distance determined by the jump covariance.

This is the principal independent metric test of the paper. The proof in Section S2.3 controls the entire discrete Lipschitz ball, not merely samples of smooth functions: interpolation gives compactness, the quadratic root constraint identifies the limiting unit gradient bound, and uniformly smoothed distance functions give the reverse estimate.

What the limit does and does not assert. The theorem compares vertices, that is, pure states. It is not by itself a convergence theorem for the full quantum state-space Lip-norms or for the Hodge operator in norm resolvent. Indeed, retaining both oriented edges gives

$$\dim \text{Ker } D_{m,c} = 2|E| - |V| + 2, \quad (23)$$

which is $11|V| + 2$ on the periodic A_3 root graph. A compatible limit retaining all these zero modes cannot have compact resolvent. The spin construction in Section 5 uses a different, additionally marked operator and does not remove this obstruction by assertion.

There is also an independent endpoint-block construction on every closed connected Riemannian manifold. Its seminorm is the ordinary graph Lipschitz seminorm and its Connes distance is exactly the graph shortest-path metric; refining sufficiently dense meshes gives uniform geodesic convergence (Theorem S4.1). This is a complementary construction, not an extension of the quadratic A_3 Hodge theorem by the same proof.

4 Spectral geometry as an information-reduced description

The spectral map is not injective. The stationary-current theorem makes the loss of directional information exact, while positive dynamical memory and marked monodromy give two complementary inverse problems. Keeping these reductions distinct is essential: they forget different data and obey different coarse-graining laws.

4.1 The stationary-current fibre

For a connected graph with fixed masses m and conductances c , write

$$\overline{\mathcal{J}}(G, c) = \{j \in \text{Ker } B : |j_e| \leq c_e\}, \quad \mathcal{J}(G, c) = \{j \in \text{Ker } B : |j_e| < c_e\}. \quad (24)$$

The closed polytope is the full nonnegative-rate stationary realization fibre. Its relative interior $\mathcal{J}(G, c)$ is the fibre with strictly positive flux in both directions, and has dimension $b_1(G) = |E| - |V| + 1$. Every point has the same metric Hodge–Dirac triple, although directed path probabilities, time-reversal asymmetry, and entropy production may differ.

For a strictly positive realization, its edge affinity is

$$a_e = \text{arctanh} \frac{j_e}{c_e} = \frac{1}{2} \log \frac{q_e}{q_{\bar{e}}}. \quad (25)$$

Adding a vertex potential $B^\top \phi$ changes the representative but not the cohomology class, or equivalently its sums around cycles. The nonlinear stationarity equation selects precisely one representative from each such class.

Theorem 4.1 (Cohomological reconstruction of directed stationary dynamics). *For fixed connected (G, m, c) with $c_e > 0$, the map*

$$\mathfrak{A} : \mathcal{J}(G, c) \longrightarrow H^1(G; \mathbb{R}), \quad j \longmapsto [\text{arctanh}(j/c)] \quad (26)$$

is a diffeomorphism. For a representative a_0 of a class α , minimize

$$\sum_e c_e \log \cosh(a_{0,e} + (B^\top \phi)_e) \quad \text{over} \quad \sum_x \phi_x = 0. \quad (27)$$

The unique minimizer gives $a_\alpha = a_0 + B^\top \phi_\alpha$, $j_\alpha = c \tanh a_\alpha$, and the complete directed generator $k_{xy} = (c_{xy} + j_{\alpha,xy})/m_x$.

The full proof, including coercivity, strict convexity, the positivity domain, and smooth dependence, is Theorem S3.7. The variational selection specializes nonlinear Hodge theory [33]; the reconstruction statement identifies that selector with the entire fixed-traffic stationary Markov fibre. The affinity is half the logarithmic forward/reverse stationary-flux ratio; this convention fixes the factors in the entropy formula.

In particular, geometry need not determine irreversibility. On a three-cycle with equal conductance c , every constant current $j_e = s$, $|s| < c$, gives the same metric triple, whereas

$$\sigma(s) = 6s \log \frac{c+s}{c-s} \quad (28)$$

changes with s . More generally $\sigma(j) = 4 \sum_e j_e a_e$, so the current-affinity pairing is invariant under adding an exact cochain. The additional cohomology class restores the directed generator on the strictly positive fibre without changing its metric triple.

4.2 Positive memory and global monodromy

Hidden positive memory gives a second inverse problem. Eliminating auxiliary degrees of freedom produces a Stieltjes self-energy

$$\Sigma_\mu(z) = \int_{(0,\infty)} \frac{1}{\lambda - z} d\mu(\lambda) = B_\mu(C_\mu - z)^{-1} B_\mu^*. \quad (29)$$

The full operator-valued measure determines the source-minimal positive dilation, whereas a static effective operator may retain only the inverse moment $\Sigma_\mu(0) = \int \lambda^{-1} d\mu(\lambda)$. Section S7 distinguishes static data from the full self-energy, including examples where all integer moments agree but the self-energies differ. Section S10 retains the explicit minimal dilation construction and proof.

The third inverse problem is global. Locally marked-isomorphic spectral triples with specified edge identifications on a connected graph are classified by

$$[\rho] \in \text{Hom}(\pi_1(\Gamma), G_S)/G_S. \quad (30)$$

Trivial monodromy gives a marked trivialization compatible with those identifications; nontrivial monodromy obstructs it. This concerns marked flat transport, not a claim that every underlying unmarked vector bundle is nontrivial. The complete classification is in Section S8.

4.3 Componentwise coarse graining

These inverse data have different natural effective maps. Current is pushed through the graph incidence chain map; a positive-memory cutoff removes the high-energy measure while retaining its static Schur correction; and monodromy descends only when the contracted cycles have trivial holonomy.

Theorem 4.2 (Coarse graining of the core inverse data). *Under the separate quotient, integrability, and descent hypotheses of Theorems S3.9, S7.7, and S8.5, current aggregation contracts entropy production, positive-memory cutoff preserves the zero-energy Schur complement, and marked monodromy descends exactly when $\text{Ker } q_* \subseteq \text{Ker } \rho$. These operations compose componentwise under compatible graph quotients, nested thresholds, and nested effective-space isometries.*

Surjectivity on linear cycle spaces does not assert surjectivity of the bounded current polytopes. Nor do these three examples alone exhaust the inverse fibre of an arbitrary predictive system. The additional joint-realization and completed-fibre framework is stated separately in Section S12, with its retained-coordinate and completion hypotheses explicit.

5 Further mathematical extensions

The preceding graph theorems and their finite framework form the common mathematical core. The extended version also retains results on continuum spin operators, indefinite memory, and joint realization spaces. Their proofs are collected in Part II of the Mathematical Supplement. None is a prerequisite for the core A_3 metric or stationary-current theorem.

5.1 From local spin discretization to a global convergence criterion

The spin problem is subtler. Local positive stencils reconstruct the inverse metric, and a density-symmetric covariant Wilson discretization removes lattice doublers while remaining $O(h)$ on sampled smooth spinors. These chartwise estimates do not by themselves guarantee that independently discretized charts assemble into one stable global operator. The mathematical supplement therefore isolates the additional global approximation properties in the notion of a *compatible stable spin discretization*.

Theorem 5.1 (Additional spin continuum results). *The following statements hold.*

- (i) *The covariant Wilson construction has one low-energy species and satisfies the local uniform graph estimates required for a spin continuum limit.*
- (ii) *On a flat periodic spin torus with constant metric and the periodic spin structure, the uniform covariant Wilson discretization satisfies the global stability and core-consistency conditions and converges in generalized norm-resolvent sense to the doubled continuum Dirac operator.*
- (iii) *Let $(M^d, g, \mathfrak{s})$ be closed. If the finite covariant Wilson operators form a compatible stable spin discretization in the precise sense of Definition S9.7, then for every $z \in \mathbb{C} \setminus \mathbb{R}$,*

$$\left\| W_h(D_h - z)^{-1} W_h^* - (\hat{D}_{M, \mathfrak{s}} - z)^{-1} \right\| \rightarrow 0, \quad \hat{D}_{M, \mathfrak{s}} = D_{M, \mathfrak{s}} \otimes \sigma_1. \quad (31)$$

The theorem separates three issues that should not be conflated: the extensive kernel of the unreduced universal graph calculus, the Wilson control of the spinor sector, and the additional atlas compatibility needed on a general closed manifold. Flat periodic spin tori provide an unconditional global spin class; the general norm-resolvent statement is a convergence criterion for compatible stable assemblies. The scalar metric approximation is unconditional. The precise local estimates, atlas definition, periodic corollary, and compactness argument are given in Section S9.

The spin lift, global spin structure, and stable reconstruction maps are additional inputs. The Wilson family is a separate discretization of the doubled spin Dirac operator; it is not identified with the unreduced one-step graph Hodge operator. Its unique low-energy lattice species is understood within that fixed doubled Clifford convention.

5.2 Indefinite memory beyond the positive branch

For normalized rational Hermitian self-energies, the coefficient-Hankel construction gives a finite source-minimal Pontryagin realization. Its ordinary dimension is the stabilized Hankel rank and its negative index is the number of negative squares of the corresponding Nevanlinna kernel (Theorem S11.2). The pole-Hankel invariants and the finite descriptor extension for polynomial parts are retained in Section S11 rather than incorporated into the positive Stieltjes theorem. They use their own indefinite inner-product and realization conventions; no arbitrary positive-memory model is assigned a negative index.

5.3 Joint realizations and completed inverse fibres

Prescribed marginal realizations need not have a common realization merely because their pairwise overlaps are admissible. Positive block correlation operators classify minimal common Gram realizations, with support rank giving the minimum common dimension and a stabilizer recording the remaining gauge freedom (Theorem S12.1). The three-block example in Proposition S12.2 makes the obstruction explicit.

The broader finite-fibre and completion statements retain this joint Gram coordinate together with the declared current, memory, marking, calibration, and monodromy data. They concern the specified fully reduced retained-coordinate class, not an unconditional exhaustive classification of all microscopic predictive dynamics. Compactification, compatible restriction maps, resolvent and weight data, and cofinal separation enter the completion hypotheses. Within that setting, the stable-image pruning lemma distinguishes finite-stage points that extend indefinitely from those that do not. The resulting inverse-limit description is kept in Section S12 with these assumptions rather than promoted to a consequence of the two graph theorems.

6 Interpretation and discussion

The results separate predictive reconstruction from operator-theoretic spectralization in a useful way. Renewal Geometry first reconstructs the Grand Tensor $\mathbb{T}_{\text{Grand},X}$, including the future-minimal predictive carrier, represented history algebra, and positive moment hierarchy. Spectralization then equips that reconstructed algebra with a compatible first-order differential datum δ_X . Once $(\mathcal{A}, \tau, \delta)$ is given, the finite Hodge–Dirac theorem is an ordinary statement about C^* -algebras, Hilbert bimodules, and derivations. The predictive reconstruction is therefore logically prior to the spectral triple without being an extra hypothesis of the abstract operator-theoretic result.

6.1 Spectral geometry as an effective description of predictive dynamics

The first consequence is a change in the logical status of the spectral triple. In Connes’ formulation, the algebra, its Hilbert-space representation, and the Dirac operator constitute the spectral data from which geometry is extracted. In the present construction these objects can instead occur downstream of the Renewal Grand Tensor:

$$\text{predictive dynamics} \longrightarrow \mathbb{T}_{\text{Grand},X} \longmapsto (\mathcal{A}_X^{\text{hist}}, \tau_X) \xrightarrow{\delta_X} (\mathcal{A}_X, \mathcal{H}_X, D_X, J_X, \gamma_X). \quad (32)$$

A spectral triple therefore need not be interpreted as the most microscopic object in the description. Within the framework considered here, it is the spectral image of a specific algebraic and differential sector of a richer predictive object.

This does not mean that an arbitrary Dirac operator is determined by the algebra. The finite Hodge construction shows precisely where additional information enters. The algebra and faithful trace do not fix an absolute metric scale, and the commutator derivation determines the full compatible Dirac operator only modulo the appropriate self-adjoint commutant. Likewise, finite marked surjectivity uses exactly $q_S = \dim_{\mathbb{R}} \mathfrak{c}_S$ independent real linear commutant coordinates: the derivation fixes the centered implementer and the anchors recover precisely the remaining compatible affine component. It is not a claim that an arbitrary target Dirac operator has been derived from the austere Hodge datum alone. At the norm-multiplicative spectral-compression level the reconstruction becomes genuinely selective, with spectral quasidiagonality providing the exact criterion.

The independent metric construction is important for the same reason. On the periodic A_3 family, the Riemannian metric reconstructed from jump second moments and the Connes distance reconstructed from the Hodge–Dirac operator are obtained by different routes and converge to the same continuum geometry. The agreement therefore tests the spectralization rather than defining the target metric by construction. For flat periodic spin tori the Wilson construction also gives unconditional norm-resolvent convergence. For general closed Riemannian manifolds the scalar metric approximation remains unconditional, while the spin norm-resolvent statement requires the compatible stable-discretization hypotheses stated above.

A common predictive origin for two geometric constructions. The independent Lorentzian construction [8] uses the same Renewal Grand Tensor $\mathbb{T}_{\text{Grand},X}$ to reconstruct Lorentzian spacetime without requiring a spectral triple as an intermediate primitive. Together, the two constructions therefore place the Grand Tensor as a common predictive object upstream of two distinct downstream descriptions, as summarized in (2).

This matters conceptually. If spectral geometry were the only downstream output, one could interpret the predictive construction merely as an alternative procedure for generating spectral triples. The independent spacetime branch instead shows that Lorentzian geometry and Connes-type spectral geometry can be different projections of a common predictive structure rather than successive fundamental layers.

The claim is structural rather than ontological. The results do not establish that Renewal Geometry is the unique microscopic theory underlying all noncommutative geometries or all physical spacetimes. They show that, within the framework studied here, neither spectral geometry nor spacetime geometry needs to be placed at the primitive entrance. Both may arise from a predictive structure defined before either metric geometry or a spectral triple is introduced.

Information loss and nonfaithfulness. The inverse results sharpen this effective interpretation. Spectralization is generically many-to-one:

$$\mathfrak{S}(\mathcal{R}_1) = \mathfrak{S}(\mathcal{R}_2) \quad \text{need not imply} \quad \mathcal{R}_1 \simeq \mathcal{R}_2, \quad (33)$$

where $\mathcal{R}_i$ denote predictive or dynamical realizations and $\mathfrak{S}$ their spectral image. A spectral triple is therefore not, in general, a complete invariant of the underlying dynamics.

The stationary-current fibre provides the simplest example. Fixing the masses and symmetric conductances fixes the metric Hodge–Dirac triple, while a divergence-free stationary current remains undetermined. Moving within this fibre changes path probabilities, time-reversal asymmetry, and entropy production without changing the spectral metric. The spectral geometry retains the symmetric metric sector while forgetting part of the directed dynamics.

The memory fibre exhibits a different form of information loss. Eliminating auxiliary degrees of freedom produces a frequency-dependent self-energy whose minimal realization is encoded by an operator-valued measure. A static spectral description may retain only selected boundary values, moments, or an effective operator, so distinct dynamical tails can have the same static spectral image. What is lost here is temporal organization rather than merely a finite parameter.

Global marked monodromy provides a third mechanism. Locally equivalent spectral packets can differ in how their marked realizations are assembled globally. Consequently spectralization can forget directional, dynamical, and global information for mathematically different reasons.

These examples motivate the description of spectral geometry as an *information-reduced effective geometry*. This should not be interpreted as a claim that noncommutative geometry is mathematically incomplete. A spectral triple may completely encode the geometric observables for which it is

designed. The point is that it need not completely encode the predictive dynamics from which it arose. A single spectral geometry can therefore represent an entire family of dynamically inequivalent microscopic realizations.

The analogy with effective descriptions in statistical physics is useful. A set of macroscopic variables may completely characterize an effective state while failing to determine a unique microscopic realization. Likewise, a spectral triple may fully characterize an effective metric and differential sector while identifying systems that remain distinct at the predictive level.

Why the quotient has a universality interpretation. The inverse fibre

$$\mathfrak{S}^{-1}(S) \tag{34}$$

is therefore more than a technical obstruction to inversion. It describes the microscopic information quotiented out in the passage to a fixed spectral packet.

Moreover, the discarded variables are themselves organized under effective reduction. Directed currents are pushed through the induced graph chain map; high-energy memory can be absorbed into a zero-energy contact correction while the retained low-energy measure survives; and marked monodromy descends exactly when the cycles removed by the quotient carry trivial holonomy. These maps compose under successive coarse grainings.

This is the sense in which the spectral packet acts as a universality-level observable. Different microscopic systems can share the same spectral geometry because distinctions that remain meaningful at the predictive level no longer affect the operator-geometric structure retained at that level of description.

The coarse-graining language should nevertheless be interpreted precisely. The results do not prove that spectralization is universally generated by a renormalization-group flow, nor that arbitrary predictive dynamics converges toward a spectral fixed point. What is established is a controlled many-to-one map, explicit inverse fibres, and composable effective maps on their principal coordinates. The universality interpretation is therefore structural rather than a claim of a universal dynamical attractor.

6.2 Inverse fibres, scope, and open problems

Once the lost information has been identified, a natural question is whether it should be retained when it is physically relevant. The present examples suggest several possible enrichments. A metric spectral triple can be supplemented by an affinity or current class to retain irreversible directed information; an operator-valued self-energy can retain frequency-dependent memory; and a marked holonomy class can retain global assembly data. Schematically,

$$(\mathcal{A}, \mathcal{H}, D, J, \gamma) \longrightarrow (\mathcal{A}, \mathcal{H}, D, J, \gamma; \text{current, memory, monodromy}, \dots). \tag{35}$$

The ordinary spectral packet is recovered by forgetting the additional coordinates.

This perspective changes the natural inverse problem. Rather than expecting one microscopic dynamics to correspond to one spectral geometry, one should generally expect a moduli space or stratified fibre of realizations. Under their declared retained-coordinate and completion hypotheses, the finite stabilizer quotient and its completed inverse-limit version provide examples of such a description. The natural question is therefore not to identify a unique dynamics represented by a spectral triple, but to classify the predictive dynamics that share it as a common effective geometry.

The independent Lorentzian construction [8] makes this inverse problem richer. Two predictive systems with the same spectral image need not have the same image under a distinct downstream

geometric reconstruction; conversely, systems may agree on a spacetime sector while differing in spectral or dynamical refinements. Joint fibres of the spectral and Lorentzian projections therefore provide a natural object for future study.

Mathematical scope. Several limitations of the present results should remain explicit. The strict spectral image would benefit from an intrinsic characterization independent of a chosen spectral projection sequence. For spin geometry, the local Wilson construction provides the required analytic estimates and the flat periodic class closes unconditionally, but an unconditional global realization theorem for every closed spin manifold would require verification of the compatibility and stability hypotheses in general. Likewise, the extensive kernel theorem shows that any cellular or relation-adapted reduction of the universal one-step graph calculus requires its own continuum certificate; no such reduction theorem is assumed here. Noncompact and genuinely semifinite geometries require further control of domains, localization, and weights. The positive Stieltjes memory classification has a finite rational indefinite extension in Part II, but the unrestricted nonrational infinite-dimensional indefinite problem is not treated here.

Nor does the finite compact-quantum-metric-space result by itself imply quantum Gromov–Hausdorff, propinquity, or spectral-propinquity convergence in the continuum limit [5, 6, 7]. Such statements require uniform control of the full state-space Lip-norms and, for spectral propinquity, compatible control of the spectral data beyond the pure-state distance convergence proved for the A_3 approximants.

Open questions and broader interpretation. The effective-geometric viewpoint raises several further questions. Which parts of the inverse fibre can be reconstructed from natural enrichments of a spectral triple? Which fibre coordinates remain stable under broader classes of coarse graining? Can additional dynamical principles select canonical representatives inside a spectral fibre? And how much of the many-to-one structure persists in continuum classes substantially larger than the finite and controlled compact-resolvent settings considered here?

The coexistence of the spectral and spacetime branches suggests a second family of questions. One may ask when the two downstream reconstructions determine one another, when one retains information absent from the other, and which microscopic coordinates are invisible to both. More generally, one can ask whether other geometric or physical structures arise as additional quotients of the same predictive object.

Within the scope established in this paper, the hierarchy is nevertheless clear. Predictive dynamics carries temporal, directional, and global information before a metric or spectral triple is introduced. The Renewal Grand Tensor $\mathbb{T}_{\text{Grand},X}$ organizes the future-minimal carrier, represented history algebra, and positive moment structure. Lorentzian spacetime can arise from one downstream sector, while on the spectral branch the history algebra becomes Connes-type spectral geometry after a compatible first-order variation is specified. The resulting spectral packet retains the metric and differential structures relevant to its level while identifying classes of underlying dynamics that remain distinct microscopically.

Thus, within this framework, Connes-type spectral geometry is naturally interpreted as a stable effective geometry of a richer predictive dynamics: powerful and potentially complete for its geometric sector, but not in general a complete description of the dynamics from which that geometry emerges.

Formal Verification and Code Availability

All definitions, theorems, and proofs in this work have been formally verified in `Lean 4` using `Mathlib` without unproven assumptions (`sorry-free`). The complete source code, build configuration, and documentation are available at <https://github.com/AI-MathPhys/renewal-geometry>.

Mathematical Supplement

The supplement is divided into two parts. Part I contains the self-contained mathematical core of the paper. It first develops the finite Hodge–Dirac construction from a real bimodule derivation, including the scale ambiguity of the resulting metric, the affine commutant fibre of compatible Dirac implementers, and the distinction between strong and norm-multiplicative spectral compression. It then treats stationary Markov graphs. For the covariance-normalized periodic A_3 family, the graph commutator seminorm is identified explicitly and the associated pure-state Connes distances are proved to converge uniformly to the flat Riemannian metric reconstructed independently from the jump covariance. At fixed vertex masses and symmetric conductances, the directed stationary realizations form a cycle-current fibre whose strictly positive interior is globally parametrized by $H^1(G; \mathbb{R})$ through the affinity class $[\operatorname{arctanh}(j/c)]$; the inverse is obtained by a weighted log cosh minimization, and graph quotients induce the corresponding current and entropy-contraction maps.

The remaining sections of Part I supply the mathematical structures supporting these results. They include an independent endpoint construction recovering Riemannian distance from finite metric triples, edge-ratio and cellular Hodge descriptions of the affinity class, operational reconstruction and recognition of the finite history algebra from complete intervention data, positive operator-valued Stieltjes memory and its static-preserving cutoff maps, and the classification and descent of marked monodromy. These results are stated with their own hypotheses and are logically separate from the proofs of the two principal graph theorems.

Part II collects further mathematical extensions that are not required for the core results. It develops positive local stencils and covariant Wilson discretizations for spin geometry, including unconditional norm-resolvent convergence on flat periodic spin tori and a general convergence theorem under explicit compatible-stability hypotheses. It also extends the positive-memory theory to finite rational Hermitian dynamics through minimal Pontryagin/descriptor realizations, and studies joint and completed inverse fibres using correlation-Gram data, stabilizer quotients, and compact finite-stage inverse systems. Thus the spin-discretization, indefinite-memory, and completed-fibre results enlarge the scope of the arXiv version without entering the assumptions or proofs of the finite Hodge–Dirac, A_3 metric, or stationary-current reconstruction theorems.

Contents of the Mathematical Supplement

Part I. Core mathematical results	21
S1 Finite Hodge–Dirac reconstruction	21
S1.1 Finite real differential data	21
S1.2 The associated spectral triple	22
S1.3 The commutant fibre of a fixed derivation	24
S1.4 Functoriality and finite spectral compressions	25
S2 Metric realization on stationary graphs	28
S2.1 Stationary graphs and the Hodge–Dirac operator	28
S2.2 The periodic A_3 lattice and its covariance normalization	30
S2.3 Uniform convergence of the Connes distance	31
S3 Stationary-current fibres and affinity coordinates	35
S3.1 Directed currents at fixed metric geometry	35

	S3.2	Cohomological coordinates of the stationary-current fibre	37
	S3.3	Coarse graining of stationary currents	39
S4		Independent endpoint metric approximation	39
S5		Edge affinities and cellular Hodge decomposition	41
	S5.1	Relative edge-density data	41
	S5.2	Cellular curvature and gauge classes	42
S6		Operational reconstruction, realization and recognition	43
	S6.1	Reconstruction from complete intervention data	43
	S6.2	Realization and recognition with fixed boundary contexts	48
S7		Positive Stieltjes memory and static-preserving cutoffs	51
	S7.1	Positive measures, self-energies, and inverse moments	51
	S7.2	Moment determinacy and indeterminacy	52
	S7.3	Memory cutoff and composition	53
	S7.4	High-energy escape and contact limits	54
S8		Marked monodromy and componentwise coarse graining	55
	S8.1	Flat bundles of marked spectral triples	55
	S8.2	Descent under graph contraction	56
	S8.3	Componentwise coarse graining	57
Part II. Further mathematical extensions			58
S9		Analytic details for compact-spin spectralization	58
	S9.1	A positive Renewal stencil for an arbitrary local inverse metric	58
	S9.2	Density-symmetric spin Dirac and finite Wilson packet	59
	S9.3	Atlas compatibility and the global resolvent argument	62
	S9.4	Independent metric approximation and the spin conclusion	64
S10		Explicit minimal realization of positive Stieltjes memory	64
S11		Finite rational indefinite memory	65
	S11.1	Effective-space compression of indefinite memory	68
S12		Joint realizations and completed inverse fibres	68
	S12.1	Joint Gram realizations and the finite stratified fibre	68
	S12.2	Quasilocal and semifinite completion	69
References			72

Part I. Core mathematical results

S1 Finite Hodge–Dirac reconstruction

We first construct a finite Hodge–Dirac triple from a real bimodule derivation and identify the ambiguity of a Dirac implementer with the same commutators. These operator-algebraic facts provide the framework for the graph metric and stationary-current theorems. The final subsection distinguishes strong spectral compression from recovery of multiplication in operator norm.

S1.1 Finite real differential data

The finite construction requires only a finite-dimensional unital C^* -algebra $\mathcal{A}$, a faithful tracial state τ , and the differential datum specified below. The Hilbert space of zero-forms is the vector space $\mathcal{A}$ with inner product

$$\mathcal{H}_0 = L^2(\mathcal{A}, \tau), \quad \langle a, b \rangle_0 = \tau(a^*b). \quad (36)$$

Faithfulness makes this inner product nondegenerate. Left and right multiplication are bounded commuting actions, with adjoints $L_a^* = L_{a^*}$ and $R_a^* = R_{a^*}$. The right action is a representation of the opposite algebra: $R_a R_b = R_{ba}$.

There is a distinguished possible choice of trace, but the results do not require it. If $\mathcal{A} \cong \bigoplus_\lambda M_{n_\lambda}(\mathbb{C})$, its normalized regular trace is

$$\widehat{\tau}_{\text{reg}}(a) = \frac{\text{Tr}_{\mathbb{C}}(L_a)}{\dim_{\mathbb{C}} \mathcal{A}} = \frac{\sum_\lambda n_\lambda \text{Tr}(a_\lambda)}{\sum_\lambda n_\lambda^2}. \quad (37)$$

This is a faithful tracial state invariant under $*$ -isomorphisms. Thus both $\mathcal{A}$ and $\widehat{\tau}_{\text{reg}}$ have ordinary finite-dimensional algebraic meanings independent of the motivating history construction.

A Hilbert-space $\mathcal{A}$ -bimodule means a Hilbert space $\mathcal{K}$ with commuting unital $*$ -representations of $\mathcal{A}$ and $\mathcal{A}^{\text{op}}$. We denote these by $\lambda_{\mathcal{K}}(a)$ and $\rho_{\mathcal{K}}(b)$, so that $a\xi b = \lambda_{\mathcal{K}}(a)\rho_{\mathcal{K}}(b)\xi$ and $\rho_{\mathcal{K}}(a)\rho_{\mathcal{K}}(b) = \rho_{\mathcal{K}}(ba)$.

Definition S1.1 (Finite real differential datum). A finite real differential datum on $(\mathcal{A}, \tau)$ consists of a Hilbert-space $\mathcal{A}$ -bimodule $\mathcal{K}$, an antiunitary involution $C_{\mathcal{K}}$ with

$$C_{\mathcal{K}}(a\xi b) = b^* C_{\mathcal{K}}(\xi) a^*,$$

and a complex-linear map $\delta : \mathcal{A} \rightarrow \mathcal{K}$ satisfying

$$\delta(\mathbf{1}) = 0, \quad \delta(ab) = a\delta(b) + \delta(a)b, \quad C_{\mathcal{K}}\delta(a) = \delta(a^*). \quad (38)$$

Only the bimodule generated by $\delta(\mathcal{A})$ is retained. No assumption is made about the origin of these algebraic data.

For completeness, the one-form construction can be performed entirely within this datum. The universal first-order calculus is

$$\Omega_u^1(\mathcal{A}) = \text{Ker}(m : \mathcal{A} \otimes \mathcal{A} \rightarrow \mathcal{A}), \quad d_u a = \mathbf{1} \otimes a - a \otimes \mathbf{1}, \quad (39)$$

with bimodule action $c(a \otimes b)d = ca \otimes bd$ and conjugate-linear reversal

$$\left(\sum_i a_i \otimes b_i \right)^{\natural} = - \sum_i b_i^* \otimes a_i^*. \quad (40)$$

Direct expansion gives $(d_u a)^\natural = d_u(a^*)$ and $d_u(ab) = a d_u b + (d_u a)b$. The map

$$T_\delta \left(\sum_i a_i \otimes b_i \right) = \sum_i a_i \delta(b_i), \quad \sum_i a_i b_i = 0,$$

is the unique bimodule map with $T_\delta d_u = \delta$. Indeed, the Leibniz rule makes it right linear because the additional term is $(\sum_i a_i b_i) \delta(c) = 0$; left linearity is immediate. Star compatibility gives $T_\delta(\omega^\natural) = C_\mathcal{K} T_\delta(\omega)$. Consequently

$$Q_\delta(\omega, \eta) = \langle T_\delta \omega, T_\delta \eta \rangle_\mathcal{K} \quad (41)$$

is positive, and its null space is exactly $\text{Ker } T_\delta$. The resulting one-form Hilbert space is

$$\mathcal{H}_1(\delta) = \Omega_u^1(\mathcal{A}) / \text{Ker } T_\delta \cong \text{Ran } T_\delta = \text{span}\{a\delta(b)c : a, b, c \in \mathcal{A}\}. \quad (42)$$

It is finite dimensional even if the initially supplied $\mathcal{K}$ is not. The displayed identification is isometric, so this quotient removes exactly the unused bimodule directions. We call the resulting one-form bimodule *minimal* in this generated-bimodule sense. Write $\partial a = [d_u a]$ for the induced derivation.

S1.2 The associated spectral triple

Set $\mathcal{H}_0 = L^2(\mathcal{A}, \tau)$ and let L_a and R_a denote left and right multiplication on $\mathcal{H}_0$. On

$$\mathcal{H} = \mathcal{H}_0 \oplus \mathcal{H}_1(\delta) \quad (43)$$

define

$$\pi(a) = L_a \oplus \lambda(a), \quad D = \begin{pmatrix} 0 & \partial^* \\ \partial & 0 \end{pmatrix}, \quad \gamma = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}. \quad (44)$$

On $\mathcal{H}_0$ put $J_0 a = a^*$, and on $\mathcal{H}_1(\delta)$ let $J_1[\omega] = [\omega^\natural]$; set $J = J_0 \oplus J_1$.

Theorem S1.2 (Finite Hodge–Dirac spectral triple). *The quintuple $(\mathcal{A}, \mathcal{H}, D, J, \gamma)$ is a real even finite spectral triple. With the standard sign convention its real-structure signs are those of KO -dimension 0:*

$$J^2 = I, \quad JD = DJ, \quad J\gamma = \gamma J. \quad (45)$$

More precisely, π is faithful, $D = D^*$ has compact resolvent, every $[D, \pi(a)]$ is bounded,

$$\gamma^* = \gamma, \quad \gamma^2 = I, \quad [\gamma, \pi(a)] = 0, \quad D\gamma = -\gamma D, \quad (46)$$

and J is antiunitary. The opposite representation is

$$\pi^o(b) = J\pi(b^*)J^{-1} = R_b \oplus \rho(b).$$

For all $a, b \in \mathcal{A}$ the order-zero and first-order conditions hold:

$$[\pi(a), \pi^o(b)] = 0, \quad [[D, \pi(a)], \pi^o(b)] = 0. \quad (47)$$

If $B_a : \mathcal{H}_0 \rightarrow \mathcal{H}_1(\delta)$ is defined by $B_a b = (\partial a)b$, then

$$[D, \pi(a)] = \begin{pmatrix} 0 & -B_{a^*}^* \\ B_a & 0 \end{pmatrix}. \quad (48)$$

In particular, for $a = a^*$,

$$\|[D, \pi(a)]\| = \|B_a\|, \quad \text{Ker}(a \mapsto \|[D, \pi(a)]\|) = \text{Ker } \partial \cap \mathcal{A}_{\text{sa}}. \quad (49)$$

Proof. Faithfulness follows from the left regular summand. Since all spaces are finite dimensional, D is self-adjoint with compact resolvent and all commutators are bounded. The grading identities in (46) follow directly from the block-diagonal representation and the off-diagonal form of D . The Leibniz rule gives

$$(\partial L_a - \lambda(a)\partial)b = \partial(ab) - a\partial b = (\partial a)b = B_a b,$$

and taking adjoints yields (48). The reversal identities make J_1 antiunitary, give $J_1^2 = I$, and imply $J_1\partial = \partial J_0$; hence the three signs in (45). Moreover $J_0 L_b^* J_0 = R_b$ and $J_1 \lambda(b^*) J_1 = \rho(b)$, so the opposite representation has the asserted form. Left and right actions commute. Finally, the lower-left block of the double commutator is $B_a R_b - \rho(b) B_a$, which vanishes because B_a is right $\mathcal{A}$ -linear. The upper-right block vanishes by taking the corresponding adjoint identity for a^* and b^* . For self-adjoint a , the two off-diagonal blocks are B_a and $-B_a^*$, so both have norm $\|B_a\|$, proving (49). $\square$

Remark S1.3. Here “real even” refers to the grading, reality, order-zero and first-order axioms listed in Theorem S1.2; orientability and Poincaré duality are not asserted. The real structure is the algebraic reversal of the differential bimodule. Additional physical interpretations of J require extra markings and are not used below.

Proposition S1.4. *Let $\delta \neq 0$ be a finite real differential datum and let $s > 0$. Replacing δ by $s\delta$ leaves $(\mathcal{A}, \tau)$ unchanged and, under the canonical identification of the one-form spaces, sends*

$$D \mapsto sD, \quad L_D(a) := \|[D, \pi(a)]\| \mapsto sL_D(a). \quad (50)$$

Whenever the Connes distance is finite, it scales as $d_D \mapsto s^{-1}d_D$. Hence the algebra and canonical trace do not determine an absolute spectral metric scale.

Proof. The inner product induced by $s\delta$ is $s^2 Q_\delta$, so the canonical unitary from the scaled one-form space to the original one sends the derivation vector $[d_u a]$ to $s[d_u a]$. Both off-diagonal blocks of D therefore acquire the factor s . The commutator and dual-distance scalings follow immediately. $\square$

The spectral seminorm also gives a compact quantum metric space, provided that zero differential means constant observable. This is the finite-dimensional form of the state-space criterion underlying Rieffel’s theory [34, 4].

Proposition S1.5. *Suppose $\text{Ker } \partial = \mathbb{C}\mathbf{1}$ and set $L_D(a) = \|[D, \pi(a)]\|$. On the real order-unit space $\mathcal{A}_{\text{sa}}$, L_D is a continuous Lip-norm. Explicitly, the distance*

$$\text{mk}_{L_D}(\varphi, \psi) = \sup\{|\varphi(a) - \psi(a)| : a = a^*, L_D(a) \leq 1\} \quad (51)$$

is finite on the state space $S(\mathcal{A})$ and induces its weak- topology. Moreover, the complex extension satisfies*

$$L_D(ab) \leq \|a\|L_D(b) + \|b\|L_D(a), \quad L_D(a^*) = L_D(a). \quad (52)$$

Conversely, if $\text{Ker } \partial \neq \mathbb{C}\mathbf{1}$, the distance in (51) is infinite for some pair of states, so it is not a compact quantum metric on the whole state space.

Proof. The commutator is a linear map between finite-dimensional normed spaces; hence L_D is continuous. The commutator product rule and the *-identity give (52). By Theorem S1.2, the kernel on $\mathcal{A}_{\text{sa}}$ is $\mathbb{R}\mathbf{1}$. On $E_\tau = \{a = a^* : \tau(a) = 0\}$ it follows that L_D is a norm, and finite-dimensional norm equivalence gives a constant C_δ such that

$$\|a - \tau(a)\mathbf{1}\| \leq C_\delta L_D(a), \quad a = a^*. \quad (53)$$

Thus $\{a \in E_\tau : L_D(a) \leq 1\}$ is norm compact. Scalar multiples of $\mathbf{1}$ do not affect differences of states, so the supremum in (51) can be taken over this compact set and is at most $2C_\delta$. Weak-* convergence of states is uniform on a norm-compact set, since states have norm one; it therefore implies convergence in mk_{L_D} . Conversely, for each self-adjoint a ,

$$|\varphi(a) - \psi(a)| \leq L_D(a) \text{mk}_{L_D}(\varphi, \psi),$$

with the scalar case immediate. Hence metric convergence implies weak-* convergence and the metric separates states.

If the kernel is larger, star compatibility gives a non-scalar self-adjoint $b \in \text{Ker } \partial$. States separate self-adjoint elements modulo scalars, so some φ, ψ have $\varphi(b) \neq \psi(b)$. Every multiple tb has zero seminorm, making their distance infinite. $\square$

The quotient used here is $\mathcal{A}_{\text{sa}}/\mathbb{R}\mathbf{1}$; the metric itself is on $S(\mathcal{A})$, not on a quotient of states. Finite dimensionality is essential to the automatic compactness argument: for an infinite-dimensional limit, a scalar kernel alone does not replace total boundedness of the normalized Lipschitz ball.

Proposition S1.6. *For every differential datum of Definition S1.1, the Hodge operator of Theorem S1.2 satisfies*

$$D(\mathbf{1}, 0) = 0. \tag{54}$$

Consequently the Hodge functor $\mathfrak{H}_{\text{fin}}$ never produces an invertible finite Dirac operator. More generally, knowledge of the commutator derivation does not determine the component of D in the compatible commutant; that residual ambiguity is characterized in Theorem S1.8.

Proof. The universal derivation satisfies $\partial\mathbf{1} = 0$, hence the lower-left block of $D(\mathbf{1}, 0)$ vanishes; the upper-right block acts on the zero one-form component and vanishes as well. The second assertion is immediate from Theorem S1.8. $\square$

S1.3 The commutant fibre of a fixed derivation

Definition S1.7 (Marked finite spectral triple and commutant coordinates). A marked finite spectral triple is

$$S = (\mathcal{A}, \mathcal{H}, \pi, D, J, \gamma; \mathfrak{m}),$$

where $\mathfrak{m}$ denotes any additional declared data preserved by isomorphisms. When a real structure is present, its Dirac sign is written $JD = \varepsilon' DJ$, $\varepsilon' \in \{\pm 1\}$; absent marks impose no condition. Define the compatible real commutant space

$$\mathfrak{c}_S = \{C \in \pi(\mathcal{A})'_{\text{sa}} : C\gamma + \gamma C = 0, \quad JC = \varepsilon' CJ\}, \tag{55}$$

with the corresponding condition omitted when γ or J is absent. An m -component *real linear commutant-coordinate map* means

$$\alpha : \mathfrak{c}_S \longrightarrow \mathbb{R}^m.$$

Its components are the scalar commutant coordinates used below. Let $P_{\mathfrak{c}_S}$ be real Hilbert–Schmidt orthogonal projection onto this space and put

$$D_{\text{der}} = D - P_{\mathfrak{c}_S} D, \quad q_S = \dim_{\mathbb{R}} \mathfrak{c}_S. \tag{56}$$

For a fixed represented algebra and fixed grading and reality signs, the elementary commutator identity identifies the remaining ambiguity exactly. The following statement records its affine structure and the sharp number of independent real linear coordinates needed to recover the operator.

Theorem S1.8 (Dirac implementers with a fixed commutator derivation). *Among self-adjoint operators with the declared grading and reality signs that implement the same commutator derivation $[D, \pi(\cdot)]$, the complete solution set is*

$$D + \mathfrak{c}_S. \quad (57)$$

The centered operator D_{der} is the unique minimum-Hilbert-Schmidt-norm representative of this affine fibre. If $C_1, \dots, C_{q_S}$ is a real Hilbert-Schmidt orthonormal basis of $\mathfrak{c}_S$ and

$$\alpha_j(D) = \langle C_j, D \rangle_{\text{HS}, \mathbb{R}}, \quad 1 \leq j \leq q_S, \quad (58)$$

then

$$D = D_{\text{der}} + \sum_{j=1}^{q_S} \alpha_j(D) C_j. \quad (59)$$

Consequently q_S independent real linear commutant coordinates are sufficient to recover every compatible implementer from its commutator derivation, and fewer coordinates cannot distinguish the full affine fibre in general. In particular, the canonical derivation-only reconstruction is exactly the centered class $P_{\mathfrak{c}_S} D = 0$.

Proof. If D' has the same commutator derivation as D , then $C = D' - D$ commutes with $\pi(\mathcal{A})$. Subtracting the grading and reality equations for D' and D gives $C \in \mathfrak{c}_S$. Conversely every $C \in \mathfrak{c}_S$ preserves the commutators and all declared linear sign relations, proving (57). Orthogonal projection gives

$$\|D + C\|_{\text{HS}}^2 = \|D_{\text{der}}\|_{\text{HS}}^2 + \|P_{\mathfrak{c}_S} D + C\|_{\text{HS}}^2,$$

so D_{der} is the unique minimum-norm representative. Expanding $P_{\mathfrak{c}_S} D$ in the orthonormal basis gives (59). By Definition S1.7, any m -component real linear commutant-coordinate map is a real linear map $\mathfrak{c}_S \rightarrow \mathbb{R}^m$. If $m < q_S$ the map has nonzero kernel, so two distinct compatible implementers remain indistinguishable. Taking the coordinate functionals in (58) gives an injective commutant-coordinate map of size q_S . $\square$

S1.4 Functoriality and finite spectral compressions

Let $\mathbf{Diff}_{\text{fin}}$ have objects $(\mathcal{A}, \tau, \delta)$ as above and morphisms the trace-preserving $*$ -isomorphisms intertwining the differential bimodule data. The construction of Theorem S1.2 is functorial: the algebra isomorphism induces unitaries on $L^2(\mathcal{A}, \tau)$ and on the minimal one-form quotient, and these unitaries intertwine π, D, J, γ . We denote this finite Hodge functor by $\mathfrak{H}_{\text{fin}}$.

Reconstruction from commutator data retains the represented algebra, the specified real structure and grading, and separating real linear coordinates on the compatible commutant. The Dirac operator is then recovered from these data. In the notation below, the superscript cal records this choice of commutant coordinates. For compact-resolvent triples, finite data are obtained by spectral compression, with two distinct convergence requirements.

Definition S1.9 (Finite reconstruction and spectral-compression data). Let $\mathbf{Spec}_{\text{fin}}$ be the category of finite marked spectral triples, with unitary equivalences intertwining the represented algebra, D , and all declared marks as isomorphisms. Let $\mathbf{Spec}_{\text{cpt}}^{\text{sep}}$ denote the corresponding separable compact-resolvent triples. For every compact-resolvent triple we fix, as part of the marked datum, a dense unital $*$ -subalgebra $\mathcal{A}_D \subseteq \mathcal{A}$ on which $[D, \pi(a)]$ is bounded. In the finite case $\mathcal{A}_D = \mathcal{A}$.

The reconstruction problem is considered in three source categories.

- (C1) In the *finite reconstruction category*, the represented finite algebra and symmetry marks are retained together with the commutator derivation and separating real linear commutant coordinates on $\mathfrak{c}_S$. The resulting reconstruction is denoted $\mathfrak{S}_{\text{fin}}^{\text{cal}}$.
- (C2) In the *strong spectral-compression category*, one uses increasing finite-rank spectral projections $P_n \uparrow I$ with $P_n D = D P_n$ and records the compressed matrices, product defects, commutators, marks, and finite commutant coordinates. After extending compressed operators by zero on $P_n^\perp \mathcal{H}$, convergence means that for every $a, b \in \mathcal{A}_D$, $z \in \mathbb{C} \setminus \mathbb{R}$, and $\xi \in \mathcal{H}$,

$$\begin{aligned} P_n \pi(a) P_n \xi &\longrightarrow \pi(a) \xi, \\ P_n [D, \pi(a)] P_n \xi &\longrightarrow [D, \pi(a)] \xi, \\ P_n (D - z)^{-1} P_n \xi &\longrightarrow (D - z)^{-1} \xi, \\ (P_n \pi(ab) P_n - P_n \pi(a) P_n \pi(b) P_n) \xi &\longrightarrow 0. \end{aligned} \tag{60}$$

The declared bounded marks are required to converge in the same strong sense. The reconstruction is denoted $\mathfrak{S}_{\text{cpt}}^{\text{strong, cal}}$.

- (C3) In the *norm-multiplicative spectral-compression category*, the same data are required in addition to satisfy

$$\|P_n \pi(ab) P_n - P_n \pi(a) P_n \pi(b) P_n\| \longrightarrow 0 \quad (a, b \in \mathcal{A}_D). \tag{61}$$

The corresponding reconstruction is $\mathfrak{S}_{\text{cpt}}^{\text{norm, cal}}$.

A compact-resolvent triple is called *spectrally quasidiagonal* when there are finite-rank symmetric spectral projections $P_n \uparrow I$ such that

$$P_n D = D P_n, \quad \|[P_n, \pi(a)]\| \longrightarrow 0 \tag{62}$$

for every $a \in \mathcal{A}_D$, equivariantly for J and γ when these are marked. Write $\mathbf{Spec}_{\text{cpt}}^{\text{sqd}}$ for this subcategory. The source classes are taken modulo unitary equivalences preserving the represented algebra, $\mathcal{A}_D$, marks, commutant-coordinate functionals, and compression maps.

Strong compression uses strong convergence of bounded operator data and exists for every compact-resolvent triple. Norm-multiplicative compression also recovers multiplication in operator norm. Skalski and Zacharias [35] study the relation between compact-type spectral triples and quasidiagonal representations, including a Toeplitz example. Our condition is more restrictive than quasidiagonality of the representation: the approximating projections must also be spectral projections of the prescribed D . The next theorem identifies this additional requirement precisely through the norm of the multiplication defect.

Proposition S1.10 (Finite recovery from commutant coordinates). *For a finite marked spectral triple S , its represented algebra, declared symmetry signs and commutator derivation determine the centered implementer D_{der} . Exactly $q_S = \dim_{\mathbb{R}} \mathfrak{c}_S$ independent real linear commutant coordinates suffice, and are in general necessary, to recover the full compatible Dirac operator.*

Proof. Let $S_0 = (\mathcal{B}, K, \pi_0, D_0, J_0, \gamma_0)$ be finite. Retain the represented algebra $\pi_0(\mathcal{B})$, the declared marks, and the commutator derivation

$$\delta_{D_0}(a) = [D_0, \pi_0(a)].$$

Choose a real Hilbert–Schmidt orthonormal basis $C_1, \dots, C_{q_{S_0}}$ of the compatible commutant and retain the coordinates $\alpha_j(D_0)$ of (58). By Theorem S1.8, the commutator data determine the centered implementer $(D_0)_{\text{der}}$ and these coordinates recover D_0 uniquely by (59). Thus the full generator need not be retained, and the same theorem proves sharpness of the number of real linear coordinates. This is a finite-dimensional linear reconstruction statement. $\square$

The compression problem is different: finite-rank compressions of a representation are generally operator systems, not representations of the original algebra. In particular, $a \mapsto P_n \pi(a) P_n$ need not be multiplicative, and the compressed matrices are not asserted to inherit every real first-order axiom. This is the operator-system distinction relevant to spectral truncations [36, 37]. The next theorem concerns the matrix data and topologies explicitly specified in Definition S1.9.

Theorem S1.11 (Strong and norm-multiplicative spectral compressions). *Every separable compact-resolvent spectral triple admits finite-rank symmetric spectral projections $P_n \uparrow I$ satisfying the strong convergence conditions (60). For $D_n = P_n D|_{P_n \mathcal{H}}$ and $a_n = P_n \pi(a)|_{P_n \mathcal{H}}$ one has*

$$[D_n, a_n] = P_n [D, \pi(a)]|_{P_n \mathcal{H}}, \quad a \in \mathcal{A}_D. \quad (63)$$

For a sequence of such projections, norm-multiplicativity (61) holds if and only if $\|[P_n, \pi(a)]\| \rightarrow 0$ for every $a \in \mathcal{A}_D$. Consequently a norm-multiplicative spectral-compression sequence exists exactly for spectrally quasidiagonal triples. This class is a proper subclass of the separable compact-resolvent triples.

Proof. Let $(\mathcal{A}, \mathcal{H}, \pi, D)$ be separable with compact resolvent and choose increasing symmetric spectral projections $P_n = \mathbf{1}_{[-R_n, R_n]}(D)$, $R_n \uparrow \infty$. Then $P_n D = D P_n$ and $P_n \rightarrow I$ strongly. The same commutation relation gives (63) by expanding the two compressed products. Symmetric cutoffs preserve the declared J and γ signs, and their bounded compressions converge strongly as well. For $a, b \in \mathcal{A}_D$,

$$P_n \pi(ab) P_n - P_n \pi(a) P_n \pi(b) P_n = P_n \pi(a) (I - P_n) \pi(b) P_n, \quad (64)$$

which tends strongly to zero because $(I - P_n) \rightarrow 0$ strongly and the remaining factors are uniformly bounded. Since $[D, \pi(a)]$ is bounded, $P_n [D, \pi(a)] P_n \rightarrow [D, \pi(a)]$ strongly. Likewise $P_n (D - z)^{-1} P_n \rightarrow (D - z)^{-1}$ strongly for $z \notin \mathbb{R}$. Thus (60) holds for the canonical spectral compressions, proving the strong-compression assertion.

If (62) holds, then from (64)

$$\|P_n \pi(a) (I - P_n) \pi(b) P_n\| \leq \|\pi(a)\| \|[P_n, \pi(b)]\|,$$

so multiplication is recovered in norm. Conversely, applying (61) to (a^*, a) gives

$$\|P_n \pi(a)^* (I - P_n) \pi(a) P_n\| = \|(I - P_n) \pi(a) P_n\|^2 \longrightarrow 0.$$

Applying the same argument to a^* controls the opposite off-diagonal block; hence $\|[P_n, \pi(a)]\| \rightarrow 0$. This proves the equivalence with spectral quasidiagonality.

Properness is witnessed by the Toeplitz representation on $\ell^2(\mathbb{N}_0)$ with number operator $N e_k = k e_k$, unilateral shift $S e_k = e_{k+1}$, and dense unital $*$ -algebra $\mathcal{A}_D = \text{alg}(I, S, S^*)$. Here $[N, S] = S$ and $[N, S^*] = -S^*$, so the commutators are bounded. Every finite-rank spectral projection of N is diagonal in the number basis, and every nontrivial finite diagonal projection has $\|[P, S]\| = 1$. This compact-resolvent triple lies in the strong-compression class but not in the norm-multiplicative class. $\square$

Corollary S1.12 (Reconstruction classes and their essential images). *For the reconstruction classes in Definition S1.9,*

$$\begin{aligned}\text{EssIm}(\mathfrak{S}_{\text{fin}}^{\text{cal}}) &= \mathbf{Spec}_{\text{fin}}, \\ \text{EssIm}(\mathfrak{S}_{\text{cpt}}^{\text{strong,cal}}) &= \mathbf{Spec}_{\text{cpt}}^{\text{sep}}, \\ \text{EssIm}(\mathfrak{S}_{\text{cpt}}^{\text{norm,cal}}) &= \mathbf{Spec}_{\text{cpt}}^{\text{sqd}} \subsetneq \mathbf{Spec}_{\text{cpt}}^{\text{sep}}.\end{aligned}\tag{65}$$

Proof. The finite equality is Proposition S1.10. The compact-resolvent equalities are Theorem S1.11, with the finite commutant coordinates retained as in the source definitions. $\square$

Remark S1.13. The essential images in Corollary S1.12 refer to the specified reconstruction data. Finite recovery uses the commutator derivation and separating commutant coordinates. The Hodge construction alone is more restrictive: it always has the unit zero mode of Proposition S1.6. For spectral compressions the additional distinction is whether multiplication is recovered strongly or in operator norm.

S2 Metric realization on stationary graphs

The finite reconstruction results identify compatible operator data. To test their metric content, we now compare the Connes distance with a metric obtained without using the Dirac operator. Stationary graph dynamics provides a transparent model: its symmetric jump covariance fixes the candidate metric normalization, while its Hodge operator supplies the commutator seminorm. The graph construction belongs to the established Hodge–Dirac framework [38, 39]; the continuum comparison is the question addressed here.

S2.1 Stationary graphs and the Hodge–Dirac operator

Definition S2.1 (Stationary finite Markov graph). Let V be finite, let $m_x > 0$ with $\sum_x m_x = 1$, and let $k_{xy} \geq 0$ for $x \neq y$. Put

$$(\mathcal{L}f)(x) = \sum_{y \neq x} k_{xy}(f(y) - f(x)), \quad q_{xy} = m_x k_{xy}, \tag{66}$$

and assume stationarity

$$\sum_y (q_{xy} - q_{yx}) = 0 \quad (x \in V). \tag{67}$$

Define

$$c_{xy} = \frac{q_{xy} + q_{yx}}{2} = c_{yx}, \quad j_{xy} = \frac{q_{xy} - q_{yx}}{2} = -j_{yx}. \tag{68}$$

Assume the undirected graph on the pairs with $c_{xy} > 0$ is connected.

Set

$$\mathcal{H}_0 = \ell^2(V, m), \quad \mathcal{H}_1 = \ell^2(E^{\text{or}}, c), \quad (\partial f)(x, y) = f(y) - f(x), \tag{69}$$

where both orientations are retained and $\|\omega\|_1^2 = \sum_{x,y} c_{xy} |\omega(x, y)|^2$. Let

$$(\lambda(a)\omega)(x, y) = a(y)\omega(x, y), \quad (\rho(a)\omega)(x, y) = a(x)\omega(x, y). \tag{70}$$

Then $\partial(fg) = \lambda(f)\partial g + \rho(g)\partial f$. The associated Hodge operator is

$$D_{m,c} = \begin{pmatrix} 0 & \partial^* \\ \partial & 0 \end{pmatrix}. \tag{71}$$

The represented algebra and real-even structure are

$$\begin{aligned}\pi(a) &= M_a \oplus \lambda(a), & \gamma &= \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, & J &= J_0 \oplus J_1, \\ (J_0 f)(x) &= \overline{f(x)}, & (J_1 \omega)(x, y) &= -\overline{\omega(y, x)}.\end{aligned}\tag{72}$$

The minus sign is required by the derivation convention: $J_1 \partial f = \partial \bar{f}$. Symmetry of c makes J_1 antiunitary, and $J_1 \lambda(a^*) J_1 = \rho(a)$. Thus these are precisely the KO-dimension-zero signs and opposite actions of Theorem S1.2.

Theorem S2.2 (Graph Hodge square and exact Lipschitz seminorm). *Let $\mathcal{L}^\dagger$ be the adjoint of $\mathcal{L}$ in $\ell^2(V, m)$ and let $\mathcal{L}_s = (\mathcal{L} + \mathcal{L}^\dagger)/2$. Then*

$$D_{m,c}^2|_{\mathcal{H}_0} = \partial^* \partial = -2\mathcal{L}_s.\tag{73}$$

For every real $f \in C(V)$,

$$L_{m,c}(f)^2 = \|[D_{m,c}, f]\|^2 = \max_{x \in V} \frac{1}{m_x} \sum_y c_{xy} |f(y) - f(x)|^2.\tag{74}$$

Thus, with the standard symmetric carré-du-champ normalization

$$\Gamma_s(f)(x) = \frac{1}{2m_x} \sum_y c_{xy} |f(y) - f(x)|^2,\tag{75}$$

one has $L_{m,c}(f)^2 = 2\|\Gamma_s(f)\|_\infty$. The kernel of $L_{m,c}$ is the constants, so the Connes formula gives a finite metric on V . On the reversible branch, $\mathcal{L} = \mathcal{L}_s$ and $D_{m,c}^2|_{\mathcal{H}_0} = -2\mathcal{L}$.

Proof. The adjoint generator has rates q_{yx}/m_x , so $\mathcal{L}_s$ has rates c_{xy}/m_x . Therefore

$$-\langle f, \mathcal{L}_s f \rangle_0 = \frac{1}{2} \sum_{x,y} c_{xy} |f(y) - f(x)|^2 = \frac{1}{2} \|\partial f\|_1^2.$$

Polarization gives (73). The lower-left commutator block satisfies

$$(B_f g)(x, y) = (f(y) - f(x))g(x).$$

Hence

$$\|B_f g\|_1^2 = \sum_x |g(x)|^2 \sum_y c_{xy} |f(y) - f(x)|^2.$$

Its norm relative to $\sum_x m_x |g(x)|^2$ is the maximum in (74). Connectedness makes $\partial f = 0$ equivalent to constancy. $\square$

The connectedness assumption has a state-space consequence as well as a pointwise one. Proposition S1.5 implies that $(C(V)_{\text{sa}}, L_{m,c})$ is a compact quantum metric space. The exact normalization (74)–(75) relates the squared commutator seminorm to the maximal local carré-du-champ of the symmetric Markov generator; the continuum result below is the uniform convergence of the resulting A_3 Connes metric to the independently reconstructed Riemannian metric. Its states are probability distributions on V , and the spectral metric on those distributions induces their usual weak-* topology. This conclusion does not require the Hodge operator itself to be invertible; it is the scalar kernel of the *commutator seminorm* that matters. The following large one-form kernel is therefore an issue for the operator continuum limit, not for the finite Lip-norm.

Proposition S2.3. *Let the connected underlying graph have n vertices and e undirected edges, and retain both orientations in $\mathcal{H}_1$. Then*

$$\dim \operatorname{Ker} D_{m,c} = 2e - n + 2. \quad (76)$$

On a periodic A_3 root graph, $e = 6n$, so the kernel has dimension $11n + 2$. Consequently, any compatible cofinal limit that retains all these zero modes has infinite-dimensional kernel and cannot have compact resolvent.

Proof. Connectedness gives $\dim \operatorname{Ker} \partial = 1$ and $\operatorname{rank} \partial = n - 1$. Since the directed one-form space has dimension $2e$, $\dim \operatorname{Ker} \partial^* = 2e - n + 1$. Because $\operatorname{Ker} D = (\operatorname{Ker} \partial) \oplus (\operatorname{Ker} \partial^*)$, the formula follows. $\square$

S2.2 The periodic A_3 lattice and its covariance normalization

Let

$$W_0 = \left\{ x \in \mathbb{R}^4 : \sum_{i=1}^4 x_i = 0 \right\}, \quad \Lambda_{A_3} = W_0 \cap \mathbb{Z}^4, \quad (77)$$

and let $\Phi_{A_3} = \{e_i - e_j : i \neq j\}$ be the twelve oriented roots.

Lemma S2.4 (A_3 tight-frame identity). *On W_0 ,*

$$\sum_{\alpha \in \Phi_{A_3}} \alpha \alpha^\top = 8I_{W_0}. \quad (78)$$

Equivalently, $\sum_{i \neq j} (v_i - v_j)^2 = 8\|v\|^2$ for $v \in W_0$.

Proof. Expand the left side. Each v_i^2 occurs six times in the two diagonal sums, while $\sum_{i \neq j} v_i v_j = (\sum_i v_i)^2 - \sum_i v_i^2 = -\sum_i v_i^2$. The total is $8\sum_i v_i^2$. Polarization gives the operator identity. $\square$

Put

$$M = W_0 / \Lambda_{A_3}, \quad V_h = h\Lambda_{A_3} / \Lambda_{A_3}, \quad h = d^{-1}, \quad d \in \mathbb{N}, \quad d \geq 3, \quad (79)$$

and assign each root edge the reversible rate and vertex mass

$$k_{\alpha,h} = \frac{1}{8h^2}, \quad m_h(x) = h^3, \quad c_{\alpha,h} = \frac{h}{8}. \quad (80)$$

We identify W_0 with its dual using the ambient Euclidean pairing. For a translation-invariant jump process with displacements v_α and rates k_α , the infinitesimal covariance tensor is $Q = \sum_\alpha k_\alpha v_\alpha \otimes v_\alpha$. When it is positive definite, we use Q as the inverse metric tensor, so that the candidate metric is $g = Q^{-1}$. This convention corresponds to a diffusion generator with principal part $\frac{1}{2}Q : \nabla^2$ and fixes the factor of two independently of the Hodge square.

Proposition S2.5 (Jump covariance and the independent metric). *For the rates and displacements in (80),*

$$Q_h = \sum_{\alpha \in \Phi_{A_3}} k_{\alpha,h} (h\alpha) \otimes (h\alpha) = \frac{1}{8} \sum_{\alpha \in \Phi_{A_3}} \alpha \otimes \alpha = I_{W_0}. \quad (81)$$

Thus $g = Q_h^{-1}$ is the ambient flat metric on W_0 / Λ_{A_3} , independent of h . Its geodesic distance is

$$d_g(x, y) = \min_{\ell \in \Lambda_{A_3}} \|\tilde{x} - \tilde{y} + \ell\|, \quad (82)$$

where $\tilde{x}, \tilde{y}$ are arbitrary lifts to W_0 .

Proof. Substitute $k_{\alpha,h} = 1/(8h^2)$ and apply Lemma S2.4. The quotient formula for the flat geodesic distance gives (82) and is independent of the lifts. No Dirac operator or commutator seminorm enters this computation. $\square$

Integrals on M below use normalized Haar measure, equivalently the flat Riemannian volume divided by its total volume. Each periodic fundamental mesh cell then has measure h^3 , consistently with the vertex masses in (80). This normalization does not change g or its gradient.

Theorem S2.6 (A_3 Hodge identities and smooth-function consistency). *The graph spectral triple has*

$$L_h(f)^2 = \max_{x \in V_h} \frac{1}{8h^2} \sum_{\alpha \in \Phi_{A_3}} |f(x + h\alpha) - f(x)|^2, \quad (83)$$

$$D_h^2|_{C(V_h)} = -2\mathcal{L}_h, \quad (\mathcal{L}_h f)(x) = \frac{1}{8h^2} \sum_{\alpha \in \Phi_{A_3}} (f(x + h\alpha) - f(x)). \quad (84)$$

For every $F \in C^2(M)$,

$$|L_h(F|_{V_h}) - \|\nabla F\|_{L^\infty(M)}| \leq C_{A_3} h \|F\|_{C^2(M)}. \quad (85)$$

Moreover $\mathcal{L}_h F \rightarrow \frac{1}{2}\Delta F$ uniformly for $F \in C^4(M)$.

Proof. The first two identities follow from Theorem S2.2 and (80). Taylor expansion gives

$$\frac{F(x + h\alpha) - F(x)}{h} = \alpha \cdot \nabla F(x) + r_{\alpha,h}(x), \quad |r_{\alpha,h}(x)| \leq Ch \|F\|_{C^2}.$$

Apply the reverse triangle inequality in $\ell^2(\Phi_{A_3})$ and Lemma S2.4, then maximize over the mesh. Since the mesh is a Ch -net and ∇F has Lipschitz constant bounded by $\|F\|_{C^2}$, the maximum of $|\nabla F|$ on the mesh differs from its supremum on M by at most $Ch\|F\|_{C^2}$. This gives (85). For the generator, the linear terms cancel by central symmetry. The quadratic term is $\frac{1}{16} \sum_{\alpha} (\alpha \otimes \alpha) : \nabla^2 F = \frac{1}{2}\Delta F$; the cubic terms cancel and the remainder is $O(h^2\|F\|_{C^4})$. $\square$

S2.3 Uniform convergence of the Connes distance

Consistency on smooth functions does not by itself control the supremum in the Connes distance formula. We first construct a fixed family of periodic interpolants and then prove compactness of the normalized discrete Lipschitz balls. The interpolation estimate gives a uniform Lipschitz bound; the quadratic edge constraint will improve the limiting bound to the sharp constant one.

Lemma S2.7 (A periodic shape-regular A_3 triangulation). *Let $b_i = e_i - e_4$ for $1 \leq i \leq 3$, and define the linear isomorphism $T : \mathbb{R}^3 \rightarrow W_0$ by $Tu_i = b_i$, where u_1, u_2, u_3 is the standard basis of $\mathbb{R}^3$. For a permutation $\sigma \in S_3$, put*

$$K_\sigma = \text{conv}\{0, u_{\sigma(1)}, u_{\sigma(1)} + u_{\sigma(2)}, u_1 + u_2 + u_3\}. \quad (86)$$

The simplices $hT(k + K_\sigma)$, $k \in \mathbb{Z}^3$, form a periodic shape-regular triangulation, with a shape-regularity constant independent of h . For $h = d^{-1}$, $d \geq 3$, it descends to a simplicial mesh on M with vertex set V_h . The endpoints of every mesh edge can be joined by at most three root steps. If $I_h f$ is the continuous piecewise-affine interpolant of $f : V_h \rightarrow \mathbb{R}$, then

$$\|\nabla I_h f\|_{L^\infty(M)} \leq \sqrt{24} L_h(f). \quad (87)$$

The cells $Q_{h,x} = x + hT[0,1]^3$, interpreted on the torus, partition M up to boundaries, have normalized Haar measure h^3 , and have diameter at most Ch . If $\hat{f}_h$ equals $f(x)$ on $Q_{h,x}$, then

$$\|\hat{f}_h - I_h f\|_\infty \leq ChL_h(f). \quad (88)$$

Proof. Every $x \in \Lambda_{A_3}$ can be written uniquely as $x = \sum_{i=1}^3 x_i b_i$, so $T\mathbb{Z}^3 = \Lambda_{A_3}$. The six simplices K_σ partition the unit cube: the simplex indexed by σ is the region $1 \geq t_{\sigma(1)} \geq t_{\sigma(2)} \geq t_{\sigma(3)} \geq 0$. On a common face of adjacent cubes the induced ordering of the remaining two coordinates agrees, so the translated subdivisions fit face to face. Applying the fixed isomorphism T preserves this compatibility. There are only six nondegenerate reference simplices, and scaling by h leaves the ratio of diameter to inradius unchanged. This proves uniform shape regularity. Translations by Λ_{A_3} correspond to translations by $d\mathbb{Z}^3$ before scaling, which gives the periodic quotient mesh.

Consecutive vertices in the defining chain of each simplex differ by $hb_{\sigma(i)}$. These are root edges, and any pair of its vertices is joined by a subchain of at most three such edges. By (83), each consecutive difference in the values of f has absolute value at most $\sqrt{8}hL_h(f)$. On one simplex let $v = \nabla I_h f$, which is constant there. The three consecutive differences therefore give $|\langle v, b_i \rangle| \leq \sqrt{8}L_h(f)$ for $i = 1, 2, 3$. Since $T^*T = I_3 + \mathbf{1}_3\mathbf{1}_3^\top$ has eigenvalues 1, 1, 4, the smallest singular value of $T^* : W_0 \rightarrow \mathbb{R}^3$ is one. Hence

$$|v|^2 \leq |T^*v|^2 = \sum_{i=1}^3 |\langle v, b_i \rangle|^2 \leq 24L_h(f)^2,$$

which proves (87). Here $\mathbf{1}_3 = (1, 1, 1)^\top$.

The image under T of the unit cube is a fundamental parallelepiped of Λ_{A_3} . Its h -scaled translates have normalized measure h^3 and uniformly bounded diameter divided by h . On $Q_{h,x}$, integrate the piecewise-constant gradient of $I_h f$ along the segment from x to the evaluation point. The preceding gradient bound gives (88). $\square$

Lemma S2.8. *Let $h_n \downarrow 0$ and let $f_n : V_{h_n} \rightarrow \mathbb{R}$ satisfy $L_{h_n}(f_n) \leq 1$ and $f_n(x_{0,n}) = 0$ for $x_{0,n} \rightarrow x_0$. Then a subsequence of the interpolants $I_{h_n}f_n$ of Lemma S2.7 converges uniformly to $F \in W^{1,\infty}(M)$ with*

$$|\nabla F| \leq 1 \quad \text{almost everywhere.} \quad (89)$$

Proof. From (83), for every oriented root α and every vertex,

$$|f_n(x + h_n\alpha) - f_n(x)| \leq \sqrt{8}h_n. \quad (90)$$

Use the triangulation of Lemma S2.7 and put $F_n = I_{h_n}f_n$. Formula (87) gives $\|\nabla F_n\|_\infty \leq \sqrt{24}$, uniformly in n . Together with $F_n(x_{0,n}) = 0$ and compactness of M , this bounds $\|F_n\|_\infty$ uniformly. Arzelà–Ascoli yields, after passing to a subsequence, uniform convergence $F_n \rightarrow F$ for a Lipschitz function F . The task is now to improve this preliminary gradient bound to one.

For every $\alpha \in \Phi_{A_3}$ define the forward difference quotient on vertices by

$$g_{n,\alpha}(x) = \frac{f_n(x + h_n\alpha) - f_n(x)}{h_n},$$

and extend it as a piecewise-constant function on the cells $Q_{h_n,x}$ of Lemma S2.7. At every cell based at a lattice vertex,

$$\frac{1}{8} \sum_{\alpha \in \Phi_{A_3}} |g_{n,\alpha}|^2 \leq 1. \quad (91)$$

Thus the vector $g_n = (g_{n,\alpha})_\alpha$ is bounded in $L^\infty(M; \mathbb{R}^{12})$. Pass to a weak-* convergent subsequence $g_n \xrightarrow{*} g$. The set

$$K = \left\{ z \in \mathbb{R}^{12} : \frac{1}{8} \sum_{\alpha} |z_{\alpha}|^2 \leq 1 \right\}$$

is closed and convex. Since each g_n takes values in K , separation of K by linear functionals shows that the weak-* limit also satisfies $g(x) \in K$ for almost every x .

It remains to identify the components of g . Let $\varphi \in C^\infty(M)$ and sample it on the lattice. Periodic discrete summation by parts gives

$$h_n^3 \sum_{x \in V_{h_n}} g_{n,\alpha}(x) \varphi(x) = -h_n^3 \sum_{x \in V_{h_n}} f_n(x) \frac{\varphi(x) - \varphi(x - h_n \alpha)}{h_n}.$$

To justify the passage to integrals, let $\widehat{f}_n$, φ_n and $\psi_{n,\alpha}$ be the cellwise-constant extensions of, respectively, $f_n(x)$, $\varphi(x)$ and $(\varphi(x) - \varphi(x - h_n \alpha))/h_n$. The cell diameter bound, Taylor's formula and (88) give

$$\begin{aligned} \|\widehat{f}_n - F_n\|_\infty &\leq Ch_n, \\ \|\varphi_n - \varphi\|_\infty &\leq Ch_n \|\varphi\|_{C^1}, \\ \|\psi_{n,\alpha} - \alpha \cdot \nabla \varphi\|_\infty &\leq Ch_n \|\varphi\|_{C^2}. \end{aligned}$$

Since each cell has normalized measure h_n^3 , the summation-by-parts identity is exactly

$$\int_M g_{n,\alpha} \varphi_n = - \int_M \widehat{f}_n \psi_{n,\alpha}.$$

The displayed estimates and the uniform bounds on $g_{n,\alpha}$ and F_n allow replacement of the cellwise test functions by their smooth limits with an $O(h_n)$ error. Weak-* convergence of $g_{n,\alpha}$ and uniform convergence of F_n therefore yield

$$\int_M g_\alpha \varphi = - \int_M F_\alpha \cdot \nabla \varphi.$$

Hence $g_\alpha = \alpha \cdot \nabla F$ distributionally, and therefore almost everywhere. Combining this identity with (91) and the tight-frame identity (78) gives

$$|\nabla F|^2 = \frac{1}{8} \sum_{\alpha} |\alpha \cdot \nabla F|^2 = \frac{1}{8} \sum_{\alpha} |g_\alpha|^2 \leq 1$$

almost everywhere. □

Lemma S2.9 (Uniform smoothing of the torus distance). *There are constants $\varepsilon_0 > 0$ and $C_M < \infty$ such that, for every $y \in M$ and $0 < \varepsilon < \varepsilon_0$, there is a function $F_{y,\varepsilon} \in C^\infty(M)$ satisfying*

$$\|F_{y,\varepsilon} - d_g(\cdot, y)\|_\infty \leq \varepsilon, \quad \|\nabla F_{y,\varepsilon}\|_\infty \leq 1, \tag{92}$$

and

$$\sup_{y \in M} \|F_{y,\varepsilon}\|_{C^2(M)} \leq C_M \varepsilon^{-1}. \tag{93}$$

All constants are independent of the centre y .

Proof. Choose a nonnegative smooth function η supported in the Euclidean unit ball of W_0 , with integral one, and put $\eta_\varepsilon(u) = \varepsilon^{-3}\eta(u/\varepsilon)$. Let $r_y(\tilde{x}) = d_g(\tilde{x} + \Lambda_{A_3}, y)$ be the periodic lift of the distance function. For any lift $\tilde{x}$ of $x \in M$, define

$$F_{y,\varepsilon}(x) = \int_{W_0} \eta_\varepsilon(u) r_y(\tilde{x} - u) du,$$

where du is Euclidean volume on W_0 . Periodicity makes this independent of the lift. The function r_y is one-Lipschitz, so

$$|F_{y,\varepsilon}(x) - d_g(x, y)| \leq \int_{W_0} \eta_\varepsilon(u) |u| du \leq \varepsilon.$$

Convolution preserves the one-Lipschitz bound, giving the gradient estimate in (92). Equivalently, $\nabla F_{y,\varepsilon} = \eta_\varepsilon * \nabla r_y$ with $|\nabla r_y| \leq 1$ almost everywhere. Differentiating this identity once more gives, componentwise, $\nabla^2 F_{y,\varepsilon} = (\nabla \eta_\varepsilon) * \nabla r_y$. Consequently

$$\|\nabla^2 F_{y,\varepsilon}\|_\infty \leq C \|\nabla \eta_\varepsilon\|_{L^1(W_0)} \leq C\varepsilon^{-1}.$$

Also $0 \leq F_{y,\varepsilon} \leq \text{diam}(M)$. Taking $\varepsilon_0 < 1$ proves (93). The bounds are uniform in y because the distance functions are translations of one another on the flat torus. $\square$

Theorem S2.10 (Connes-distance convergence on the periodic A_3 lattice). *Let d_h be the pure-state Connes distance defined by (83), and let d_g be the flat geodesic distance of the metric determined by the jump covariance in Proposition S2.5. Then*

$$\sup_{x,y \in V_h} |d_h(x, y) - d_g(x, y)| \longrightarrow 0. \quad (94)$$

Proof. We prove the lower and upper bounds uniformly in the vertices.

For the lower bound, fix $0 < \varepsilon < \varepsilon_0$ and use the functions supplied by Lemma S2.9. The smooth-function consistency estimate in Theorem S2.6 gives

$$L_h(F_{y,\varepsilon}|_{V_h}) \leq 1 + Ch/\varepsilon,$$

uniformly in y . Dividing by this factor makes the sampled function admissible in the Connes dual formula. Formula (92) and the uniform diameter bound then give

$$\begin{aligned} d_h(x, y) &\geq \frac{F_{y,\varepsilon}(x) - F_{y,\varepsilon}(y)}{1 + Ch/\varepsilon} \\ &\geq \frac{d_g(x, y) - 2\varepsilon}{1 + Ch/\varepsilon} \geq d_g(x, y) - C' \left(\varepsilon + \frac{h}{\varepsilon} \right). \end{aligned} \quad (95)$$

The constants are independent of x, y, h and ε in the specified range. First let $h \downarrow 0$ and then $\varepsilon \downarrow 0$ to obtain the uniform lower bound.

For the upper bound, suppose the uniform upper estimate failed. Then for some $\varepsilon > 0$ there would be $h_n \downarrow 0$ and vertices x_n, y_n such that

$$d_{h_n}(x_n, y_n) \geq d_g(x_n, y_n) + \varepsilon.$$

Because the finite Lipschitz ball modulo constants is compact, the dual supremum is attained; choose f_n with $L_{h_n}(f_n) \leq 1$, normalize $f_n(y_n) = 0$, and arrange

$$f_n(x_n) \geq d_g(x_n, y_n) + \varepsilon. \quad (96)$$

After a subsequence, compactness of M gives $x_n \rightarrow x$ and $y_n \rightarrow y$. Lemma S2.8, applied with base points y_n , gives uniform convergence of the corresponding interpolants to a function F with $|\nabla F| \leq 1$ almost everywhere. The uniform edge Lipschitz bounds imply $F(y) = 0$ and $f_n(x_n) \rightarrow F(x)$. Continuity of d_g gives $d_g(x_n, y_n) \rightarrow d_g(x, y)$. Passing to the limit in (96) yields

$$F(x) - F(y) \geq d_g(x, y) + \varepsilon,$$

contradicting the fact that $|\nabla F| \leq 1$ implies that F is one-Lipschitz for the flat Riemannian distance. This proves the uniform upper bound and hence (94). $\square$

Each finite A_3 approximant is a compact quantum metric space, but the theorem above compares only pure states, namely vertices. A quadratic graph commutator seminorm need not equal the ordinary Lipschitz seminorm of its pure-state metric [4]. Consequently, the displayed distance convergence is not by itself a quantum Gromov–Hausdorff, propinquity, or spectral-propinquity theorem [5, 6, 7]. The stronger approximation results of [40, 36] control additional state-space or operator data. Lemma S2.8 supplies the scalar precompactness needed here; an upgrade would also require an explicit comparison of the full Lip-norm structures. The extensive one-form kernel remains a separate obstruction to retaining the full Hodge operator in a compact-resolvent limit. The operator theorem of Miranda–Parra [41] concerns a different differential complex with higher-degree cochains; it is complementary to this pure-state metric limit.

Remark S2.11 (Comparison with endpoint triples). Section S4 gives an independent construction on arbitrary closed connected Riemannian manifolds. A two-point spectral block on each short mesh edge has the graph Lipschitz seminorm exactly, so its Connes distance is the shortest-path metric. That direct approximation should be distinguished from the quadratic, covariance-normalized Hodge seminorm used in Theorem S2.10.

S3 Stationary-current fibres and affinity coordinates

The graph Hodge–Dirac construction depends only on symmetric jump data and therefore forgets the stationary current. We first describe this fibre as a cycle-current polytope. The principal result of this section then identifies its strictly positive relative interior with $H^1(G; \mathbb{R})$ through the affinity class, giving a global coordinate for the directed generator at fixed metric data. The section ends with the behaviour of currents and entropy production under graph quotients.

S3.1 Directed currents at fixed metric geometry

Fix a connected undirected graph $G = (V, E)$, positive masses m_x , and positive conductances c_e . Choose one orientation of each undirected edge and let $B : \mathbb{R}^E \rightarrow \mathbb{R}^V$ be the incidence matrix. An oriented current is extended to the reverse orientation by $j_{\bar{e}} = -j_e$.

Definition S3.1 (Cycle-current polytope and its positive interior). Define the closed cycle-current polytope

$$\overline{\mathcal{J}}(G, c) = \{j \in \text{Ker } B : |j_e| \leq c_e \text{ for every } e \in E\}, \quad (97)$$

and its relative interior

$$\mathcal{J}(G, c) = \{j \in \text{Ker } B : |j_e| < c_e \text{ for every } e \in E\}. \quad (98)$$

The closed polytope allows vanishing directed rates. The open fibre $\mathcal{J}(G, c)$ consists precisely of currents for which both directional fluxes $c_e \pm j_e$ are strictly positive on every retained edge; it is the natural domain of the affinity coordinate $\text{arctanh}(j/c)$ used below.

The decomposition $q = c + j$ is the usual separation of traffic and net current. For comparison, the stationary coordinates in [42] have node probability $\pi_x = m_x$, edge traffic $\tau_{xy} = 2c_{xy}$ and net flux $J_{xy} = 2j_{xy}$, so their formula $k_{xy} = (\tau_{xy} + J_{xy})/(2\pi_x)$ is exactly $k_{xy} = (c_{xy} + j_{xy})/m_x$. The next theorem records what this established parametrization means for a fixed graph Hodge–Dirac triple.

Theorem S3.2 (Stationary Markov realization fibre of a metric graph triple). *Let $\mathfrak{T}(G, m, c) = (C(V), \mathcal{H}_{m,c}, D_{m,c}, J, \gamma)$ be the metric graph spectral triple of Theorem S2.2. Then:*

(R1) *it has the reversible Markov realization*

$$k_{xy}^{\text{rev}} = \frac{c_{xy}}{m_x}; \quad (99)$$

(R2) *every $j \in \overline{\mathcal{J}}(G, c)$ gives a stationary Markov realization*

$$q_{xy} = c_{xy} + j_{xy}, \quad k_{xy} = \frac{c_{xy} + j_{xy}}{m_x}; \quad (100)$$

currents in $\mathcal{J}(G, c)$ are exactly those for which both directions on every retained edge have positive flux;

(R3) *conversely, every stationary Markov generator with masses m and symmetric conductances c is uniquely of the form (100) with $j \in \overline{\mathcal{J}}(G, c)$;*

(R4) *every realization in the fibre has the same metric spectral triple.*

Thus, up to unobserved parallel-channel refinements,

$$\text{RReal}_{\text{graph}}(\mathfrak{T}(G, m, c)) \cong \overline{\mathcal{J}}(G, c), \quad (101)$$

while $\mathcal{J}(G, c)$ is its strictly positive relative interior.

Proof. For (99), the stationary flux is $q_{xy} = c_{xy} = q_{yx}$, so detailed balance holds. For (100), nonnegativity follows from $|j_{xy}| \leq c_{xy}$; strict inequality is equivalent to positive flux in both directions. Moreover

$$\sum_y (q_{xy} - q_{yx}) = 2 \sum_y j_{xy} = 0$$

because $Bj = 0$, so m is stationary. Conversely, for a stationary continuous-time Markov generator define $j = (q - q^T)/2$. Stationarity gives $Bj = 0$, positivity gives the edge inequalities, and $q = c + j$ is unique. The graph spectral triple uses only m and $(q + q^T)/2 = c$. $\square$

Corollary S3.3. *For connected G , both the closed polytope $\overline{\mathcal{J}}(G, c)$ and its relative interior $\mathcal{J}(G, c)$ have affine dimension*

$$|E| - |V| + 1 = b_1(G). \quad (102)$$

The reversible realization is the only stationary realization exactly when G is a tree. If $b_1(G) > 0$, the metric passage to spectral data is not faithful.

Proof. The incidence matrix of a connected graph has rank $|V| - 1$, so $\dim \text{Ker } B = |E| - |V| + 1$. The strict inequalities cut out an open neighborhood of zero in this cycle space. A connected graph has zero first Betti number exactly when it is a tree. $\square$

The stationary entropy-production functional used below is the standard network expression for continuous-time Markov chains [43].

Proposition S3.4. *For $j \in \mathcal{J}(G, c)$, the steady entropy production is*

$$\sigma(j) = 2 \sum_{e \in E} j_e \log \frac{c_e + j_e}{c_e - j_e} \geq 0. \quad (103)$$

It vanishes exactly at $j = 0$. On the boundary of $\overline{\mathcal{J}}(G, c)$ the same formula is interpreted in the extended sense, with value $+\infty$ when a positive-conductance edge has a vanishing directional flux.

Proof. Start from $\frac{1}{2} \sum_{x,y} (q_{xy} - q_{yx}) \log(q_{xy}/q_{yx})$, pair the two orientations of each edge, and substitute $q_e = c_e + j_e$, $q_{\bar{e}} = c_e - j_e$. Each summand is nonnegative because j_e and the logarithm have the same sign; it vanishes only for $j_e = 0$. $\square$

Example S3.5 (A stationary family on the three-cycle). On an oriented three-cycle with equal conductance c , every constant current $j_e = s$, $|s| < c$, is divergence free. All values of s give one metric spectral triple, whereas

$$\sigma(s) = 6s \log \frac{c+s}{c-s}$$

varies continuously from zero and diverges as $|s| \uparrow c$.

Thus whenever $b_1(G) > 0$, the map from stationary generators with fixed (G, m, c) to the metric graph triple is noninjective on a $b_1(G)$ -dimensional open set. Quotienting by the finite automorphism group preserving (m, c) does not remove this positive-dimensional ambiguity. This consequence will be refined below by replacing the current coordinate with an intrinsic affinity class.

S3.2 Cohomological coordinates of the stationary-current fibre

For $j \in \mathcal{J}(G, c)$ define the real affinity

$$a_e = \operatorname{arctanh} \frac{j_e}{c_e} = \frac{1}{2} \log \frac{c_e + j_e}{c_e - j_e}. \quad (104)$$

Exact one-cochains are $\operatorname{Ran} B^\top$, so a has a class $[a] \in H^1(G; \mathbb{R}) = \mathbb{R}^E / \operatorname{Ran} B^\top$.

Lemma S3.6. *If $j \in \mathcal{J}(G, c)$ is divergence free and $[a] = 0$, then $j = 0$.*

Proof. Write $a = B^\top \phi$. Since $j_e = c_e \tanh((B^\top \phi)_e)$ and $Bj = 0$,

$$0 = \langle \phi, Bj \rangle = \sum_e c_e (B^\top \phi)_e \tanh((B^\top \phi)_e).$$

Every summand is nonnegative and vanishes only when the edge difference is zero. Connectedness makes ϕ constant, hence $a = j = 0$. $\square$

The preceding variational theory selects a nonlinear coclosed representative of a cohomology class [33, Proposition V.7]. Here $\psi(t) = \log \cosh t$ has $\psi'(t) = \tanh t$ and $\psi''(t) = \operatorname{sech}^2 t > 0$. Its conductance-weighted Euler equation is exactly the stationarity condition $B(c \tanh a) = 0$. The theorem identifies these representatives with all currents satisfying $|j_e| < c_e$ and establishes the smooth global coordinate needed to recover the directed Markov generator. We include the variational proof with the weights, positivity domain and smooth dependence made explicit.

Theorem S3.7 (Affinity coordinates for a fixed-traffic Markov fibre). *The map*

$$\mathfrak{A} : \mathcal{J}(G, c) \longrightarrow H^1(G; \mathbb{R}), \quad j \longmapsto [\operatorname{arctanh}(j/c)] \quad (105)$$

is a diffeomorphism. More explicitly, for a class α and representative a_0 , define on mean-zero vertex potentials

$$\mathcal{F}_\alpha(\phi) = \sum_{e \in E} c_e \log \cosh(a_{0,e} + (B^\top \phi)_e). \quad (106)$$

This functional is coercive and strictly convex. Its unique minimizer ϕ_α gives

$$a_\alpha = a_0 + B^\top \phi_\alpha, \quad j_{\alpha,e} = c_e \tanh(a_{\alpha,e}), \quad B j_\alpha = 0, \quad (107)$$

and this is the unique current with affinity class α .

Proof. If $E = \emptyset$, connectedness gives $|V| = 1$ and both spaces are singletons, so the assertion is immediate. Assume henceforth that $E \neq \emptyset$. Write $V_0 = \{\phi \in \mathbb{R}^V : \sum_x \phi_x = 0\}$ and $c_* = \min_e c_e > 0$. Connectedness gives $\|\phi\| \leq C \|B^\top \phi\|$ on V_0 . The inequality $\log \cosh t \geq |t| - \log 2$ therefore gives

$$\mathcal{F}_\alpha(\phi) \geq c_* \|B^\top \phi\|_1 - \sum_e c_e |a_{0,e}| - (\log 2) \sum_e c_e \longrightarrow \infty$$

as $\|\phi\| \rightarrow \infty$ in V_0 . Its Hessian is

$$B \operatorname{diag} \left(c_e \operatorname{sech}^2(a_{0,e} + (B^\top \phi)_e) \right) B^\top,$$

which is positive definite on V_0 : all edge weights are positive and $\operatorname{Ker} B^\top$ consists of constants. Thus the functional is coercive and strictly convex and has a unique minimizer. The Euler equation is

$$0 = \sum_e c_e \tanh(a_{\alpha,e}) (B^\top \psi)_e = \langle B j_\alpha, \psi \rangle \quad (\psi \in V_0).$$

Since $B j_\alpha$ has zero vertex sum, it follows that $B j_\alpha = 0$. Also $|\tanh a_{\alpha,e}| < 1$, so $j_\alpha \in \mathcal{J}(G, c)$.

Replacing a_0 by $a_0 + B^\top \chi$, with $\chi \in V_0$, replaces the minimizer by $\phi_\alpha - \chi$ and leaves a_α and j_α unchanged. The construction therefore depends only on the class α . Conversely, if $j \in \mathcal{J}(G, c)$ and $a = \operatorname{arctanh}(j/c)$, stationarity makes zero a critical point of (106) with representative a . Strict convexity makes it the unique minimizer. Hence the two constructions are mutual inverses.

Finally choose a linear section of $\mathbb{R}^E \rightarrow H^1(G; \mathbb{R})$ and use it to select a_0 smoothly. The stationarity equation, as a map to V_0 , has derivative in ϕ equal to the displayed positive-definite Hessian. The implicit-function theorem gives smooth local dependence on α ; uniqueness makes these local solutions agree globally. The forward map is smooth because componentwise $\operatorname{arctanh}(j/c)$ is smooth on the open current polytope. This proves the diffeomorphism. $\square$

Corollary S3.8. *For fixed (G, m, c) , the graph triple together with its affinity class*

$$\mathfrak{S}_{\text{aff}}(j) = (C(V), \mathcal{H}_{m,c}, D_{m,c}, J, \gamma; [a]) \quad (108)$$

determines the complete stationary generator through

$$[a] \longmapsto a_\alpha \longmapsto j = c \tanh a_\alpha \longmapsto q = c + j \longmapsto k_{xy} = q_{xy}/m_x.$$

Thus the affinity class determines the directed generator throughout the strictly positive fibre.

Proof. Apply Theorem S3.7 and Theorem S3.2. $\square$

The entropy production is the current–affinity pairing

$$\sigma(j) = 4 \sum_{e \in E} j_e a_e, \quad (109)$$

and is unchanged by $a \mapsto a + B^\top \phi$ because $\langle j, B^\top \phi \rangle = \langle B j, \phi \rangle = 0$.

S3.3 Coarse graining of stationary currents

The affinity coordinate reconstructs each current at fixed conductances. Under a graph quotient, however, the natural operation is to sum the directional fluxes, and hence to sum currents and conductances. The following result records this operation and the associated entropy bound.

Write $B_G = B : \mathbb{R}^E \rightarrow \mathbb{R}^V$ for the incidence matrix and $\mathcal{Z}(G) = \text{Ker } B_G$. A vertex quotient $q : V \rightarrow \bar{V}$ induces maps Q_V and Q_E by summing vertex and oriented edge coefficients over fibres.

Theorem S3.9 (Graph quotients and entropy contraction). *The incidence maps satisfy*

$$B_{\bar{G}} Q_E = Q_V B_G, \quad (110)$$

so $Q_E \mathcal{Z}(G) \subseteq \mathcal{Z}(\bar{G})$. If $\bar{c}_{\bar{e}} = \sum_{e \mapsto \bar{e}} c_e$, then $Q_E \mathcal{J}(G, c) \subseteq \mathcal{J}(\bar{G}, \bar{c})$. When the vertex fibres are connected and the coarse graph retains every inter-fibre edge class, the induced map on cycle spaces is onto and

$$0 \longrightarrow \text{Ker}(Q_E|_{\mathcal{Z}(G)}) \longrightarrow \mathcal{Z}(G) \xrightarrow{Q_E} \mathcal{Z}(\bar{G}) \longrightarrow 0, \quad (111)$$

with

$$\dim \text{Ker}(Q_E|_{\mathcal{Z}(G)}) = b_1(G) - b_1(\bar{G}). \quad (112)$$

The stationary entropy production satisfies

$$\sigma_{\bar{G}}(Q_E j) \leq \sigma_G(j). \quad (113)$$

Proof. For a coarse vertex, the divergence of the summed inter-fibre current equals the sum of fine divergences; internal fine edges cancel in pairs. This is exactly (110). The conductance inequality $|(Q_E j)_{\bar{e}}| \leq \sum_{e \mapsto \bar{e}} |j_e| < \bar{c}_{\bar{e}}$ gives the fibre map. Connected fibres allow every coarse cycle to be lifted and completed inside fibres, proving surjectivity. The dimension formula follows from the Betti-number formula for connected graphs. Finally, on every coarse edge the log-sum inequality for the aggregated forward and reverse fluxes gives entropy contraction; summing over coarse edges yields (113). $\square$

The surjectivity assertion concerns the linear cycle spaces. It does not assert surjectivity of the bounded current polytopes. The graph quotient therefore defines a map of admissible current fibres without identifying every coarse current with an admissible fine current.

Section S5 develops the edge-ratio and cellular Hodge interpretations of the same affinity class, including its curvature and holonomy data.

S4 Independent endpoint metric approximation

The periodic A_3 metric theorem uses the Hodge–Dirac operator associated with a stationary graph and therefore has a quadratic local commutator seminorm. The following construction is independent of that choice. It assigns one elementary spectral block to each sufficiently short edge of a dense mesh. Edge-based triples recovering graph geodesic distance are standard; see, for example, the auxiliary edge construction in [44]. Here we record the resulting uniform approximation on a closed manifold. The commutator seminorm is exactly the ordinary graph Lipschitz seminorm, so the comparison is explicit on all pure states.

Let M be closed and connected. For each $h \downarrow 0$, let $V_h \subset M$ be an h -net, so every point of M lies within distance h of a vertex. Choose a mesoscopic radius $\rho_h \downarrow 0$ with $h/\rho_h \rightarrow 0$ and, for all

sufficiently small h, ρ_h below the injectivity radius. Join every pair $x, y \in V_h$ with $d_g(x, y) \leq \rho_h$ and assign that edge the length

$$\ell_h(x, y) = d_g(x, y), \quad (114)$$

or, more generally, a positive length satisfying $\sup_{E_h} |\ell_h/d_g - 1| \rightarrow 0$. For an unoriented edge $e = \{x, y\}$ set

$$\mathcal{H}_e = \mathbb{C}^2, \quad \pi_e(f) = \begin{pmatrix} f(x) & 0 \\ 0 & f(y) \end{pmatrix}, \quad D_e = \frac{1}{\ell_h(x, y)} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (115)$$

Let

$$\mathcal{H}_h^{\text{met}} = \bigoplus_{e \in E_h} \mathcal{H}_e, \quad \pi_h = \bigoplus_{e \in E_h} \pi_e, \quad D_h^{\text{met}} = \bigoplus_{e \in E_h} D_e. \quad (116)$$

This is the finite endpoint metric triple used below.

Theorem S4.1 (Endpoint spectral triples and geodesic convergence). *Under the preceding mesh assumptions, for $f \in C(V_h)$,*

$$L_h^{\text{met}}(f) = \|[D_h^{\text{met}}, \pi_h(f)]\| = \max_{\{x, y\} \in E_h} \frac{|f(x) - f(y)|}{\ell_h(x, y)}. \quad (117)$$

The pure-state Connes distance is exactly the graph shortest-path metric. On every closed Riemannian manifold,

$$\sup_{x, y \in V_h} |d_h^{\text{met}}(x, y) - d_g(x, y)| \rightarrow 0. \quad (118)$$

For smooth f , $L_h^{\text{met}}(f|_{V_h}) \rightarrow \|\nabla_g f\|_\infty$.

Proof. The commutator of one endpoint block has off-diagonal entry $(f(y) - f(x))/\ell_h(x, y)$, so the operator norm gives (117). Telescoping along graph paths gives one inequality in the Connes dual formula; the distance-to-a-fixed-vertex function has seminorm one and gives equality.

When $\ell_h = d_g$ on edges, every graph path is a chain of geodesic segments, so the triangle inequality gives $d_g \leq d_h^{\text{met}}$; the same conclusion holds with an $o(1)$ multiplicative error for the allowed approximate lengths. Conversely, fix $x, y \in V_h$ and partition a minimizing geodesic from x to y into pieces of length at most $\rho_h/2$. Replace each interior partition point by a vertex of the h -net. For small h , consecutive replacement vertices are at distance at most $\rho_h/2 + 2h < \rho_h$ and are therefore connected. The number of pieces is $O(1 + d_g(x, y)/\rho_h)$, so the accumulated replacement error is $O(h + h/\rho_h)$ uniformly on the compact manifold. Allowing the final partition remainder gives, uniformly,

$$d_h^{\text{met}}(x, y) \leq d_g(x, y) + C \left(\rho_h + \frac{h}{\rho_h} \right) + o(1).$$

Hence (118) follows.

For smooth f , the upper bound on $L_h^{\text{met}}(f|_{V_h})$ follows from the geodesic mean-value estimate. For the reverse bound, choose p where $|\nabla f|$ is within $o(1)$ of its maximum, move a distance $\rho_h/2$ from p in the gradient direction, and approximate both endpoints by vertices of the h -net. Taylor expansion and $h/\rho_h \rightarrow 0$ give the matching lower bound. Thus $L_h^{\text{met}}(f|_{V_h}) \rightarrow \|\nabla_g f\|_\infty$. $\square$

For these endpoint metric triples the finite Lip-norm is exactly the ordinary Lipschitz seminorm of the graph shortest-path distance. The metric on $S(C(V_h))$, identified with probability measures

on V_h , is therefore the Kantorovich transport distance for d_h^{met} [4]. If p, q are probability measures supported on V_h , the same transport plans give

$$\left| \text{mk}_{L_h^{\text{met}}}(p, q) - W_{1, d_g}(p, q) \right| \leq \sup_{x, y \in V_h} |d_h^{\text{met}}(x, y) - d_g(x, y)|.$$

Here W_{1, d_g} denotes the transport distance between the same measures viewed on M . Unlike the general quadratic graph seminorm, this particular endpoint construction thus yields an immediate comparison on all states, not only on vertices.

S5 Edge affinities and cellular Hodge decomposition

This section records two auxiliary descriptions of the affinity used in Section S3: forward/reverse edge-density ratios and their cellular Hodge decomposition. They relate the stationary-current coordinate to transport, local curvature and global holonomy.

S5.1 Relative edge-density data

Let E^{or} contain both orientations of every edge, with reversal $e \mapsto \bar{e}$, source $s(e)$, terminal $t(e) = s(\bar{e})$, and positive directed stationary flux q_e . On $\mathcal{H}_E = \ell^2(E^{\text{or}})$, define

$$(J_E \xi)(e) = \overline{\xi(\bar{e})}, \quad (\Delta_q \xi)(e) = \frac{q_e}{q_{\bar{e}}} \xi(e) = e^{2a_e} \xi(e), \quad S_q = J_E \Delta_q^{1/2}. \quad (119)$$

Using the same left/right endpoint convention as in (70), put $(\lambda_E(f)\xi)(e) = f(t(e))\xi(e)$ and $(\rho_E(f)\xi)(e) = f(s(e))\xi(e)$. The involution J_E is auxiliary edge reversal; it is not the signed one-form real structure in (72).

Theorem S5.1 (Edge reversal and relative density ratios). *The edge data (119) satisfy*

$$J_E^2 = I, \quad J_E \Delta_q J_E = \Delta_q^{-1}, \quad S_q^2 = I, \quad (120)$$

$$J_E \lambda_E(f^*) J_E = \rho_E(f), \quad S_q \lambda_E(f^*) S_q^{-1} = \rho_E(f), \quad \log \Delta_q(e) = 2a_e. \quad (121)$$

The reversible branch is exactly $\Delta_q = I$.

Proof. Reversal preserves counting measure, so J_E is antiunitary and squares to the identity. Since $a_{\bar{e}} = -a_e$, conjugating the diagonal multiplier e^{2a_e} by J_E gives its inverse; hence $S_q^2 = I$. Reversal exchanges the terminal and source of an edge, proving the opposite-action identities. The logarithm statement is the definition of a_e . $\square$

Remark S5.2 (Scope of the modular terminology). Here Δ_q is the multiplication operator given by the density ratio of the forward and reversed edge measures. No von Neumann algebra with a cyclic separating vector is specified. The displayed reversal identities are therefore not asserted to construct a Tomita–Takesaki modular operator.

For an oriented cycle C ,

$$\text{Hol}_{\Delta}(C) = \prod_{e \in C} \left(\frac{q_e}{q_{\bar{e}}} \right)^{\epsilon_{C,e}} = \exp \left(2 \sum_{e \in C} \epsilon_{C,e} a_e \right). \quad (122)$$

S5.2 Cellular curvature and gauge classes

Let

$$C^0 \xrightarrow{d_0} C^1 \xrightarrow{d_1} C^2 \xrightarrow{d_2} C^3 \quad (123)$$

be a finite real Hilbert complex and let $\mathcal{H}^1 = \text{Ker } d_1 \cap \text{Ker } d_0^*$.

Proposition S5.3. *For $a \in C^1$, define*

$$h_a = P_{\mathcal{H}^1} a, \quad F_a = d_1 a. \quad (124)$$

Then a has the unique orthogonal decomposition

$$a = d_0 \phi + h_a + d_1^* \psi. \quad (125)$$

The three orthogonal components are unique; the potentials are unique only after requiring $\phi \perp \text{Ker } d_0$ and $\psi \perp \text{Ker } d_1^$. Its coexact part is reconstructed from the curvature by*

$$P_{\text{Ran } d_1^*} a = d_1^* (d_1 d_1^*)^\dagger F_a. \quad (126)$$

Consequently

$$C^1 / \text{Ran } d_0 \longrightarrow \mathcal{H}^1 \oplus \text{Ran } d_1, \quad [a]_{\text{gauge}} \longmapsto (h_a, F_a) \quad (127)$$

is injective and onto the displayed product. A canonical representative is

$$a_{h,F} = h + d_1^* (d_1 d_1^*)^\dagger F. \quad (128)$$

Moreover $d_2 F_a = 0$ and

$$\|a\|^2 = \|d_0 \phi\|^2 + \|h_a\|^2 + \left\langle F_a, (d_1 d_1^*)^\dagger F_a \right\rangle. \quad (129)$$

Proof. Finite Hodge decomposition gives $C^1 = \text{Ran } d_0 \oplus \mathcal{H}^1 \oplus \text{Ran } d_1^*$, proving (125). Since d_1 annihilates the first two summands, $F_a = d_1 d_1^* \psi$; Moore–Penrose inversion on $\text{Ran } d_1$ gives (126). If two cochains have the same (h, F) , their difference has zero harmonic projection and lies in $\text{Ker } d_1$. Its coexact part then has zero norm, so the difference is exact. This proves injectivity; (128) proves surjectivity. Orthogonality gives the norm identity, and $d_2 d_1 = 0$ gives the Bianchi identity. $\square$

Corollary S5.4. *For a cellular affinity a :*

- (M1) *if $F_a \neq 0$, the edge-ratio transport has local face curvature;*
- (M2) *if $F_a = 0$ but $h_a \neq 0$, it is locally flat with nontrivial global positive holonomy;*
- (M3) *if $F_a = h_a = 0$, then $a = d_0 \phi$ and the edge-ratio transport is trivialized by a positive vertex gauge.*

On the periodic three-torus, the global flat sector has dimension three.

Theorem S5.5. *For every oriented two-cell p ,*

$$\prod_{e \subset \partial p} \left(\frac{q_e}{q_{\bar{e}}} \right)^{\epsilon_{p,e}} = \exp(2F_a(p)). \quad (130)$$

Thus the edge-ratio transport defines a flat positive line transport on the cell complex exactly when its curvature vanishes on every two-cell.

Proof. By the cellular coboundary formula, $F_a(p) = \sum_e \epsilon_{p,e} a_e$. Substitute $q_e/q_{\bar{e}} = e^{2a_e}$ and multiply. $\square$

S6 Operational reconstruction, realization and recognition

This section develops an operational source for the finite algebra in Section S1. Complete intervention statistics first give a future-separated history C^* -algebra through causal Choi tomography, a prefix-minimal coherent realization, and a contextual quotient. Finite word-Gram data then reconstruct its algebraic operations. The converse realization and recognition results preserve representation multiplicities within a boundary-relative convention. The construction recovers the algebraic input; the differential and its scale remain independent data.

Quantum combs and process tensors provide the multitime realization framework [23, 45, 25]; future-equivalence quotients provide the predictive-state background [30, 46, 32].

S6.1 Reconstruction from complete intervention data

The causal Choi constraints and sequential-dilation ingredients used here are standard in quantum-comb/process-tensor theory [23, 25]; the future-equivalence quotient is the finite typed analogue of minimal predictive-state constructions [30, 46, 32]. The new object used below is the future-separated represented C^* -history algebra and its finite flat word-Gram reconstruction.

Fix a finite cutoff X . The operational input consists of a finite typed event grammar with finite reachable histories. If a history h ends at cut type x , let $\mathbf{F}_{X,x}$ be the finite admissible future words beginning at x . The conditional event law assigns normalized one-step probabilities and therefore cylinder probabilities

$$p_X(w \mid h) = \prod_{j=1}^k p_X(a_j \mid ha_1 \cdots a_{j-1}), \quad w = a_1 \cdots a_k, \quad (131)$$

with $p_X(\epsilon_x \mid h) = 1$. No persistent state space, algebra, Hilbert space, metric, or Dirac operator is assumed at this stage.

Two histories of the same cut type are *future equivalent* when no declared finite continuation distinguishes them:

$$h \sim_X h' \iff p_X(w \mid h) = p_X(w \mid h') \text{ for every } w \in \mathbf{F}_{X,x}. \quad (132)$$

Histories of different cut type are never identified. The relation is a typed *partial* right congruence on reachable branches. Indeed, if $h \sim_X h'$ and $a : x \rightarrow y$ satisfies $p_X(a \mid h) > 0$, then the one-letter future gives $p_X(a \mid h) = p_X(a \mid h') > 0$, and for every continuation v ,

$$p_X(v \mid ha) = \frac{p_X(av \mid h)}{p_X(a \mid h)} = \frac{p_X(av \mid h')}{p_X(a \mid h')} = p_X(v \mid h'a).$$

Thus $ha \sim_X h'a$. No update state is assigned to a zero-probability extension; its conditional continuation law is operationally undefined and is not needed in the reachable predictive state space.

Define the minimal predictive state space by

$$Z_{X,x}^{\min} := \mathbf{H}_{X,x} / \sim_X, \quad q_{X,x} : \mathbf{H}_{X,x} \rightarrow Z_{X,x}^{\min}. \quad (133)$$

For an admissible event $a : x \rightarrow y$, define

$$\kappa_{X,a}([h]) = p_X(a \mid h), \quad \text{Dom}(T_{X,a}) = \{[h] : \kappa_{X,a}([h]) > 0\}, \quad T_{X,a}([h]) = [ha]. \quad (134)$$

Future equivalence makes $\kappa_{X,a}$ well defined on every class and $T_{X,a}$ well defined on its displayed domain. Thus persistence is reconstructed from the conditional future law rather than introduced

as an independent coordinate. Moreover, this quotient is universal among reachable deterministic predictive realizations, where updates are required only on positive-probability branches. Equality of two realization states forces equality of all their future cylinder probabilities, and therefore there is a unique surjection

$$\pi_{X,x} : S_{X,x} \twoheadrightarrow Z_{X,x}^{\min} \quad (135)$$

intertwining branch probabilities and all defined updates. Hence $Z_X^{\min}$ is the unique coarsest reachable deterministic predictive realization in this partial-transition sense.

To reconstruct the operator-algebraic history object, assume in addition that the finite multitime intervention tables are specified on tomographically complete finite operator frames, are affine under randomization and additive under coarse graining, and obey normalization, prefix compatibility and causal-discard conditions. Positivity here means positivity of the reconstructed multitime Choi tensors, as required by completely positive process realization; nonnegative entries in a product-frame table alone are not a substitute for this condition. These finite conditions will be stated explicitly in (137).

For intervention slot k , let

$$\mathbf{E}_k = \text{Herm}(H_k^{\text{out}} \otimes H_k^{\text{in}})$$

be the real Choi-coordinate space. Physical completely positive branches span $\mathbf{E}_k$: a Hermitian element is the difference of its positive and negative parts, after separately rescaling those parts so that their output partial traces are bounded by the identity. Choose physical bases $M_{k,\alpha}$ and Hilbert–Schmidt duals $D_{k,\alpha}$. Write $\mathbf{p}_X$ for the resulting multilinear intervention functional. From the N -slot probability table define

$$R_X^{(N)} = \sum_{\alpha_1, \dots, \alpha_N} \mathbf{p}_X(M_{1,\alpha_1}, \dots, M_{N,\alpha_N}) D_{N,\alpha_N} \otimes \dots \otimes D_{1,\alpha_1}. \quad (136)$$

We use the following standard finite-dimensional process realization as an input [23, 45, 25]. It is stated in the present frame notation to make the subsequent quotient construction unambiguous.

Proposition S6.1 (Standard finite process realization). *The complete finite intervention table determines a unique frame-independent Hermitian tensor $R_X^{(N)}$ representing every multilinear intervention probability. It represents a deterministic causal process exactly when its prefixes satisfy*

$$R_X^{(k)} \succeq 0, \quad \text{Tr}_{H_k^{\text{out}}} R_X^{(k)} = I_{H_k^{\text{in}}} \otimes R_X^{(k-1)}, \quad R_X^{(0)} = 1. \quad (137)$$

Moreover, with $A_{X,0} = \mathbb{C}$ and $A_{X,k}$ the conjugate support of $R_X^{(k)}$, there is a coherent sequential realization with $\dim A_{X,k} = \text{rank } R_X^{(k)}$ and isometries

$$V_{X,k} : H_k^{\text{in}} \otimes A_{X,k-1} \longrightarrow H_k^{\text{out}} \otimes A_{X,k} \quad (138)$$

whose sequential Choi link gives $R_X^{(N)}$. Here support minimality means that the cumulative isometry through each cut is a minimal Stinespring dilation of the corresponding prefix channel. Any two such coherent sequential realizations, with the same external port identifications and $U_{X,0} = I$, are related by unique memory unitaries $U_{X,k}$ satisfying

$$V'_{X,k} = (I \otimes U_{X,k}) V_{X,k} (I \otimes U_{X,k-1}^*). \quad (139)$$

Consequently the reduced primitive branch maps, their Hilbert–Schmidt adjoints, all mixed history words, and their word Grams are transported by canonical unitary equivalence.

Proof. Because the physical frame is a basis of the real Hermitian Choi space, its tensor products determine the multilinear functional uniquely. Expansion in the dual frame gives (136); uniqueness makes the resulting tensor independent of the frame. The positive causal-prefix criterion is the standard comb realization theorem [23]. Existence of a sequential isometric realization with $\dim A_{X,k} = \text{rank } R_X^{(k)}$ is [45, Theorem 2]. We give the compatibility argument for uniqueness explicitly.

Suppress X and put

$$I_k = \bigotimes_{j=1}^k H_j^{\text{in}}, \quad O_k = \bigotimes_{j=1}^k H_j^{\text{out}}.$$

Let $W_k : I_k \rightarrow O_k \otimes A_k$ be the cumulative isometry through cut k , and let $W'_k : I_k \rightarrow O_k \otimes A'_k$ be that of a second realization. Both give the channel with Choi matrix $R^{(k)}$ after tracing out the last memory. Prefix minimality means

$$\text{span}\{(\langle \eta | \otimes I)W_k \xi : \eta \in O_k, \xi \in I_k\} = A_k. \quad (140)$$

Minimal Stinespring uniqueness [47] therefore gives a unique unitary $U_k : A_k \rightarrow A'_k$ with

$$W'_k = (I_{O_k} \otimes U_k)W_k. \quad (141)$$

Uniqueness follows directly from the spanning condition (140), not from a choice of purification coordinates.

The sequential factorizations, with the natural tensor permutations understood, are

$$W_k = (I_{O_{k-1}} \otimes V_k)(W_{k-1} \otimes I_{H_k^{\text{in}}})$$

and the analogous identity for W'_k . Substituting (141) at cuts k and $k-1$ shows that

$$V'_k(I_{H_k^{\text{in}}} \otimes U_{k-1}) - (I_{H_k^{\text{out}}} \otimes U_k)V_k$$

annihilates all vectors obtained by contracting the earlier output legs of W_{k-1} , tensored with arbitrary vectors of H_k^{in} . These vectors span $H_k^{\text{in}} \otimes A_{k-1}$ by (140). The displayed difference is therefore zero, which proves (139) on the whole memory domain. For $k=1$ the same argument uses $A_0 = \mathbb{C}$ and $U_0 = I$.

The maps $X \mapsto U_k X U_k^*$ are unitaries on the Hilbert–Schmidt memory spaces. They intertwine the branch maps obtained from the fixed external interventions, and hence also their Hilbert–Schmidt adjoints, mixed words and word Grams. $\square$

Remark S6.2 (Scope of sequential uniqueness). The uniqueness assertion concerns coherent isometric realizations that are minimal at every prefix. It is not a uniqueness theorem for arbitrary mixed-state, measured-memory or environment-discarding implementations with a small physical memory. The canonical process representation below uses precisely the prefix-minimal coherent class.

For each typed boundary object ξ , put $\mathcal{X}_{X,\xi} = \mathcal{B}(A_{X,\xi})$ with its Hilbert–Schmidt inner product and let $\mathcal{X}_X = \bigoplus_{\xi} \mathcal{X}_{X,\xi}$. If a primitive branch $a : \xi \rightarrow \eta$ has reduced Schrödinger superoperator $T_{X,a}$, write

$$\rho_X(a) = P_{\eta} T_{X,a} P_{\xi}$$

and adjoin a formal reverse $a^{\sharp}$ represented by $\rho_X(a)_{\text{HS}}^*$. The reverse is an algebraic adjoint and need not be an executable physical channel. Let $\mathfrak{F}_X^{\text{op}}$ be the finite-object free path $*$ -algebra on these primitive branches and their formal reverses.

Theorem S6.3. *The represented process algebra*

$$\mathcal{A}_X^{\text{proc}} := \mathfrak{F}_X^{\text{op}} / \text{Ker } \rho_X \cong C^*(\rho_X(a) : a \text{ primitive}) \subseteq \mathcal{B}(\mathcal{X}_X) \quad (142)$$

is a finite-dimensional C^* -algebra. Its represented $*$ -isomorphism class is independent of the tomographic frame, Kraus coordinates, purification coordinates, and the chosen support-minimal sequential realization.

Proof. Typed concatenation is sent to block composition, object units to orthogonal projections, and $\sharp$ to Hilbert–space adjoint, so ρ_X is a $*$ -homomorphism and its kernel is a two-sided $*$ -ideal. The image is finite-dimensional. The unitary covariance of Proposition S6.1 conjugates the complete word representation, giving the asserted represented $*$ -isomorphism. $\square$

The standard dilation theorems ensure that finite CP branches can be implemented on finite enlarged Hilbert spaces, but an operator corner is not automatically a multiplicative representation of its parent algebra. The following elementary test, which will also motivate the separation between the operational input and the algebraic essential-image theorem above, makes this distinction explicit [47, 48].

Proposition S6.4. *Every finite family of finite-dimensional completely positive trace-nonincreasing maps admits a simultaneous finite unitary completion after adjoining finite ancillas and a finite control register. Moreover, let $\mathcal{B} \subseteq \mathcal{B}(H)$ be a finite represented unital C^* -algebra, let $V : K \rightarrow H$ be an isometry, put $P = VV^*$ and $Q = I - P$, and set $\kappa(b) = V^*bV$. Then*

$$\kappa(ab) - \kappa(a)\kappa(b) = V^*aQbV, \quad \kappa(a^*a) - \kappa(a)^*\kappa(a) = (QaV)^*(QaV) \succeq 0. \quad (143)$$

Consequently κ is a $*$ -homomorphism on a unital $*$ -subalgebra $\mathcal{C} \subseteq \mathcal{B}$ if and only if P reduces $\mathcal{C}$.

Proof. Choose Stinespring contractions for the finitely many CP maps, dilate each contraction to its Julia unitary, and take their finite controlled direct sum. For the compression identity, insert $I = P + Q$ between a and b . The second formula is the case (a^*, a) . If P reduces $\mathcal{C}$, the defect vanishes and compression is multiplicative and $*$ -preserving. Conversely, if the restriction is a $*$ -homomorphism, applying the positive square defect to c and c^* gives $QcP = Qc^*P = 0$ for every $c \in \mathcal{C}$, hence $[P, c] = 0$. $\square$

The process algebra removes coordinate ambiguity in the minimal sequential dilation, but it can still contain algebraic directions that no admitted future experiment detects. The next quotient removes exactly those observationally null directions. Thus support minimality and future separation solve two different minimality problems.

Let $\mathcal{R}_X$ denote the finite family of scalar readout functionals generated by the declared future continuations and the minimal predictive record reads. Define

$$\mathcal{N}_X^{\text{fut}} = \{q \in \mathcal{A}_X^{\text{proc}} : R(uqv) = R(uq^*v) = 0 \text{ for every } R \in \mathcal{R}_X \text{ and } u, v \in \mathcal{A}_X^{\text{proc}}\}. \quad (144)$$

If $\pi : \mathcal{A}_X^{\text{proc}} \twoheadrightarrow B$ is a C^* -quotient through which every $R \in \mathcal{R}_X$ factors as $R = \tilde{R} \circ \pi$, call B *future separated* when

$$\tilde{R}(ubv) = \tilde{R}(ub^*v) = 0 \quad \text{for all admitted } \tilde{R} \text{ and } u, v \in B \implies b = 0. \quad (145)$$

Theorem S6.5 (Universal future-separated history algebra). *The space $\mathcal{N}_X^{\text{fut}}$ is a two-sided $*$ -ideal and*

$$\mathcal{A}_X^{\text{hist}} := \mathcal{A}_X^{\text{proc}} / \mathcal{N}_X^{\text{fut}} \quad (146)$$

is future separated and carries every admitted readout. It is universal with these properties: if a C^ -quotient B of $\mathcal{A}_X^{\text{proc}}$ carries every admitted readout and is future separated in the sense of (145), then the quotient maps identify B canonically with $\mathcal{A}_X^{\text{hist}}$.*

Proof. For $q \in \mathcal{N}_X^{\text{fut}}$ and $c, d \in \mathcal{A}_X^{\text{proc}}$, $R(u(cqd)v) = R((uc)q(dv)) = 0$ in every admitted context; applying the same argument to adjoints makes the space $*$ -stable. Hence it is a closed two-sided $*$ -ideal. Every admitted readout vanishes on $\mathcal{N}_X^{\text{fut}}$ and therefore factors through $\mathcal{A}_X^{\text{hist}}$. By construction, an element of the quotient annihilated in all two-sided readout contexts has a representative in $\mathcal{N}_X^{\text{fut}}$, so the quotient is future separated.

Now let $\pi : \mathcal{A}_X^{\text{proc}} \rightarrow B$ carry every admitted readout. If $q \in \text{Ker } \pi$, then every admitted two-sided readout context vanishes on q and q^* , hence $\text{Ker } \pi \subseteq \mathcal{N}_X^{\text{fut}}$. Thus there is a canonical surjection $B \rightarrow \mathcal{A}_X^{\text{hist}}$. Conversely, if $q \in \mathcal{N}_X^{\text{fut}}$, then $\pi(q)$ is annihilated by every induced two-sided readout context in B ; future separation gives $\pi(q) = 0$. Hence $\mathcal{N}_X^{\text{fut}} \subseteq \text{Ker } \pi$, the two kernels coincide, and the canonical surjection is a $*$ -isomorphism. $\square$

On every finite-dimensional C^* -algebra there is a canonical normalized regular trace. For $\mathcal{A}_X^{\text{hist}} \cong \bigoplus_{\lambda} M_{n_{\lambda}}(\mathbb{C})$, set

$$\hat{\tau}_{X,\text{reg}}(a) = \frac{\text{Tr}_{\mathbb{C}}(La)}{\dim_{\mathbb{C}} \mathcal{A}_X^{\text{hist}}} = \frac{\sum_{\lambda} n_{\lambda} \text{Tr}(a_{\lambda})}{\sum_{\lambda} n_{\lambda}^2}. \quad (147)$$

If $[w]$ denotes the represented class of a finite mixed history word, define

$$\mathbb{K}_{X,r}(v, w) = \hat{\tau}_{X,\text{reg}}([v]^*[w]), \quad |v|, |w| \leq r. \quad (148)$$

The regular trace is faithful, so the null space of $\mathbb{K}_{X,r}$ is exactly the word-relation space at depth r .

Theorem S6.6 (Finite flat reconstruction of the history algebra). *There is a least finite depth r_X for which*

$$\text{rank } \mathbb{K}_{X,r_X} = \text{rank } \mathbb{K}_{X,r_X+1}. \quad (149)$$

At this depth the represented words of length at most r_X span all of $\mathcal{A}_X^{\text{hist}}$, and the panel through depth $r_X + 1$ determines the unit, multiplication, involution, and every represented mixed history. More generally, if two finite source-generated history realizations have equal depth- $(r + 1)$ word Grams and are both flat at depth r , then there is a unique unital $$ -isomorphism between them carrying every source word to its namesake.*

Proof. The word spans form an increasing sequence of subspaces of the finite-dimensional algebra, so their dimensions stabilize. At the first rank equality the depth- r and depth- $(r + 1)$ spans coincide. Left multiplication by every primitive source preserves the common span; since it contains the unit, it contains every generated word and hence the whole algebra. The additional word layer records left multiplication by each primitive source on a spanning family, while reversing word order and taking formal adjoints recovers the involution. For two realizations with the same flat Gram panel, the map sending one represented word to the namesake word in the second is well defined and isometric because the Gram kernels have the same null space and inner products. Flatness makes it onto; the one-step word identities show that it preserves multiplication and involution. Uniqueness follows because the saturated words span. $\square$

Proposition S6.7. *Under the finite operational assumptions above, the complete intervention statistics determine, up to their natural isomorphisms,*

- (i) *the minimal predictive state space $Z_X^{\min}$, its branch probabilities, and its partial update maps;*
- (ii) *the support-minimal process representation and the universal future-separated history algebra $\mathcal{A}_X^{\text{hist}}$;*
- (iii) *the normalized regular trace and the positive word-Gram hierarchy;*
- (iv) *a finite flat depth r_X at which the history algebra, multiplication and involution are reconstructible from the word panel.*

Thus the finite history algebra and its source representation are reconstructed from the operational table without an independently specified hidden-state or Hilbert-space realization.

Proof. The predictive quotient and its universal property were proved above. Proposition S6.1 reconstructs the positive causal process; Proposition S6.1 fixes its support-minimal sequential class; Theorems S6.3 and S6.5 give the canonical future-separated history algebra; and Theorem S6.6 proves finite saturation and reconstruction of its algebraic operations. The normalized regular trace is intrinsic to the resulting finite-dimensional C^* -algebra and therefore so are the word Grams. These constructions are invariant under the natural unitary and represented $*$ -isomorphisms, proving the claim. $\square$

The proposition isolates the finite operational hypotheses used for the predictive application. The spectral reconstruction in Section S1 depends only on the resulting finite C^* -algebra, trace, and differential datum.

S6.2 Realization and recognition with fixed boundary contexts

The physical memory-dimension minimality used in this subsection is distinct from the coherent prefix-minimality used for the canonical process realization above. The two notions serve different purposes and are not identified.

The preceding reconstruction starts from a given finite intervention table. We now prove a converse realization statement for the represented algebra. The category is deliberately boundary-relative: preparations and terminal readouts are external contexts and are not adjoined as internal generators. This convention is substantive, as shown below. Measure-and-prepare maps belong to the standard theory of entanglement-breaking channels [49].

Theorem S6.8 (Finite operational realization with multiplicities). *Let $\pi : \mathcal{A} \rightarrow \mathcal{B}(K)$ be a faithful unital representation of a finite-dimensional C^* -algebra. Then there are a finite physical memory H , a finite family of normalized completely positive instruments, boundary preparations, and terminal readouts such that the reducing source representation of the internal history algebra is unitarily equivalent to $(\mathcal{A}, K, \pi)$. In particular all representation multiplicities are preserved.*

Proof. After a unitary change of coordinates write

$$K = \bigoplus_{\lambda=1}^s \mathbb{C}^{n_\lambda} \otimes \mathbb{C}^{m_\lambda}, \quad \pi(\mathcal{A}) = \bigoplus_{\lambda=1}^s M_{n_\lambda}(\mathbb{C}) \otimes I_{m_\lambda}.$$

Take H with orthonormal basis $e_{\lambda,a,r}$ and put $p_{\lambda,a,r} = |e_{\lambda,a,r}\rangle\langle e_{\lambda,a,r}|$. For $1 \leq a, b \leq n_\lambda$ define

$$L_{ab,r}^\lambda = |e_{\lambda,a,r}\rangle\langle e_{\lambda,b,r}|, \quad \Phi_{ab}^\lambda(X) = \sum_{r=1}^{m_\lambda} L_{ab,r}^\lambda X (L_{ab,r}^\lambda)^*.$$

Each map is CP and trace nonincreasing, and direct multiplication of the rank-one Kraus operators gives

$$\Phi_{ab}^\lambda \Phi_{cd}^\mu = \delta_{\lambda\mu} \delta_{bc} \Phi_{ad}^\lambda, \quad (\Phi_{ab}^\lambda)^*_{\text{HS}} = \Phi_{ba}^\lambda.$$

Let $P(X) = \sum_{\lambda,a,r} p_{\lambda,a,r} X p_{\lambda,a,r}$ be the dephasing channel. For each success branch use the complementary failure branch

$$\Psi_b^\lambda = P - \Phi_{bb}^\lambda.$$

The pair $(\Phi_{ab}^\lambda, \Psi_b^\lambda)$ is a normalized two-outcome instrument, and adjoining the failure branches adds no generator beyond the matrix-unit algebra and the identity. The diagonal Hilbert–Schmidt subspace

$$\mathcal{V} = \text{span}\{p_{\lambda,a,r}\}$$

is reducing for every branch and every Hilbert–Schmidt adjoint. Under the unitary $e_{\lambda,a,r} \mapsto p_{\lambda,a,r}$, the restricted matrix units act as $E_{ab}^{(\lambda)} \otimes I_{m_\lambda}$. Boundary preparations $p_{\lambda,a,r}$ and the corresponding terminal effects separate all source matrix coefficients, so the future-null ideal vanishes on this source representation. Hence the future-separated internal history representation is exactly $\pi(\mathcal{A})$ with the stated multiplicities. $\square$

Lemma S6.9 (Rank-one forcing from a deterministic transition panel). *Let $\rho_1, \dots, \rho_N$ be density matrices on a finite Hilbert space H that are perfectly distinguished by effects $E_1, \dots, E_N$, so that $\text{Tr}(E_i \rho_j) = \delta_{ij}$. Then*

$$\sum_{j=1}^N \text{rank } \rho_j \leq \dim H. \quad (150)$$

In particular, if $\dim H = N$, the ρ_j are rank-one states with mutually orthogonal supports. Let Φ be completely positive and suppose that, on these rank-one states, a retained one-step panel prescribes

$$\Phi(\rho_j) = w_j \rho_{\sigma(j)}, \quad w_j > 0, \quad (151)$$

for j in a domain J , while $\text{Tr } \Phi(\rho_j) = 0$ for $j \notin J$. If $\sigma : J \rightarrow \{1, \dots, N\}$ is injective, then the diagonal Hilbert–Schmidt span $\mathcal{V} = \text{span}\{\rho_j\}$ is invariant under both Φ and Φ_{HS}^ .*

Proof. Perfect discrimination implies that the supports of the ρ_j are mutually orthogonal: $\text{Tr}(E_i \rho_i) = 1$ and $0 \leq E_i \leq I$ force $E_i = I$ on $\text{supp } \rho_i$, whereas $\text{Tr}(E_i \rho_j) = 0$ forces $E_i = 0$ on $\text{supp } \rho_j$ for $j \neq i$. The rank bound follows. At equality $\dim H = N$, every support is one dimensional; write $\rho_j = |e_j\rangle\langle e_j|$.

Choose Kraus operators K_ν for Φ . If $j \notin J$, then $0 = \text{Tr } \Phi(\rho_j) = \sum_\nu \|K_\nu e_j\|^2$, hence every $K_\nu e_j$ vanishes. If $j \in J$, the positive operator $\sum_\nu |K_\nu e_j\rangle\langle K_\nu e_j|$ has one-dimensional support $\mathbb{C}e_{\sigma(j)}$, so every $K_\nu e_j$ lies in that line. Thus Φ preserves $\mathcal{V}$. For the Hilbert–Schmidt adjoint,

$$\langle e_k, \Phi^*(\rho_i) e_j \rangle = \sum_\nu \overline{\langle e_i, K_\nu e_k \rangle} \langle e_i, K_\nu e_j \rangle.$$

A nonzero term requires simultaneously $\sigma(k) = i = \sigma(j)$; injectivity of σ therefore forces $j = k$. Hence $\Phi^*(\rho_i)$ is diagonal in the same basis and $\mathcal{V}$ is reducing. $\square$

Proposition S6.10 (Finite recognition from the one-step panel). *For the realization in Theorem S6.8, retain the finite preparation–branch–readout panel consisting of the perfect boundary discrimination data and the one-step success and complementary failure transitions, including the zero probabilities outside each branch domain. Any realization of this panel whose total physical memory dimension is minimal has a future-separated source representation unitarily equivalent to $(\mathcal{A}, K, \pi)$.*

Proof. There are $N = \dim K$ retained source preparations. By Lemma S6.9, any realization needs physical memory dimension at least N . In a dimension-minimal realization equality holds, so the preparations are rank-one states $p_i = |e_i\rangle\langle e_i|$ on an orthonormal basis, uniquely up to a unitary and phase choices that disappear at the level of the projections.

Fix the retained success branch indexed by (λ, a, b) . Its domain contains all (λ, b, r) , $1 \leq r \leq m_\lambda$, with successor (λ, a, r) ; the panel records zero probabilities outside this domain. The deterministic outputs and the zero probabilities put it in the situation of Lemma S6.9. For a fixed branch the successor rule is a partial injection, so the diagonal source space

$$\mathcal{V} = \text{span}\{p_{\lambda,a,r}\}$$

is reducing for that branch and its Hilbert–Schmidt adjoint. On $\mathcal{V}$ the retained transition probabilities determine the partial matrix unit $E_{ab}^{(\lambda)} \otimes I_{m_\lambda}$. The complementary failure branches have a partial-identity successor rule on the remaining basis states, so the same lemma makes $\mathcal{V}$ reducing for them as well. Their restrictions already belong to the generated block algebra. Composition of the retained one-step branches therefore recovers

$$\bigoplus_{\lambda} M_{n_\lambda}(\mathbb{C}) \otimes I_{m_\lambda}$$

with exactly the prescribed multiplicities.

Finally, the retained terminal readouts perfectly distinguish the basis projections and, after composition with the matrix-unit branches, separate every matrix coefficient of this represented algebra. An element annihilated by all two-sided retained readout contexts is therefore zero on the source representation, so the contextual future-null ideal vanishes there. The resulting future-separated source representation is consequently unitarily equivalent to $(\mathcal{A}, K, \pi)$. $\square$

Proposition S6.11. *Let $N = \dim K \geq 2$ and let $p_1, \dots, p_N$ be the distinguished source preparations. If the reset maps*

$$\mathcal{R}_u(X) = \text{Tr}(X)p_u$$

are adjoined to the internal algebra, then on the source span their products with Hilbert–Schmidt adjoints generate $M_N(\mathbb{C})$. Thus a proper block algebra $\bigoplus_{\lambda} M_{n_\lambda}(\mathbb{C}) \otimes I_{m_\lambda}$ can be destroyed by promoting boundary preparations to internal generators.

Proof. On the diagonal source basis the reset $\mathcal{R}_u$ is the rank-one map $|e_u\rangle\langle \mathbf{1}|$. Hence $\mathcal{R}_u\mathcal{R}_v^* = N|e_u\rangle\langle e_v|$. These are all matrix units of $M_N(\mathbb{C})$. $\square$

Theorem S6.12. *Let $\rho_1, \dots, \rho_r$ be linearly independent reproducible conditional memory states and put $Lx = \sum_j x_j \rho_j$, $G = L^*L$. Let R be the retained boundary response map and set*

$$H_\alpha = L^*\Phi_\alpha L, \quad K_\alpha = (\Phi_\alpha L)^*(\Phi_\alpha L), \quad B = RL, \quad C_\alpha = R\Phi_\alpha L.$$

Assume B has full row rank. If, for every branch α ,

$$K_\alpha - H_\alpha^* G^{-1} H_\alpha = 0, \quad C_\alpha (I - B^\dagger B) = 0,$$

then $\mathcal{V} = \text{Ran } L$ is forward invariant and, with $A_\alpha = C_\alpha B^\dagger$,

$$R\Phi_\alpha X = A_\alpha RX \quad (X \in \mathcal{V}), \quad R\Phi_w L = A_w B$$

for every finite word w . Hence finitely many successor-state Gram and response measurements certify the full forward-response quotient. These enriched overlap measurements are additional resources and are not inferred from the original boundary probability table.

Proof. The orthogonal projection onto $\mathcal{V}$ is $P_\mathcal{V} = LG^{-1}L^*$, so

$$K_\alpha - H_\alpha^* G^{-1} H_\alpha = ((I - P_\mathcal{V})\Phi_\alpha L)^*((I - P_\mathcal{V})\Phi_\alpha L).$$

The first condition is therefore equivalent to forward invariance and gives $\Phi_\alpha L = LG^{-1}H_\alpha$. The second says precisely that C_α annihilates $\ker B$, hence factors uniquely through the response coordinates: $C_\alpha = A_\alpha B$. Combining these identities and iterating proves the word formula. The overlaps entering G, H_α, K_α can, when the whole memory is experimentally accessible, be measured on two copies using the swap identity $\text{Tr}(F(X \otimes Y)) = \text{Tr}(XY)$ [50]. $\square$

Corollary S6.13. *For every finite marked spectral triple, its represented algebra component admits the operational realization of Theorem S6.8. After independently supplying its commutator derivation, declared marks, and the sharp real-linear commutant coordinates of Theorem S1.8, the full compatible Dirac operator is reconstructed. No claim is made that the ordinary operational probabilities alone determine the differential datum or its scale.*

S7 Positive Stieltjes memory and static-preserving cutoffs

This section studies the inverse data lost by a static reduction. Eliminating auxiliary degrees of freedom produces a resolvent self-energy, whereas its value at zero retains only a static Schur correction. The distinction between these two data is described using classical Naimark dilation and matrix-valued Herglotz–Stieltjes theory [48, 51, 52]. We record the minimal realization relative to the effective space, moment determinacy and indeterminacy, static-preserving cutoffs, and high-energy contact limits. These results distinguish the full dynamical memory from the static correction retained by the effective operator.

S7.1 Positive measures, self-energies, and inverse moments

Definition S7.1 (Positive operator-valued tail measure). Let H be finite dimensional. A bounded positive operator-valued tail measure is a weakly countably additive map

$$\mu : \mathfrak{B}((0, \infty)) \rightarrow \mathcal{B}(H), \quad \mu(E) \succeq 0, \quad \|\mu((0, \infty))\| < \infty.$$

Its Stieltjes self-energy is

$$\Sigma_\mu(z) = \int_{(0, \infty)} \frac{1}{\lambda - z} d\mu(\lambda), \quad z \in \mathbb{C} \setminus [0, \infty). \quad (152)$$

By the standard Naimark dilation and matrix-valued Herglotz–Stieltjes realization theory [48, 51, 52], every such measure has a source-minimal realization

$$\Sigma_\mu(z) = B_\mu(C_\mu - z)^{-1}B_\mu^*, \quad C_\mu \geq 0, \quad (153)$$

unique up to a unitary intertwining C_μ and B_μ^* . We use this classical realization theorem as an input rather than reproving it. The inverse-fibre questions below concern which features of μ survive passage to the static self-energy and how the remaining dynamical coordinate behaves under coarse graining.

Theorem S7.2 (Static self-energy, inverse moments, and positive Euclidean memory). *For $\text{Im } z > 0$,*

$$\text{Im } \Sigma_\mu(z) = (\text{Im } z) \int \frac{1}{|\lambda - z|^2} d\mu(\lambda) \succeq 0. \quad (154)$$

The finite static self-energy

$$Q_0 = \lim_{s \downarrow 0} \Sigma_\mu(-s)$$

exists exactly when the inverse first moment is bounded, and then

$$Q_0 = \int \lambda^{-1} d\mu(\lambda). \quad (155)$$

If the corresponding inverse moment is finite, the r -th boundary jet is

$$\Sigma_{\mu,-}^{(r)}(0) = r! \int \lambda^{-(r+1)} d\mu(\lambda). \quad (156)$$

The Euclidean memory kernel

$$K_\mu(t) = \int e^{-t\lambda} d\mu(\lambda) = B_\mu e^{-tC_\mu} B_\mu^* \quad (157)$$

is positive and

$$\Sigma_\mu(-s) = \int_0^\infty e^{-st} K_\mu(t) dt.$$

Proof. The scalar identity $\text{Im}(\lambda - z)^{-1} = (\text{Im } z)|\lambda - z|^{-2}$ gives the sign after positive operator integration. Monotone convergence along $z = -s$ gives (155). Differentiating $(\lambda - z)^{-1}$ gives the higher inverse moments. The spectral calculus gives the second equality in (157); Tonelli's theorem applied to scalarizations gives the Laplace relation. $\square$

Remark S7.3 (Positive memory is not full OS positivity). Positivity of $K_\mu(t)$ is a two-point positive memory statement. It does not by itself supply a Euclidean field law, a positive-time product algebra, or Osterwalder–Schrader positivity for reflected words.

The Stieltjes transform is the full dynamical coordinate. Static data or finitely many moments need not determine it. The next subsection gives sufficient determinacy hypotheses and an example in which even all positive and inverse integer moments fail to determine the self-energy.

S7.2 Moment determinacy and indeterminacy

Theorem S7.4. *If $\text{supp } \mu$ is compact, the complete nonnegative moment sequence $M_n = \int \lambda^n d\mu(\lambda)$ determines μ uniquely. The same conclusion holds on unbounded support if $\int e^{a\lambda} d\text{Tr } \mu(\lambda) < \infty$ for some $a > 0$.*

Proof. On compact support, equality of all moments gives equality against all polynomials; Stone–Weierstrass gives equality against every continuous function, and polarization gives equality of the operator measures.

Under an exponential moment, each scalarized Laplace transform $\int e^{-t\lambda} d\langle \xi, \mu \eta \rangle$ extends holomorphically to a half-plane containing the origin. Its Taylor coefficients are the moments. Equality of all moments therefore gives equality of the Laplace transforms near zero and hence throughout the connected half-plane. Uniqueness of the Laplace transform and polarization recover μ . $\square$

Remark S7.5 (Scope of the memory classification). The results in this subsection concern positive operator-valued Stieltjes memory. Indefinite generalized-Nevanlinna or Pontryagin-space realizations require additional realization theory and are outside the scope of this section; the finite rational extension is developed separately in Section S11.

Proposition S7.6. *For $|\varepsilon| \leq 1$ define*

$$d\mu_\varepsilon(x) = [1 + \varepsilon \sin(2\pi \log x)] \frac{e^{-(\log x)^2/2}}{x\sqrt{2\pi}} dx. \quad (158)$$

Then for every integer $n \in \mathbb{Z}$,

$$\int_0^\infty x^n d\mu_\varepsilon(x) = e^{n^2/2} \quad (159)$$

independently of ε . Nevertheless $\varepsilon \neq \varepsilon'$ implies that the Stieltjes transforms differ for some, hence an open set of, $z \in \mathbb{C} \setminus [0, \infty)$. Consequently all positive moments, all inverse integer moments, and every finite Hankel panel can agree while the dynamic self-energies are inequivalent.

Proof. Set $y = \log x$. The perturbation of the n -th moment is

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{ny - y^2/2} \sin(2\pi y) dy = \operatorname{Im} \exp\left(\frac{(n + 2\pi i)^2}{2}\right) = e^{(n^2 - 4\pi^2)/2} \sin(2\pi n) = 0$$

for every $n \in \mathbb{Z}$. The unperturbed lognormal moment is $e^{n^2/2}$. Distinct ε give distinct measures.

If a finite signed measure on $[0, \infty)$ had identically zero Stieltjes transform on the negative real axis, the identity

$$\int \frac{1}{x + s} d\sigma(x) = \int_0^\infty e^{-st} \left(\int e^{-tx} d\sigma(x) \right) dt$$

and two applications of uniqueness of the Laplace transform would force $\sigma = 0$. Hence distinct measures have distinct Stieltjes transforms. $\square$

S7.3 Memory cutoff and composition

A cutoff removes high-energy memory while incorporating its static contribution into the effective operator. The following identities separate exact preservation at zero energy from the error at nonzero spectral parameter.

For an effective operator A and positive memory measure μ , write

$$F_{A,\mu}(z) = A - z - \Sigma_\mu(z), \quad Q_{>\Lambda} = \int_{(\Lambda, \infty)} \lambda^{-1} d\mu(\lambda), \quad (160)$$

and define

$$\operatorname{RG}_\Lambda(A, \mu) = (A - Q_{>\Lambda}, \mu|_{(0, \Lambda]}). \quad (161)$$

Theorem S7.7 (Static-preserving memory cutoffs). *Assume the static inverse moment $Q_0 = \int \lambda^{-1} d\mu(\lambda)$ is finite, and interpret $F_{A,\mu}(0)$ as the limit from the negative real axis. If $(A_\Lambda, \mu_\Lambda) = \text{RG}_\Lambda(A, \mu)$, then*

$$F_{A_\Lambda, \mu_\Lambda}(0) = F_{A, \mu}(0). \quad (162)$$

For $z \in \mathbb{C} \setminus [0, \infty)$ with $|z| \leq r < \Lambda$,

$$\|F_{A, \mu}(z) - F_{A_\Lambda, \mu_\Lambda}(z)\| \leq \frac{r}{\Lambda - r} \|Q_{>\Lambda}\|. \quad (163)$$

Moreover, if $0 < \Lambda_2 \leq \Lambda_1$,

$$\text{RG}_{\Lambda_2} \circ \text{RG}_{\Lambda_1} = \text{RG}_{\Lambda_2}. \quad (164)$$

Proof. Split the Stieltjes integral at Λ . At $z = 0$ the removed high-energy part is exactly $Q_{>\Lambda}$ and is canceled by the effective local correction. Away from zero,

$$\frac{1}{\lambda - z} - \frac{1}{\lambda} = \frac{z}{\lambda(\lambda - z)},$$

whose norm is at most $r[(\Lambda - r)\lambda]^{-1}$ for $\lambda > \Lambda$. Integrating gives (163). Two successive cutoffs simply split the same measure into disjoint shells; their inverse-moment local corrections add, proving the semigroup law. $\square$

S7.4 High-energy escape and contact limits

The same static correction can also survive when the memory measure escapes to infinite energy. The limiting object then contains an instantaneous term, as the following two results show.

Theorem S7.8 (High-energy memory escape and instantaneous contact terms). *Let μ_n be positive tail measures supported in $[m, \infty)$ for one $m > 0$ and define*

$$d\nu_n(\lambda) = \lambda^{-1} d\mu_n(\lambda), \quad Q_n = \nu_n([m, \infty)) = \Sigma_n(0).$$

If $\sup_n \text{Tr } Q_n < \infty$, every sequence has a subsequence and a positive operator-valued measure on the one-point compactification of the form

$$\nu_\infty = \nu_{\text{fin}} + Q_{\text{esc}} \delta_\infty, \quad Q_{\text{esc}} \succeq 0,$$

such that, locally uniformly off $[m, \infty)$,

$$\Sigma_n(z) \longrightarrow Q_{\text{esc}} + \int_{[m, \infty)} \frac{\lambda}{\lambda - z} d\nu_{\text{fin}}(\lambda). \quad (165)$$

The Euclidean memory measures satisfy

$$K_n(t) dt \longrightarrow Q_{\text{esc}} \delta_0 + K_{\text{fin}}(t) dt. \quad (166)$$

Proof. The matrix entries of ν_n are uniformly bounded finite measures on the compact space $[m, \infty]$. Weak-* compactness and polarization give a positive operator-valued limit, which decomposes into finite and point-at-infinity parts. For $z \notin [m, \infty)$, $g_z(\lambda) = \lambda/(\lambda - z)$ extends continuously to infinity with $g_z(\infty) = 1$, proving (165). For $\varphi \in C_c([0, \infty))$,

$$h_\varphi(\lambda) = \int_0^\infty \varphi(t) \lambda e^{-\lambda t} dt$$

extends continuously with $h_\varphi(\infty) = \varphi(0)$; integration against ν_n gives (166). $\square$

Proposition S7.9. *For $Q \succeq 0$ and $\lambda_n \rightarrow \infty$, let*

$$\mu_n = \lambda_n Q \delta_{\lambda_n}.$$

Then

$$\Sigma_n(0) = Q, \quad \Sigma_n(z) = \frac{\lambda_n}{\lambda_n - z} Q \longrightarrow Q, \quad (167)$$

whereas $\lambda_n e^{-\lambda_n t} Q \rightarrow 0$ for every $t > 0$ and converges distributionally to $Q\delta_0$. The same static local correction is therefore compatible with an intrinsic instantaneous term or with memory escaping to infinite energy.

Proof. The resolvent identities are immediate, and the exponential kernels form the standard positive approximate identity at the origin. $\square$

S8 Marked monodromy and componentwise coarse graining

Locally isomorphic spectral triples need not admit a compatible global trivialization. This section records the standard graph-holonomy description of that ambiguity and the condition for descent under graph contraction. It also distinguishes the holonomy of a full marked triple from the information detected by its one-dimensional characters.

S8.1 Flat bundles of marked spectral triples

For a marked spectral triple S in the sense of Definition S1.7, let

$$G_S = \text{Aut}(S) \quad (168)$$

be the compact group of unitary automorphisms of the marked triple preserving all declared marks and the chosen algebra automorphism convention. In this subsection the auxiliary marks may include orientation, modular, calibration, frame, or determinant-line data.

Proposition S8.1. *Let Γ be connected, let each vertex triple be marked-isomorphic to S , and let each oriented edge carry a marked unitary isomorphism U_e with $U_{\bar{e}} = U_e^{-1}$. Then:*

(B1) *after choosing a base vertex and spanning tree, chord holonomies define a homomorphism*

$$\rho_U : \pi_1(\Gamma) \longrightarrow G_S; \quad (169)$$

(B2) *changing vertex frames conjugates ρ_U by one element of G_S ;*

(B3) *a global marked trivialization compatible with all edge identifications exists if and only if ρ_U is trivial;*

(B4) *in general the isomorphism class of the flat marked spectral bundle is*

$$[\rho_U] \in \text{Hom}(\pi_1(\Gamma), G_S)/G_S. \quad (170)$$

Since $\pi_1(\Gamma)$ is free of rank $b_1(\Gamma)$, the bundle is determined by $b_1(\Gamma)$ elements of G_S modulo simultaneous conjugation.

Proof. Use the edge maps along a spanning tree to identify every vertex triple with the base triple. Each chord gives the ordered product around its fundamental cycle, an element of G_S . These cycle values extend uniquely to a homomorphism from the free fundamental group. Changing the base frame conjugates every cycle value by the same automorphism. Trivial cycle values make the tree identification path independent and hence give one global marked frame. Conversely, a global frame makes every cycle product telescope. This is precisely the clutching classification (170) on a graph. $\square$

Corollary S8.2. *If the local triples are marked-isomorphic and the edge identifications satisfy the declared compatibility relations, nontrivial $[\rho_U]$ obstructs a global marked trivialization compatible with those identifications.*

Theorem S8.3. *Let $\chi_1, \dots, \chi_m : G_S \rightarrow U(1)$ be determinant, charge, or other complex line characters and let $\epsilon_1, \dots, \epsilon_r : G_S \rightarrow \{\pm 1\}$ be real Pfaffian-sign characters. Trivial frame holonomy implies triviality of all induced line holonomies. The converse holds for all such graph bundles exactly when*

$$\bigcap_j \text{Ker } \chi_j \cap \bigcap_k \text{Ker } \epsilon_k = \{1\}. \quad (171)$$

Proof. Every character sends the identity to the identity. If (171) holds and all character holonomies are trivial, every cycle value of ρ_U lies in the displayed intersection and is therefore the identity. Conversely, if the intersection contains $g \neq 1$, assign g as monodromy on a one-cycle graph. Every listed character is trivial although the marked bundle is not. $\square$

Proposition S8.4. *Let a marked spectral triple have scalar algebra, zero Dirac operator, and a two-dimensional multiplicity space so that its marked automorphism group contains $SU(2)$. On a one-cycle graph choose monodromy*

$$g = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}, \quad \theta \notin 2\pi\mathbb{Z}. \quad (172)$$

Then the determinant-line holonomy is one although the marked Hilbert bundle has nontrivial monodromy.

Proof. Every $SU(2)$ element has determinant one, while the displayed matrix is not the identity. Apply Proposition S8.1 and Theorem S8.3. $\square$

S8.2 Descent under graph contraction

Let $\rho : \pi_1(\Gamma, v_0) \rightarrow G_S$ be marked monodromy and let a graph contraction $q : \Gamma \rightarrow \bar{\Gamma}$ induce a surjection q_* on fundamental groups.

Theorem S8.5 (Exact marked-monodromy descent). *There is a unique $\bar{\rho} : \pi_1(\bar{\Gamma}) \rightarrow G_S$ satisfying*

$$\rho = \bar{\rho} \circ q_* \quad (173)$$

if and only if

$$\text{Ker } q_* \subseteq \text{Ker } \rho. \quad (174)$$

The conjugacy class $[\bar{\rho}]$ is gauge independent and functorial under further contractions. A determinant or Pfaffian character may satisfy its own descended kernel condition even when (174) fails, so line descent does not imply descent of the full marked spectral triple.

Proof. If (173) holds, every element of $\text{Ker } q_*$ lies in $\text{Ker } \rho$. Conversely, if (174) holds, define $\bar{\rho}(q_*\gamma) = \rho(\gamma)$. The kernel condition makes the definition independent of the chosen lift. Surjectivity of q_* gives uniqueness and the quotient-group universal property gives functoriality. The character statement follows by applying the same argument to a nonfaithful one-dimensional representation of G_S . $\square$

S8.3 Componentwise coarse graining

Current aggregation, memory cutoff, and monodromy descent act on different kinds of data. The following corollary combines their respective composition laws while retaining the separate admissibility hypotheses for currents, positive memory and marked holonomy.

Corollary S8.6 (Componentwise coarse graining of the combined inverse data). *Collect*

$$\mathfrak{U}(G, c; A, \mu; \rho) = (\mathcal{J}(G, c), (A, \mu), [\rho]). \quad (175)$$

For a connected-fibre graph quotient, a memory threshold Λ , a finite static inverse moment as in Theorem S7.7, and a compatible isometric effective-space compression V , define

$$\mathfrak{U} \longmapsto (Q_E \mathcal{J}(G, c), (V^*(A - Q_{>\Lambda})V, V^*\mu|_{(0, \Lambda]}V), [\bar{\rho}]). \quad (176)$$

When the monodromy kernel condition holds, the three components are canonical. For composable graph quotients and nested memory thresholds they compose in the corresponding componentwise sense. For effective-space compression we require nested isometries: if $V_1 : H_1 \rightarrow H_0$ and $V_2 : H_2 \rightarrow H_1$, the composite compression is defined by $V_1 V_2 : H_2 \rightarrow H_0$. Along these maps entropy production cannot increase; the zero-energy Schur complement is preserved before effective-space compression; minimal positive-memory dilation rank cannot increase; and marked monodromy descends exactly when collapsed cycles lie in its kernel.

Proof. The current and entropy statements are Theorem S3.9; the memory statement and cutoff composition are Theorem S7.7; and the monodromy part is Theorem S8.5. For nested effective-space isometries,

$$V_2^* V_1^*(\cdot) V_1 V_2 = (V_1 V_2)^*(\cdot) (V_1 V_2),$$

and compression of the operator-valued measure obeys the same identity. Thus the compressed memory coordinate also composes under the stated compatibility. A compression of a positive minimal memory-dilation space cannot increase its source rank. $\square$

Part II. Further mathematical extensions

The following sections extend, rather than replace, the preceding core. The spin construction retains its local analytic estimates, the unconditional flat periodic result, and the conditional global stable-atlas theorem. An explicit positive dilation proof complements the shorter realization input used in Part I. Rational indefinite memory and joint/completed realization spaces are treated in their own declared categories.

S9 Analytic details for compact-spin spectralization

The periodic A_3 construction in Section S2 supplies the explicit metric model. This section records the analytic estimates used in the compact-spin convergence criterion. The chartwise construction is verified below; the global theorem additionally assumes the atlas stability conditions of Definition S9.7.

S9.1 A positive Renewal stencil for an arbitrary local inverse metric

Let $d \geq 2$ and put

$$\mathcal{R}_d^+ = \{e_i : 1 \leq i \leq d\} \cup \{e_i + e_j, e_i - e_j : 1 \leq i < j \leq d\}. \quad (177)$$

There are exactly d^2 unoriented directions.

Lemma S9.1 (Positive d^2 -direction decomposition). *Let $A = (A_{ij}) \in \text{Sym}_d(\mathbb{R})$ and suppose that for some $\eta > 0$,*

$$|A_{ij}| < \eta \quad (i \neq j), \quad A_{ii} > (d-1)\eta. \quad (178)$$

Define

$$w_i = A_{ii} - (d-1)\eta, \quad w_{ij}^\pm = \frac{1}{2}(\eta \pm A_{ij}). \quad (179)$$

Then every coefficient is strictly positive and

$$A = \sum_i w_i e_i e_i^\top + \sum_{i < j} \left[w_{ij}^+ (e_i + e_j)(e_i + e_j)^\top + w_{ij}^- (e_i - e_j)(e_i - e_j)^\top \right]. \quad (180)$$

Proof. Positivity follows immediately from (178). In the right-hand side of (180), the (i, j) off-diagonal entry is $w_{ij}^+ - w_{ij}^- = A_{ij}$. Every pair containing i contributes $w_{ij}^+ + w_{ij}^- = \eta$ to the i th diagonal, so the diagonal coefficient is $w_i + (d-1)\eta = A_{ii}$. $\square$

A linear coordinate normalization makes g^{-1} equal to I_d at the centre of a chart. By continuity, after shrinking the chart, the inverse metric remains inside one window (178). Compactness gives a finite good cover by such charts.

Let $\rho = \sqrt{\det g}$ and write the decomposition coefficients of g^{-1} as $w_r(x)$, $r \in \mathcal{R}_d^+$. Put $a_r(x) = \rho(x)w_r(x)$. On $Q_h = h\mathbb{Z}^d/\mathbb{Z}^d$ take

$$m_h(x) = h^d \rho(x), \quad c_h(x, r) = \frac{h^{d-2}}{4} (a_r(x) + a_r(x + hr)). \quad (181)$$

The two directed rates are the conductance divided by the corresponding endpoint mass.

Theorem S9.2 (Reversible coframe Renewal process). *The packet (181) is reversible with stationary mass m_h . Its predictable coordinate bracket is*

$$\mathcal{B}_h(x) = \frac{1}{4\rho(x)} \sum_{r \in \mathcal{R}_d^+} [a_r(x + hr) + 2a_r(x) + a_r(x - hr)] rr^\top, \quad (182)$$

and satisfies

$$\|\mathcal{B}_h - g^{-1}\|_{L^\infty} \leq Ch^2. \quad (183)$$

For every $f \in C^4$,

$$\mathcal{L}_h f = \frac{1}{2} \Delta_g f + O(h^2 \|f\|_{C^4}) \quad (184)$$

uniformly on the chart, and the Dirichlet forms converge to

$$-\langle f_h, \mathcal{L}_h f_h \rangle_{m_h} \longrightarrow \frac{1}{2} \int g^{ij} \partial_i f \partial_j \bar{f} \rho \, dx. \quad (185)$$

Proof. Symmetry of c_h gives detailed balance. The bracket follows by summing the two orientations of every direction through a vertex. Taylor expansion of $a_r(x \pm hr)$ cancels odd terms and, together with (180), gives (183). Expanding $f(x \pm hr)$ to fourth order, the first-order terms cancel and the second-order terms contract with the same tensor decomposition. The derivative of the physical density and of the conductances produces the divergence-form first-order part of Δ_g , yielding (184). Summation by parts gives the finite Dirichlet form; the coefficient convergence and Riemann sums give (185). $\square$

Proposition S9.3 (The metric bracket does not select the spin marking). *The positive bracket determines g^{-1} but not an oriented orthonormal coframe, a spin lift, or a global spin structure. A smooth rotation $R(x) \in \text{SO}(d)$ changes the coframe and spin connection without changing g^{-1} or the scalar Renewal generator, and a nontrivial $\mathbb{Z}_2$ spin holonomy is invisible to every local metric bracket.*

S9.2 Density-symmetric spin Dirac and finite Wilson packet

Let S_d be a complex spin module, let γ_a be Hermitian Clifford matrices, and let E_a^j be the inverse oriented orthonormal coframe. Set

$$c^j = E_a^j \gamma_a, \quad c^j c^k + c^k c^j = 2g^{jk} I, \quad (186)$$

and let Ω_j be the Levi-Civita spin connection. On $L^2(Q, S_d; \rho dx)$,

$$D_g = -i \sum_j c^j (\partial_j + \Omega_j). \quad (187)$$

Lemma S9.4 (Exact density-symmetric Dirac identity). *With $U_\rho \psi = \rho^{1/2} \psi$ and $\nabla_j = \partial_j + \Omega_j$,*

$$U_\rho D_g U_\rho^{-1} = -\frac{i}{2} \sum_{j=1}^d (M_{c^j} \nabla_j + \nabla_j M_{c^j}). \quad (188)$$

Proof. Metric compatibility of Clifford multiplication gives

$$\partial_j c^k + [\Omega_j, c^k] + \Gamma_{j\ell}^k c^\ell = 0.$$

Setting $k = j$ and using $\Gamma_{j\ell}^j = \partial_\ell \log \rho$ gives

$$\partial_j c^j + [\Omega_j, c^j] + c^j \partial_j \log \rho = 0.$$

Conjugating (187) by $\rho^{1/2}$ and applying this contracted identity yields (188). $\square$

Introduce one fixed two-dimensional normal factor and put

$$\widehat{S}_d = S_d \otimes \mathbb{C}^2, \quad \widehat{c}^j = c^j \otimes \sigma_1, \quad \Gamma_\perp = I \otimes \sigma_2, \quad \widehat{\gamma} = I \otimes \sigma_3. \quad (189)$$

Let $T_{j,h}^\Omega$ be exact inverse spin parallel transport followed by a lattice shift and define

$$P_{j,h}^\Omega = \frac{T_{j,h}^\Omega - (T_{j,h}^\Omega)^*}{2ih}, \quad W_h^\Omega = \frac{1}{2h} \sum_{j=1}^d (2I - T_{j,h}^\Omega - (T_{j,h}^\Omega)^*). \quad (190)$$

The local doubled Wilson operator is the density-conjugated version of

$$\widetilde{D}_{g,h}^W = \frac{1}{2} \sum_j (M_{\widehat{c}_h^j} P_{j,h}^\Omega + P_{j,h}^\Omega M_{\widehat{c}_h^j}) + \varpi \Gamma_\perp W_h^\Omega. \quad (191)$$

For the local estimates, extend the coefficients smoothly to a fixed periodic coordinate box, preserving uniform ellipticity, and use the density-conjugated ℓ^2 norm with cell weight h^d . Such an extension is an auxiliary local calculation, not a boundary condition on a chart of M . Let $S_{j,h}$ be the ordinary coordinate shift. In a smooth spin frame, $T_{j,h}^\Omega - S_{j,h} = O(h)$ in operator norm. Define

$$\|u\|_{1,h}^2 = \|u\|_h^2 + \sum_j \|h^{-1}(S_{j,h} - I)u\|_h^2. \quad (192)$$

The analogous covariant norm is uniformly equivalent to this one, since the connection coefficients are bounded. All constants below may depend on the fixed atlas and finitely many coefficient bounds, but not on h .

Theorem S9.5 (Frozen symbol, unique species, and uniform graph estimate). *For fixed $\varpi > 0$, the frozen principal symbol of (191) is*

$$q_{h,x_0}(\theta) = \frac{1}{h} \left[\sum_j \widehat{c}^j(x_0) \sin \theta_j + \varpi \Gamma_\perp \sum_j (1 - \cos \theta_j) \right]. \quad (193)$$

It satisfies

$$q_{h,x_0}(\theta)^2 = \frac{1}{h^2} \left[g^{jk}(x_0) \sin \theta_j \sin \theta_k + \varpi^2 \left(\sum_j (1 - \cos \theta_j) \right)^2 \right] I. \quad (194)$$

Its only zero is $\theta = 0$ on the Brillouin torus, within the fixed doubled spin module. On the fixed smooth periodic extension, and for sections localized in the interior of the original chart, there is a cutoff-independent bound

$$\|u\|_{1,h} \leq C(\|\widetilde{D}_{g,h}^W u\|_h + \|u\|_h). \quad (195)$$

The symmetric placement of coefficients makes the finite operator self-adjoint; the fixed doubled Clifford convention supplies the grading. A declared real structure is retained when its compatibility is supplied as in Definition S9.7.

Proof. The Clifford relations eliminate the mixed terms in the square. Put $s(\theta) = \sum_j (1 - \cos \theta_j)$ and $\ell_h(\theta) = 2h^{-2}s(\theta)$. Uniform ellipticity and the elementary identity

$$\sum_j \sin^2 \theta_j + s(\theta)^2 = 2s(\theta) + 2 \sum_{i < j} (1 - \cos \theta_i)(1 - \cos \theta_j) \geq 2s(\theta)$$

give constants $0 < c \leq C < \infty$ such that

$$c\ell_h(\theta)I \preceq q_{h,x_0}(\theta)^2 \preceq C\ell_h(\theta)I. \quad (196)$$

For the upper bound use $s \leq 2d$ and $\sum_j \sin^2 \theta_j \leq 2s$. The lower bound controls all forward differences, including modes near the nonzero Brillouin corners. Fourier transformation proves the frozen graph estimate.

To pass to variable coefficients, choose a partition with supports of fixed small diameter r , independent of h . On one support the difference of the variable principal part and its frozen value is bounded by $C\omega_c(r)\|u\|_{1,h} + C_r\|u\|_h$, where ω_c is the modulus of continuity of the coframe coefficients. Derivatives of coefficients and the replacement of ordinary shifts by smooth spin links contribute only the second term. Choose r once so that the first term is absorbed by the frozen estimate. For every fixed smooth cutoff χ ,

$$\|[\tilde{D}_{g,h}^W, M_{\chi,h}]\| \leq C\|\nabla\chi\|_\infty. \quad (197)$$

Indeed, a shift commutator contains $\chi(x + he_j) - \chi(x) = O(h)$, which cancels the h^{-1} factor in both the Dirac and Wilson differences. Apply the local estimate to χu , sum the squared bounds over a fixed finite partition with $\sum \chi^2 = 1$, and use the same discrete product rule for the forward differences in (192). This proves (195). The fixed localization scale is important: no cutoff derivative diverging as $h \rightarrow 0$ is absorbed into a uniform constant. $\square$

Let $\mathcal{J}_h^0$ be isometric piecewise-constant reconstruction and take $S_h = (\mathcal{J}_h^0)^*$, the cell-average sampling map in the chosen density and spin frame. This avoids a dimension-dependent point-evaluation assumption on Sobolev functions. A piecewise-affine reconstruction I_h on a fixed shape-regular refinement of the coordinate cells satisfies

$$\|I_h u\|_{H^1} \leq C\|u\|_{1,h}, \quad \|I_h u - \mathcal{J}_h^0 u\|_{L^2} \leq Ch\|u\|_{1,h}. \quad (198)$$

These inequalities follow cell by cell from the affine interpolation formula and scaling to a unit cell. Thus it is $I_h u$, not the discontinuous piecewise-constant reconstruction itself, that has a uniform H^1 bound.

Lemma S9.6 (Smooth-core consistency and the Wilson term). *For every smooth doubled spinor supported in the coordinate interior,*

$$\left\| \mathcal{J}_h^0 \tilde{D}_{g,h}^W S_h \psi - U_\rho \hat{D}_g U_\rho^{-1} \psi \right\|_{L^2} \leq Ch\|\psi\|_{H^3}. \quad (199)$$

In particular, on this core,

$$\|W_h^\Omega S_h \psi\|_h \leq Ch\|\psi\|_{H^2}. \quad (200)$$

This is a core estimate, not an operator-norm smallness assertion on the full lattice space.

Proof. Taylor's formula with integral remainder, applied to spin-parallel translated cell averages, gives the covariant centered derivative $-i\nabla_j$ and the second-difference expansion $W_h^\Omega = -(h/2) \sum_j \nabla_j^2$ on a smooth core, with the corresponding Sobolev remainders. Cell averaging and reconstruction add $O(h)$ errors controlled by one further derivative. Symmetric multiplication by the smooth coefficients and Lemma S9.4 give (199); applying the second-difference estimate alone gives (200). $\square$

The two roles of the Wilson term must be distinguished. On smooth sections it vanishes as $h \rightarrow 0$, whereas at nonzero lattice corners its size is of order h^{-1} and removes the spurious low-energy species. It is therefore not a bounded perturbation tending to zero in operator norm. This distinction is also central in rigorous continuum-limit studies of lattice Dirac operators [53, 54, 55]. Moreover, for any bounded sequence u_h and fixed smooth ψ , self-adjointness gives

$$|\langle W_h^\Omega u_h, S_h \psi \rangle_h| = |\langle u_h, W_h^\Omega S_h \psi \rangle_h| \leq Ch \|u_h\|_h \|\psi\|_{H^2}.$$

Hence the regularizer leaves no additional distributional term in the limiting Dirac equation. On a closed manifold the intended self-adjoint realization is the closure of D_g on smooth spinors, with domain $H^1(M; S)$ [56]; no chart-boundary domain is introduced.

S9.3 Atlas compatibility and the global resolvent argument

Local consistency and a positive overlap Gram do not, by themselves, imply global norm-resolvent convergence. The following formulation states the required global approximation properties explicitly, so that the analytic conclusion is separate from a choice of mesh and overlap realization.

Definition S9.7 (Compatible stable spin discretization). Let $\hat{D} = D_{M, \mathfrak{s}} \otimes \sigma_1$ on $\mathcal{H} = L^2(M; S \otimes \mathbb{C}^2)$, where M is closed. A compatible stable spin discretization is a sequence of finite Hilbert spaces $\mathcal{H}_h$, self-adjoint operators D_h , and isometries $W_h : \mathcal{H}_h \rightarrow \mathcal{H}$ satisfying:

(A1) $W_h W_h^* \rightarrow I$ strongly. There are discrete first-order norms $\|\cdot\|_{1,h}$ and reconstructions $I_h : \mathcal{H}_h \rightarrow H^1(M; S \otimes \mathbb{C}^2)$ with

$$\|I_h u\|_{H^1} \leq C \|u\|_{1,h}, \quad \|I_h u - W_h u\| \leq \varepsilon_h \|u\|_{1,h}, \quad \varepsilon_h \rightarrow 0. \quad (201)$$

(A2) The assembled, rather than merely chartwise, operators satisfy

$$\|u\|_{1,h} \leq C (\|D_h u\|_h + \|u\|_h). \quad (202)$$

(A3) For every smooth doubled spinor ψ there are samples $S_h \psi$ with

$$W_h S_h \psi \rightarrow \psi, \quad W_h D_h S_h \psi \rightarrow \hat{D} \psi. \quad (203)$$

The global samples use the transition functions of the fixed spin bundle on shared degrees of freedom, including coordinate and density changes. When real and grading structures are asserted, their finite actions intertwine these reconstructions and converge to the declared continuum actions.

Here is the cancellation that a proposed atlas assembly must respect. Choose fixed smooth real cutoffs χ_α supported in chart interiors with $\sum_\alpha \chi_\alpha^2 = 1$. After transporting local spinors and their density factors to the same bundle, the continuum first-order identity is

$$\sum_\alpha \chi_\alpha \hat{D} \chi_\alpha = \hat{D}, \quad \sum_\alpha \chi_\alpha [\hat{D}, \chi_\alpha] = \frac{1}{2} [\hat{D}, \sum_\alpha \chi_\alpha^2] = 0. \quad (204)$$

Thus the apparent order-one derivatives of the cutoffs cancel; one must not assert that they individually tend to zero. For a finite-range Wilson operator on a common coordinate mesh the analogous localization identity is

$$\sum_\alpha M_{\chi_\alpha, h} W_h^\Omega M_{\chi_\alpha, h} - W_h^\Omega = \frac{1}{2} \sum_\alpha [M_{\chi_\alpha, h}, [W_h^\Omega, M_{\chi_\alpha, h}]], \quad (205)$$

and its norm is at most $Ch \sum_{\alpha} \|\nabla \chi_{\alpha}\|_{\infty}^2$. Each matrix entry contains two cutoff differences and only one factor h^{-1} . Fixed collars keep all such finite-range stencils away from auxiliary chart boundaries for sufficiently small h .

On a shared geometric link $x \rightarrow y$, spin transition matrices satisfy the exact covariance identity

$$U_{\alpha\beta}(x) T_{xy}^{\beta} U_{\alpha\beta}(y)^{-1} = T_{xy}^{\alpha}.$$

It removes frame mismatch on that link. It does not identify different coordinate-grid links or supply a stable interpolation between unrelated grids. For independently meshed charts, such interpolation and its first-order error bounds must be verified in (201)–(203). In particular, positivity of a Gram matrix is not a substitute for the assembled graph estimate (202). The local Wilson estimates above establish the local ingredients; the existence of a global assembly satisfying all these properties is an additional construction requirement.

Theorem S9.8 (Norm-resolvent convergence for compatible stable atlases). *For a compatible stable spin discretization in the sense of Definition S9.7, and every $z \notin \mathbb{R}$,*

$$\left\| W_h(D_h - z)^{-1} W_h^* - (\widehat{D} - z)^{-1} \right\| \rightarrow 0. \quad (206)$$

The same argument on a fixed smooth periodic coordinate extension gives

$$\left\| \mathcal{J}_h^0(D_{g,h}^W - z)^{-1} (\mathcal{J}_h^0)^* - (\widehat{D}_g - z)^{-1} \right\| \rightarrow 0. \quad (207)$$

This local statement is about that self-adjoint periodic extension, not an unspecified self-adjoint operator on an open chart. Isolated bounded-energy spectral projections converge, and compatible real and grading structures are retained as specified in Definition S9.7.

Proof. Write $R_h(z) = W_h(D_h - z)^{-1} W_h^*$ and $R(z) = (\widehat{D} - z)^{-1}$. Self-adjointness gives

$$\|(D_h - z)^{-1}\| \leq |\operatorname{Im} z|^{-1}, \quad \|D_h(D_h - z)^{-1}\| \leq 1 + |z| |\operatorname{Im} z|^{-1}. \quad (208)$$

For $\|f\| \leq 1$ put $u_h = (D_h - z)^{-1} W_h^* f$. The global graph estimate bounds $\|u_h\|_{1,h}$ uniformly. Therefore $I_h u_h$ lies in a bounded H^1 set, while $\|W_h u_h - I_h u_h\| \rightarrow 0$ uniformly on this resolvent unit ball. Rellich compactness proves collective compactness of the embedded resolvents along the refining sequence. This step uses the reconstruction estimate and does not assign an H^1 norm to a piecewise-constant function.

For fixed f , let $W_h u_h \rightarrow u$ along a subsequence. Test the discrete resolvent equation against $S_h \psi$, with ψ any smooth doubled spinor. Self-adjointness and (203) pass the equation to $(\widehat{D} - z)u = f$ in the distributional sense. Since smooth spinors are a core for the self-adjoint operator $\widehat{D}$, this places u in its adjoint domain, which is H^1 , and gives $u = R(z)f$. Every convergent subsequence has this limit, so $R_h(z) \rightarrow R(z)$ strongly. Applying the same argument at $\bar{z}$ also gives strong convergence of the adjoints.

For clarity, this last fact is needed for the upgrade to norm convergence. If that convergence failed, choose unit vectors f_h with $\|(R_h(z) - R(z))f_h\| \geq \epsilon$. Passing to subsequences, let $f_h \rightharpoonup f$ and use collective compactness to make $R_h(z)f_h$ converge in norm. For each fixed v , strong adjoint convergence implies

$$\langle v, R_h(z)f_h \rangle = \langle R_h(\bar{z})v, f_h \rangle \rightarrow \langle R(\bar{z})v, f \rangle.$$

Hence its norm limit is $R(z)f$. Compactness of $R(z)$ also gives $R(z)f_h \rightarrow R(z)f$, a contradiction. This proves (206). On the periodic extension, (198), (195), and (199) provide the same hypotheses. The spectral-projection conclusion is the usual consequence of self-adjoint norm-resolvent convergence for isolated spectral clusters. $\square$

Corollary S9.9 (Unconditional periodic flat-spin norm-resolvent convergence). *Let $M = \mathbb{R}^d/\mathbb{Z}^d$ carry a constant flat metric and the periodic spin structure. Use the uniform periodic lattice $Q_h = h\mathbb{Z}^d/\mathbb{Z}^d$, constant Clifford coefficients, trivial spin parallel transport, and the Wilson operator (191) with fixed $\varpi > 0$. With the isometric piecewise-constant embedding $\mathcal{J}_h^0$,*

$$\left\| \mathcal{J}_h^0 (D_{g,h}^W - z)^{-1} (\mathcal{J}_h^0)^* - (\widehat{D}_M - z)^{-1} \right\| \longrightarrow 0, \quad z \in \mathbb{C} \setminus \mathbb{R}. \quad (209)$$

Thus the compatible-stability hypotheses are automatically satisfied on this periodic flat class.

Proof. There is no atlas-gluing problem. The constant-coefficient Fourier estimate (196) is global and gives (202); the periodic affine and piecewise-constant reconstructions satisfy (201); and Lemma S9.6 gives (203) globally on smooth periodic spinors. Moreover $\mathcal{J}_h^0 (\mathcal{J}_h^0)^*$ is the cellwise conditional expectation and converges strongly to the identity on $L^2(M; S \otimes \mathbb{C}^2)$. Hence all conditions of Definition S9.7 hold and Theorem S9.8 gives (209). $\square$

S9.4 Independent metric approximation and the spin conclusion

The endpoint-block construction and its complete uniform geodesic-convergence proof are given in Section S4. They are independent of the spin-assembly hypotheses and are not repeated here.

Corollary S9.10 (Conditional stable compact-spin approximation). *A closed marked Riemannian spin manifold admitting a compatible stable spin discretization has a finite-dimensional approximation of its doubled spin Dirac operator in generalized norm resolvent, together with convergent independent metric heads. Consequently, essential surjectivity of this particular atlas-based spin construction follows if such compatible stable discretizations are supplied for all target manifolds.*

Proof. Apply Theorem S9.8 to the spin operators and Theorem S4.1 to the independent metric heads. The last assertion records explicitly the additional existence requirement. $\square$

S10 Explicit minimal realization of positive Stieltjes memory

Section S7 uses the classical source-minimal positive realization theorem. For completeness, this extension retains the explicit construction and uniqueness proof. It starts from the same bounded positive operator-valued measure on $(0, \infty)$ as Definition S7.1; no additional static or moment determinacy hypothesis is imposed here.

Construction S10.1 (Canonical minimal Stieltjes dilation). Let $\mathcal{S}_\mu$ be the finite H -valued Borel simple functions and set

$$\left\langle \sum_j \mathbf{1}_{E_j} \xi_j, \sum_k \mathbf{1}_{F_k} \eta_k \right\rangle_\mu = \sum_{j,k} \langle \xi_j, \mu(E_j \cap F_k) \eta_k \rangle.$$

Quotient by the null space and complete to K_μ . Multiplication by $\mathbf{1}_E$ defines a projection-valued measure $P_\mu(E)$, and

$$C_\mu = \int \lambda \, dP_\mu(\lambda), \quad V_\mu \xi = [\mathbf{1}_{(0,\infty)} \xi], \quad B_\mu = V_\mu^*.$$

Theorem S10.2 (Operator-measure realization and minimality). *Every bounded positive operator-valued measure satisfies*

$$\Sigma_\mu(z) = B_\mu(C_\mu - z)^{-1}B_\mu^*. \quad (210)$$

Moreover

$$K_\mu = \overline{\text{span}}\{P_\mu(E)B_\mu^*\xi : E \in \mathfrak{B}((0, \infty)), \xi \in H\}. \quad (211)$$

Conversely, if $C \geq 0$ is self-adjoint on K with $E_C(\{0\}) = 0$, $B : K \rightarrow H$ is bounded, and

$$K = \overline{\text{span}}\{E_C(E)B^*H\},$$

then $\mu(E) = BE_C(E)B^*$ is a bounded positive operator-valued measure with self-energy $B(C-z)^{-1}B^*$. Two minimal realizations of the same μ are related by a unique unitary U satisfying

$$UC_1 = C_2U, \quad UB_1^* = B_2^*. \quad (212)$$

Proof. The spectral calculus gives

$$B_\mu(C_\mu - z)^{-1}B_\mu^* = V_\mu^* \left(\int \frac{1}{\lambda - z} dP_\mu(\lambda) \right) V_\mu = \int \frac{1}{\lambda - z} d\mu(\lambda).$$

Every simple function is a finite sum of vectors $P_\mu(E)V_\mu\xi$, proving (211).

For the converse, the spectral theorem gives positivity and countable additivity of $BE_C(E)B^*$. On the algebraic span of spectral-source vectors define

$$U_0 \left(\sum_j E_C(E_j)B^*\xi_j \right) = \sum_j P_\mu(E_j)V_\mu\xi_j.$$

The squared norms on the two sides are both

$$\sum_{j,k} \langle \xi_j, BE_C(E_j \cap E_k)B^*\xi_k \rangle,$$

so U_0 is well defined and isometric. Source minimality makes its extension unitary. It intertwines every spectral projection, hence C , and sends B^* to V_μ . Applying the same construction to two realizations proves existence and uniqueness of (212). $\square$

S11 Finite rational indefinite memory

The positive Stieltjes theory in Section S7 is not the only finite realization class considered here. A finite Krein/Pontryagin tail is instead controlled by the number of negative squares of its generalized Nevanlinna kernel.

Definition S11.1 (Normalized rational Hermitian dynamic function). Let H be finite dimensional and let Q be an $\text{End}(H)$ -valued rational function satisfying

$$Q(\bar{z})^* = Q(z), \quad \lim_{z \rightarrow \infty} Q(z) = 0. \quad (213)$$

Write its Laurent expansion at infinity as

$$Q(z) = - \sum_{n \geq 0} M_n z^{-n-1}, \quad M_n = M_n^*. \quad (214)$$

Let $\mathcal{K}_Q$ be the span of Hankel columns $c_j(h) = (M_{j+r}h)_{r \geq 0}$ and define

$$[c_i(u), c_j(v)]_Q = \langle u, M_{i+j}v \rangle. \quad (215)$$

After quotienting the radical of the Hermitian form, let $A_Q c_j(h) = c_{j+1}(h)$ and $\Gamma_Q h = c_0(h)$.

Theorem S11.2 (Complete normalized rational generalized-Nevanlinna realization). *The construction above is a minimal finite Pontryagin realization and*

$$Q(z) = \Gamma_Q^+ (A_Q - z)^{-1} \Gamma_Q. \quad (216)$$

Its ordinary dimension is the stabilized coefficient-Hankel rank

$$\dim \mathcal{K}_Q = \text{McM}(Q), \quad (217)$$

and its negative index is the inertia of (215). The Nevanlinna kernel

$$N_Q(z, w) = \frac{Q(z) - Q(w)^*}{z - \bar{w}} \quad (218)$$

has exactly that many negative squares. Every other minimal finite Pontryagin realization of Q is related to this one by a unique source-fixing Pontryagin unitary.

Proof. By construction, the output functional L_Q on the Hankel column model obeys $L_Q A_Q^n \Gamma_Q = M_n$. For large z ,

$$L_Q (A_Q - z)^{-1} \Gamma_Q = - \sum_{n \geq 0} \frac{M_n}{z^{n+1}} = Q(z),$$

and rational continuation proves (216). The power-cyclic span of $\Gamma_Q H$ is all of $\mathcal{K}_Q$ by definition. In finite dimension, the resolvent-source span equals the power-cyclic span: expansion at infinity gives one inclusion and a finite Vandermonde inversion at sufficiently many regular spectral parameters gives the other. Thus the realization is source minimal.

The dimension is the stabilized Hankel rank by construction. Choosing a Hilbert majorant and a congruence which writes the nondegenerate Hankel form as a fundamental symmetry J_Q gives $A_Q^* J_Q = J_Q A_Q$ and $Q(z) = \Gamma_Q^* J_Q (A_Q - z)^{-1} \Gamma_Q$. The standard resolvent identity gives

$$N_Q(z, w) = \Gamma_Q^* J_Q (A_Q - z)^{-1} (A_Q - \bar{w})^{-1} \Gamma_Q.$$

Every finite kernel Gram is therefore a compression of the Pontryagin form, so it has at most its negative index. Source minimality makes the resolvent-source vectors span the whole state space and attains the full index. Finally, if two minimal realizations have the same Q , their complete Hankel moments agree. Mapping $A_Q^j \Gamma_Q h$ to the corresponding cyclic vector in the second realization preserves the indefinite form, is onto, fixes the source, and intertwines the state operators. Uniqueness follows from cyclicity. $\square$

Theorem S11.3 (Pole-Hankel invariants). *For a normalized rational Hermitian Q :*

- (P1) *the principal part at each pole determines the local McMillan degree and all Jordan block sizes through shifted principal-part Hankel ranks;*
- (P2) *at each real pole, the principal-part Hankel Gram determines the complete local Pontryagin inertia;*

- (P3) *every nonreal conjugate pole pair forms a neutral paired root space, and its upper-half-plane local McMillan degree contributes equally to positive and negative index;*
- (P4) *the sum of the local contributions equals the minimum global negative index.*

Proof. At a pole λ , multiply by a sufficiently high power of $(z - \lambda)$ and read the finite principal-part coefficient sequence. The ranks of its shifted Hankel matrices are the controllability/observability ranks of the nilpotent local state operator and their successive differences recover the Jordan chain lengths. For a real pole the local Hankel form is Hermitian and Sylvester inertia is invariant under every minimal state-coordinate change, hence gives the local Pontryagin signature. Hermitian symmetry pairs a nonreal root space with its conjugate; their indefinite form is neutral off-diagonal, so the two signs occur with equal multiplicity. The primary decomposition is Pontryagin orthogonal, and the global index is the sum of local indices. $\square$

A general finite rational Hermitian function has a unique split

$$Q(z) = P(z) + Q_0(z), \quad \lim_{z \rightarrow \infty} Q_0(z) = 0, \quad (219)$$

with a Hermitian polynomial P . The following descriptor statement uses the minimal regular, source-fixing Pontryagin descriptor convention and its finite/infinity primary decomposition; it is separate from the explicit strictly proper Hankel construction above.

Corollary S11.4 (All finite rational Hermitian dynamics). *The strictly proper part Q_0 is realized by [Theorem S11.2](#). The nonconstant polynomial jet P has a canonical finite infinity-chain/regular descriptor realization. Their primary direct sum gives an equivalence, up to source-fixing descriptor Pontryagin unitary,*

$$\begin{array}{c} \text{finite rational Hermitian dynamic functions} \\ \longleftrightarrow \\ \text{minimal regular Pontryagin descriptor realizations.} \end{array} \quad (220)$$

This includes finite real and nonreal poles, arbitrary finite Jordan chains, and all polynomial jets at infinity.

Proof. The pencil of any regular finite descriptor realization has a primary decomposition at infinity and at finite spectral points. On the finite primary block the transfer function is strictly proper and is minimized by [Theorem S11.2](#). On the generalized zero eigenspace of an anchored descriptor pencil, the resolvent expansion is polynomial and is represented by the canonical nilpotent infinity chain. The direct sum attains both minimum state dimensions and indices. Any second minimal realization splits into the same invariant primary blocks, and the unique source-fixing unitaries on the finite and infinity blocks combine to the unique descriptor equivalence. $\square$

Remark S11.5 (Positive Stieltjes memory is the index-zero branch). A positive Stieltjes memory packet has zero negative-square index. A nonzero physical Pontryagin index requires an independently occurring fundamental-symmetry or signed packet. Neither OS reflection, temporal orientation, charge conjugation, nor a determinant/Pfaffian line determines that sign by itself. The unrestricted nonrational infinite-dimensional indefinite essential image is not claimed here.

S11.1 Effective-space compression of indefinite memory

Proposition S11.6 (Negative-square index under effective-space compression). *Let Q be a normalized rational Hermitian dynamic function and $V : H' \rightarrow H$ an isometry. Put $Q_V(z) = V^*Q(z)V$. Then the negative-square index of N_{Q_V} does not exceed that of N_Q , and the minimal finite realization dimension of Q_V does not exceed that of Q . Nested isometries give the same compressed function as their composite.*

Proof. For any finite set of regular spectral parameters, the kernel Gram of $N_{Q_V}(z, w) = V^*N_Q(z, w)V$ is a congruence of the corresponding kernel Gram of N_Q . Its number of negative eigenvalues therefore cannot increase. A source-minimal realization of Q gives a realization of Q_V on the same state space by replacing its source map Γ by ΓV ; minimizing cannot increase the state dimension. Finally, $V_2^*V_1^*Q(z)V_1V_2 = (V_1V_2)^*Q(z)(V_1V_2)$ gives composition. $\square$

S12 Joint realizations and completed inverse fibres

This extension concerns the declared retained-coordinate realization model. The elementary joint Gram classification and compact inverse-system pruning are distinguished from the broader fibre descriptions, which use the full-reduction and compatibility hypotheses stated below. In particular, these hypotheses are not derived from the stationary-current theorem, and the section does not assert that current, memory, and monodromy exhaust every possible predictive realization.

S12.1 Joint Gram realizations and the finite stratified fibre

Let $E_1, \dots, E_m$ be finite Hilbert spaces representing minimal marginal realization spaces, normalized so that their marginal Gram operators are the identity. A joint realization is encoded by a block correlation operator

$$\mathcal{K} = [K_{ab}]_{a,b=1}^m, \quad K_{aa} = I_{E_a}, \quad \mathcal{K} \succeq 0. \quad (221)$$

Let Corr_X denote this correlation spectrahedron and let $\mathcal{G}_X$ be the product of the unitary gauge groups of the marginal realization spaces.

Theorem S12.1 (Joint Gram classification). *Every minimal joint realization of the retained marginal Gram blocks is classified, up to a unitary on the common realization space, by a point of $\text{Corr}_X / \mathcal{G}_X$. The support rank of $\mathcal{K}$ is the minimum possible joint realization dimension. A representative $\mathcal{K}$ has residual gauge group exactly $\text{Stab}_{\mathcal{G}_X}(\mathcal{K})$.*

Proof. A joint realization has a Gram factorization $\mathcal{K} = S^*S$, with the identity diagonal expressing the chosen normalization of the marginal blocks. Conversely every positive block operator with identity diagonal admits such a factorization. Restricting the realization space to $\overline{\text{Ran } S}$ gives the minimal realization, whose dimension is $\text{rank } \mathcal{K}$ in the finite case. Two minimal Gram factorizations of the same positive operator are related by a unique unitary on the common range. Independent changes of marginal frames act by block congruence, and the residual action is precisely the stated stabilizer. $\square$

Proposition S12.2 (Pairwise positivity need not give a joint realization). *Pairwise admissible Gram blocks need not assemble into one positive joint Gram operator. For example,*

$$\begin{pmatrix} 1 & -3/4 & -3/4 \\ -3/4 & 1 & -3/4 \\ -3/4 & -3/4 & 1 \end{pmatrix} \quad (222)$$

has every 2×2 principal block positive but has eigenvalue $-1/2$. Thus pairwise compatibility does not imply global Gram compatibility.

Let $\mathcal{J}_X$ be the current fibre, $\mathcal{M}_X$ the complete dynamic-memory fibre over the fixed spectral data, $\mathcal{P}_X$ the retained auxiliary-marking fibre, $\mathcal{C}_X^{\text{cal}}$ the residual calibration class, and $\mathcal{R}_X$ the marked monodromy representation space. We call a finite target *fully reduced* when all null and spectator directions have been quotiented out, the domain, weight, orientation, determinant-line, normalization, and auxiliary-marking conventions are fixed, every retained Gram block is minimal, and a specified finite family of bounded separating functionals (equivalently, finite-rank defect operators) has trivial common kernel on any additional retained direction. This last condition is the standard separation hypothesis that replaces the programme-specific “zero-exhaustion” terminology.

Theorem S12.3 (Finite stratified fibre in the fully reduced coordinate model). *For a completed fully reduced finite target $\mathbf{S}^\sharp$,*

$$\text{RReal}_X^{\sharp, \min}(\mathbf{S}) \cong \bigsqcup_{[\mathcal{K}] \in \text{Corr}_X / \mathcal{G}_X} \frac{\mathcal{J}_X \times \mathcal{M}_X \times \mathcal{P}_X \times \mathcal{C}_X^{\text{cal}} \times \mathcal{R}_X}{\text{Stab}_{\mathcal{G}_X}(\mathcal{K})}. \quad (223)$$

The factors vary, respectively, directed dynamics, dynamical memory, auxiliary markings, calibration, and global bundle data while preserving the fixed spectral packet. In general the quotient is stratified and not a Cartesian product.

Proof. The declared coordinate-completeness part of full reduction restricts the realization class to the listed retained directions after the null quotients. The separating family distinguishes those directions; it does not establish that every predictive model belongs to this class. The graph inverse theorem classifies the directed-current freedom; the positive and Pontryagin/descriptor theorems classify returned dynamic memory; the retained auxiliary markings give $\mathcal{P}_X$; the calibration data give $\mathcal{C}_X^{\text{cal}}$; and marked bundle descent gives $\mathcal{R}_X$. Theorem S12.1 classifies all compatible joint Gram realizations and identifies their residual stabilizers. Equivalence of two fully reduced realizations is therefore simultaneous equivalence in these factors under the common stabilizer, yielding (223). Proposition S12.2 explains why the general fibre cannot be replaced by a product of independently quotiented factors. $\square$

S12.2 Quasilocal and semifinite completion

For completion one must distinguish the finite inverse fibre relevant to strictly positive realizations from a compact finite-stage carrier used in the projective limit. For the current factor define the closed polytope

$$\overline{\mathcal{J}}(G, c) = \{j \in \text{Ker } B : |j_e| \leq c_e\}, \quad (224)$$

whose relative interior is the strict-rate fibre $\mathcal{J}(G, c)$. It is compact. For positive memory, a finite-stage compactification is obtained by placing the operator-valued tail measure on the one-point compactification of the retained energy range and imposing a uniform bound on its total trace mass; in finite source dimension this set is weak-* compact. Correlation spectrahedra are compact, unitary gauge groups are compact, and for a finite incidence graph with compact packet automorphism group the marked-monodromy space is a compact quotient of a finite product of that group. We therefore formulate the completion theorem on compact finite-stage strata, rather than silently assuming compactness of the open physical fibres.

Definition S12.4 (Compact finite-stage inverse system). A compact finite-stage inverse system consists of increasing finite-trace screens $P_n \uparrow I$ and nonempty compact Hausdorff fully reduced strata obtained as the diagonal compact-group quotient

$$\bar{\mathfrak{F}}_n := \frac{\text{Corr}_n \times \bar{\mathcal{J}}_n \times \bar{\mathcal{M}}_n \times \bar{\mathcal{P}}_n \times \bar{\mathcal{C}}_n^{\text{cal}} \times \bar{\mathcal{R}}_n}{\mathcal{G}_n}. \quad (225)$$

Equivalently, its orbit-type decomposition is

$$\bar{\mathfrak{F}}_n \cong \bigsqcup_{[\mathcal{K}_n] \in \text{Corr}_n / \mathcal{G}_n} \frac{\bar{\mathcal{J}}_n \times \bar{\mathcal{M}}_n \times \bar{\mathcal{P}}_n \times \bar{\mathcal{C}}_n^{\text{cal}} \times \bar{\mathcal{R}}_n}{\text{Stab}(\mathcal{K}_n)}.$$

The displayed disjoint union is an orbit-type stratification, not a disjoint-union topology. The factors use the closed current polytope (224), weak-* compact tight memory sets with uniform trace-mass bounds, compact marking/calibration sets, and compact monodromy carriers, with the natural continuous $\mathcal{G}_n$ actions. Since $\mathcal{G}_n$ is compact, (225) is compact Hausdorff. There are continuous restriction maps $r_{m,n} : \bar{\mathfrak{F}}_m \rightarrow \bar{\mathfrak{F}}_n$ for $m \geq n$. Compatible self-adjoint resolvent data and a compatible normal semifinite weight are retained at each stage. The restriction maps satisfy $r_{n,n} = \text{id}$ and $r_{\ell,n} = r_{m,n} r_{\ell,m}$ for $\ell \geq m \geq n$. The operator-level completion statement additionally assumes that the transported limiting resolvent family satisfies the resolvent identity, adjoint symmetry, injectivity, and dense range, and that the retained finite-trace weights admit the compatible normal semifinite extension used below. These requirements are distinct from compactness of the coordinate inverse system.

For $m \geq n$ put

$$\bar{\mathfrak{F}}_n^{[m]} = r_{m,n}(\bar{\mathfrak{F}}_m), \quad \bar{\mathfrak{F}}_n^\infty = \bigcap_{m \geq n} \bar{\mathfrak{F}}_n^{[m]}. \quad (226)$$

Finally, choose compatible finite families $\mathcal{F}_n$ of bounded separating functionals or finite-rank defect operators such that the cofinal family $\bigcup_n \mathcal{F}_n$ has trivial common kernel on every additional retained quasilocal direction after quotienting by the displayed coordinates.

Lemma S12.5 (Canonical pruning of compact finite-stage fibres). *For every system of Definition S12.4, each stable image $\bar{\mathfrak{F}}_n^\infty$ is nonempty and compact, and*

$$r_{n+1,n}(\bar{\mathfrak{F}}_{n+1}^\infty) = \bar{\mathfrak{F}}_n^\infty. \quad (227)$$

Moreover $\bar{\mathfrak{F}}_n^\infty$ is exactly the level- n projection of $\varprojlim_m \bar{\mathfrak{F}}_m$ and is the largest subsystem in which every retained finite-stage point extends indefinitely.

Proof. For fixed n , the sets $\bar{\mathfrak{F}}_n^{[m]}$ are nonempty compact and decrease with m , so their intersection is nonempty compact. The forward inclusion in (227) is immediate. For the reverse inclusion, fix $a \in \bar{\mathfrak{F}}_n^\infty$ and consider

$$B_m = \{b \in \bar{\mathfrak{F}}_{n+1}^{[m]} : r_{n+1,n}(b) = a\}, \quad m \geq n+1.$$

Each B_m is nonempty compact and the family is decreasing, so its intersection contains a point of $\bar{\mathfrak{F}}_{n+1}^\infty$ mapping to a . The projection and maximality statements follow directly from the definition of the stable images. $\square$

Theorem S12.6 (Completed retained-coordinate inverse-fibre theorem). *For a compact finite-stage inverse system satisfying the compatible resolvent, weight, transport, tightness, and cofinal-separation hypotheses of Definition S12.4, the retained completed inverse fibre is*

$$\boxed{\text{RReal}_{\infty}^{\sharp, \min}(\mathcal{S}) \cong \varprojlim_n \overline{\mathfrak{F}}_n^{\infty}.} \quad (228)$$

The inverse limit is nonempty and compact. It contains no additional retained quasilocal coordinate beyond the displayed finite-stage factors. Its strict-rate physical part is obtained by intersecting the compatible threads with the relative interiors $\mathcal{J}_n \subset \overline{\mathcal{J}}_n$ and with the corresponding nondegenerate interiors of the other compactified factors. On a factor-preserving stratum the limit may be written componentwise before the compatible compact gauge quotient; without factor-preserving transport the stratified form is essential.

Proof. Lemma S12.5 replaces any possibly nonsurjective finite-stage system by its canonical stable subsystem, whose bonding maps are surjective. The product $\prod_n \overline{\mathfrak{F}}_n^{\infty}$ is compact, and the compatibility equations define closed subsets with the finite-intersection property; hence the inverse limit is nonempty compact.

Tightness and the weak-* compact memory completion prevent positive dynamic mass from disappearing without being represented by either a finite-energy tail or the explicit point-at-infinity contact component. Compatible resolvents determine a single closed self-adjoint operator by the resolvent identity, adjoint symmetry, injectivity and dense range. Compatible finite-trace weights increase to a unique lower-semicontinuous normal semifinite weight by monotone convergence, and compatible bundle transport fixes the global marked data. Finally, if an additional retained quasilocal coordinate existed, the cofinal separating family would contain a functional or defect operator nonzero on it, contradicting the trivial-common-kernel hypothesis. The finite classification is Theorem S12.3; taking its indefinitely extendible compact threads proves (228). $\square$

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